Upskilling Week
Barcelona

Mobile
WorldCapital
Barcelona

15:00h - 18:30h

Visión por Computador con Arduino UNO Q y Edge Impulse

Amb la col·laboració de

Generalitat de Catalunya

Barcelona

Barcelona Activa

WiFi

SSID: Cortesia
PSK: Cortesia022026

Before we start!

1. Download Arduino AppLab
   https://bit.ly/arduino-applab
2. Sign-up to Edge Impulse
   https://edgeimp.com/ta-july26
3. And sign-up to Arduino with **the same e-mail**
   https://bit.ly/arduino-account

Slides here: https://bit.ly/summer-ta26

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EDGE IMPULSE
a Qualcomm company

Computer Vision with Arduino UNO Q and Edge Impulse

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Agenda

1. **What is the AI, ML and Edge AI.**
2. **Train an Image Classification ML model.**
3. **Re-train an Object Detection ML model.**
4. **Test with the Arduino UNO Q.**
5. **Train a Visual Anomaly Detection ML model.**
6. **Learn about Model Cascading from Object Detection to VLM**

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What does the real-world look like?

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Machine Learning

Supervised learning

Task-driven

- Regression
- Classification
- Object detection

Unsupervised learning

Data-driven

- Clustering
- Segmentation
- Anomaly detection

Reinforcement learning

Learn from experience

- Robotics
- Games
- Recommender systems

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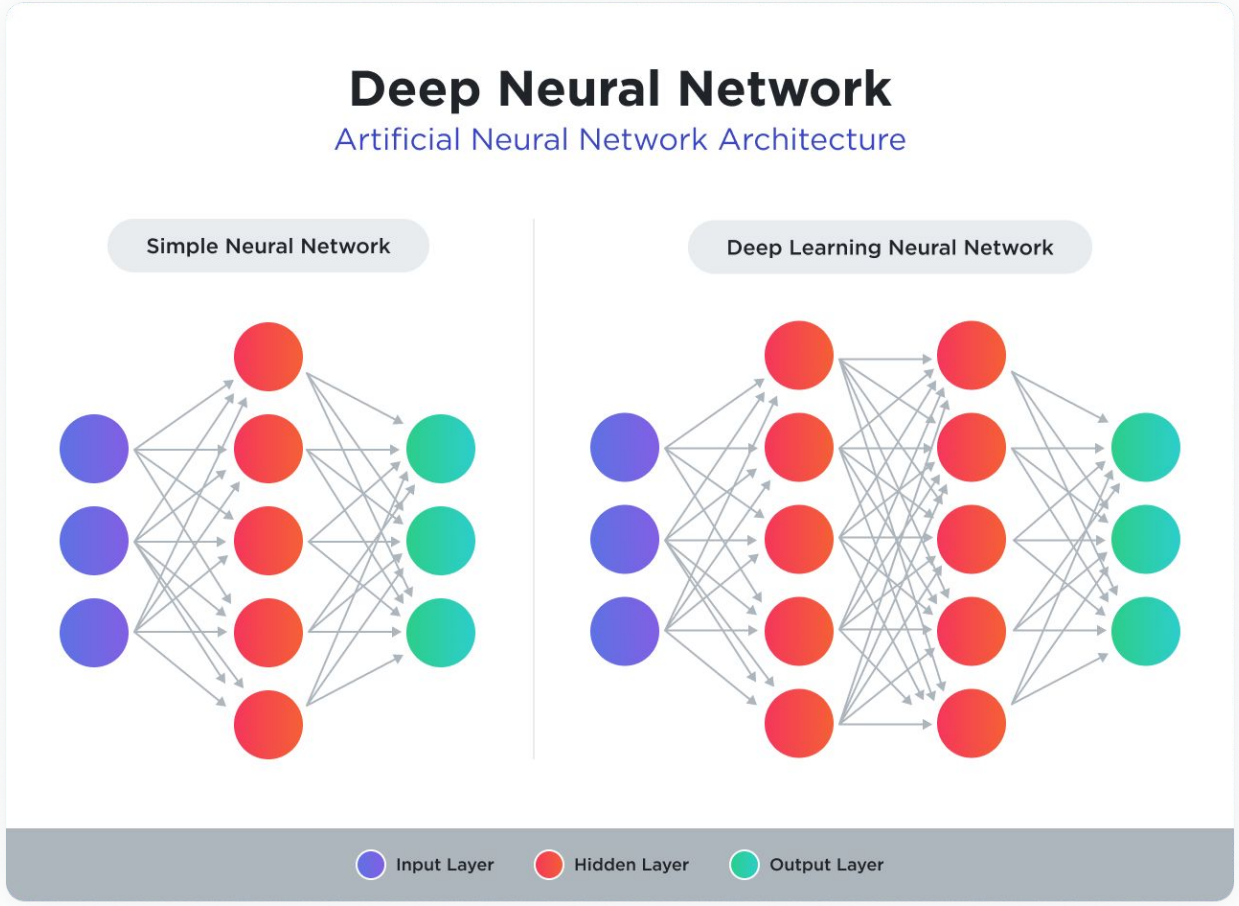
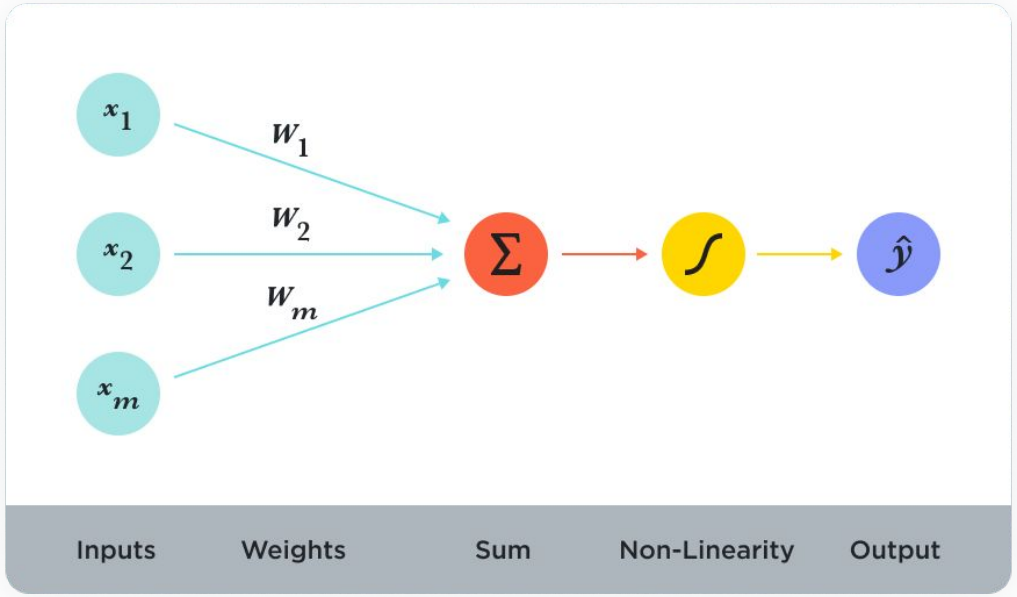
Neural Networks



Single Neuron



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What is Edge Impulse?

The End-to-End ML Ops platform for model development & model deployment to the edge:

Ingest & Build Data Sets

- Supports sensor, audio, and image data
- Collect and visualize datasets from real-world environments

Iterate and Develop Models

- Extract features and design model architectures
- Rapid iteration with built-in tools for labeling, training, and tuning

Optimize & Deploy

- Profile model performance for target hardware
- Compress, quantize, and export to MCUs, NPUs, and edge devices

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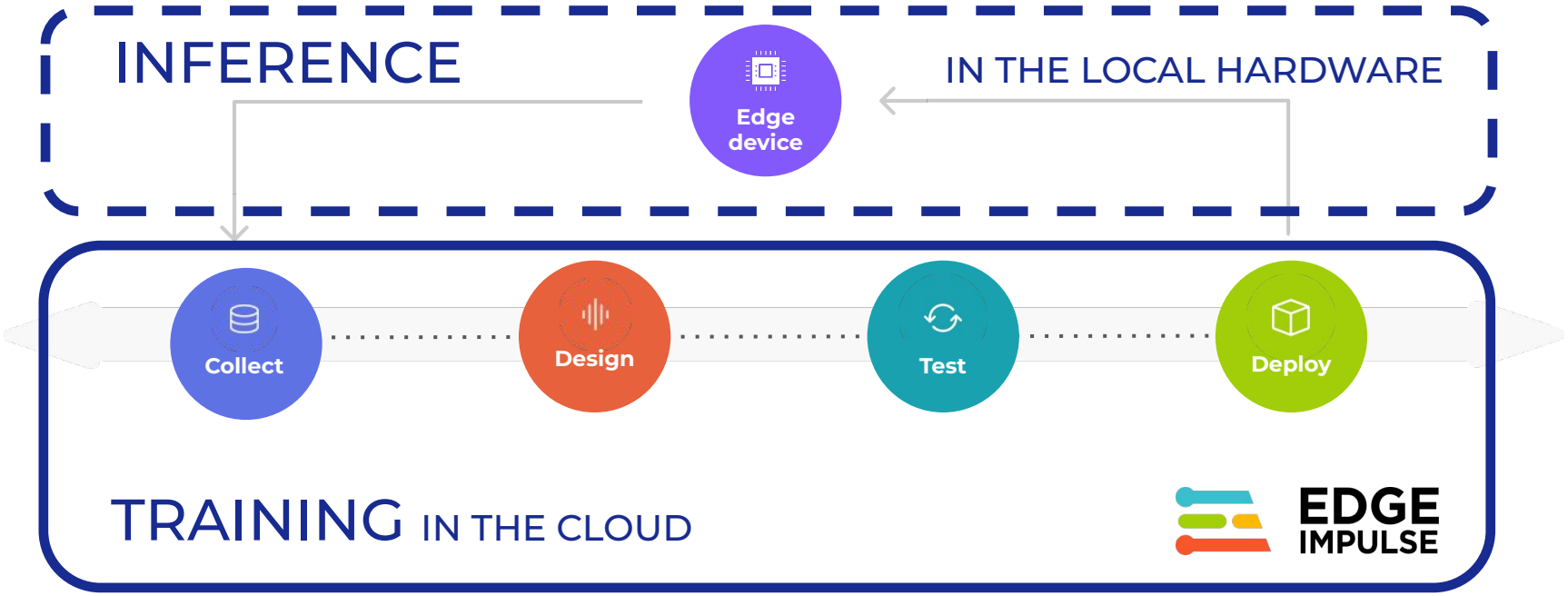
Edge AI



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Train and deploy Image Classification ML model

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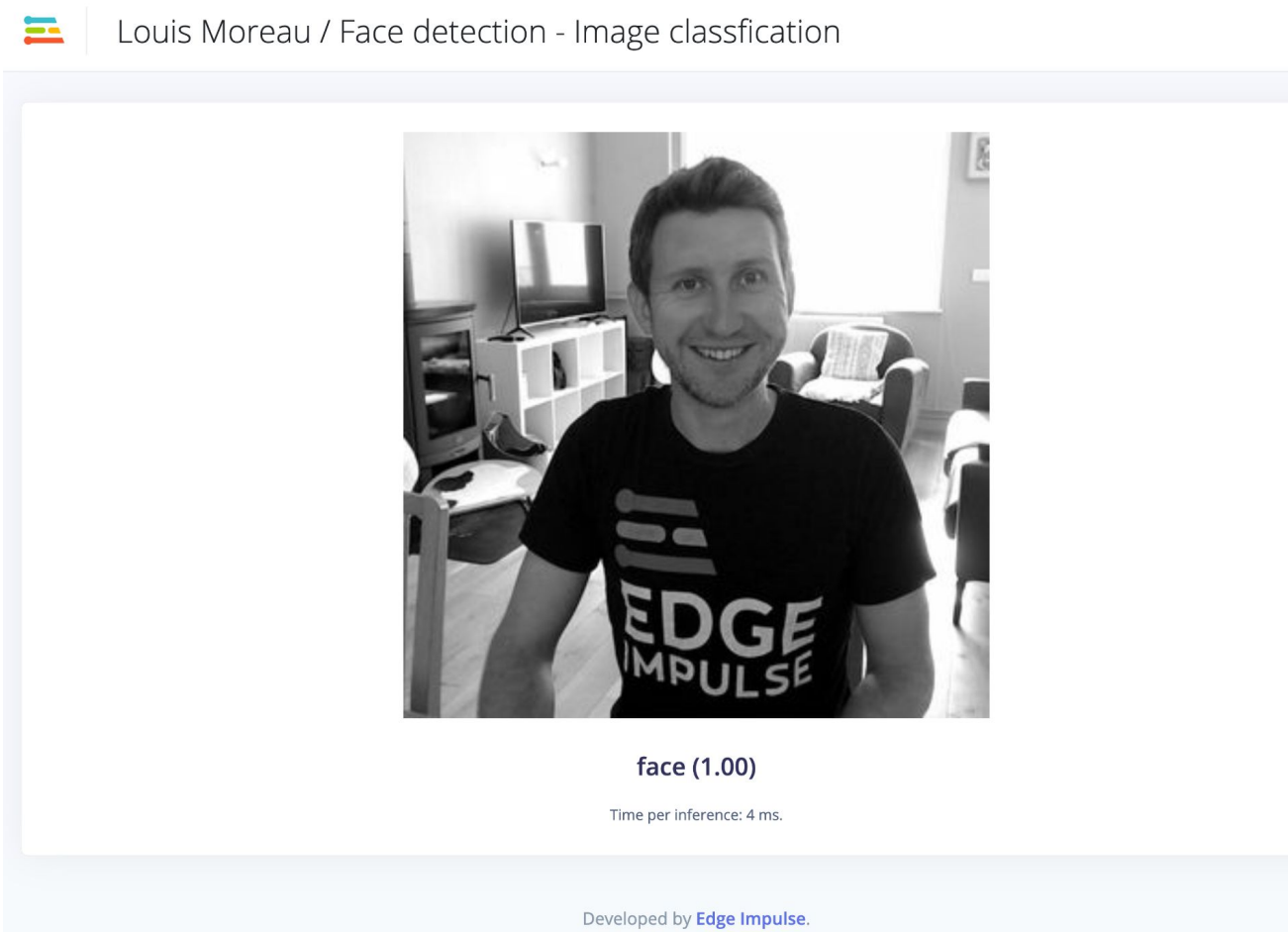
Image classification

The question the model is trying to answer is:

“Is there a face or not in the image?”



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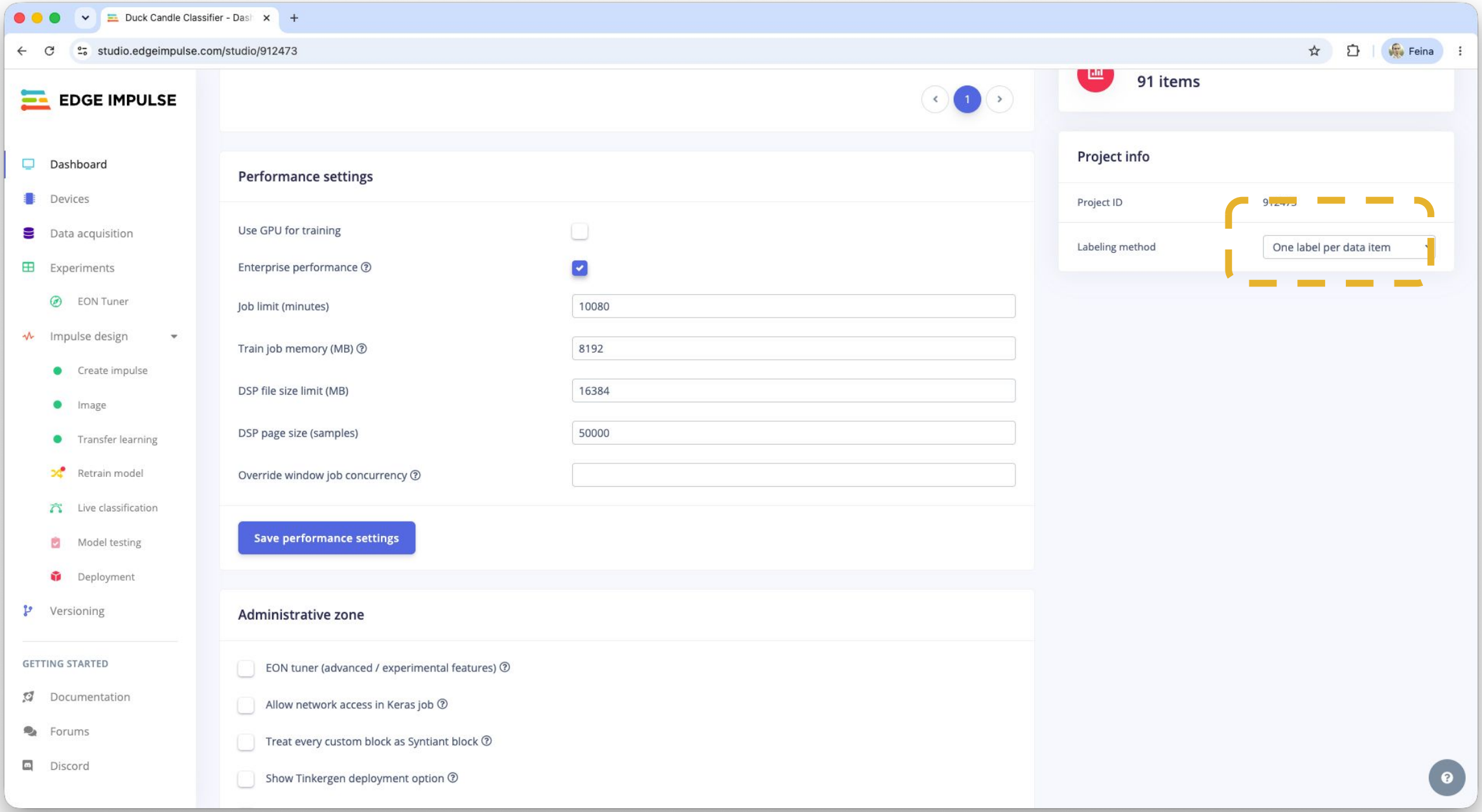

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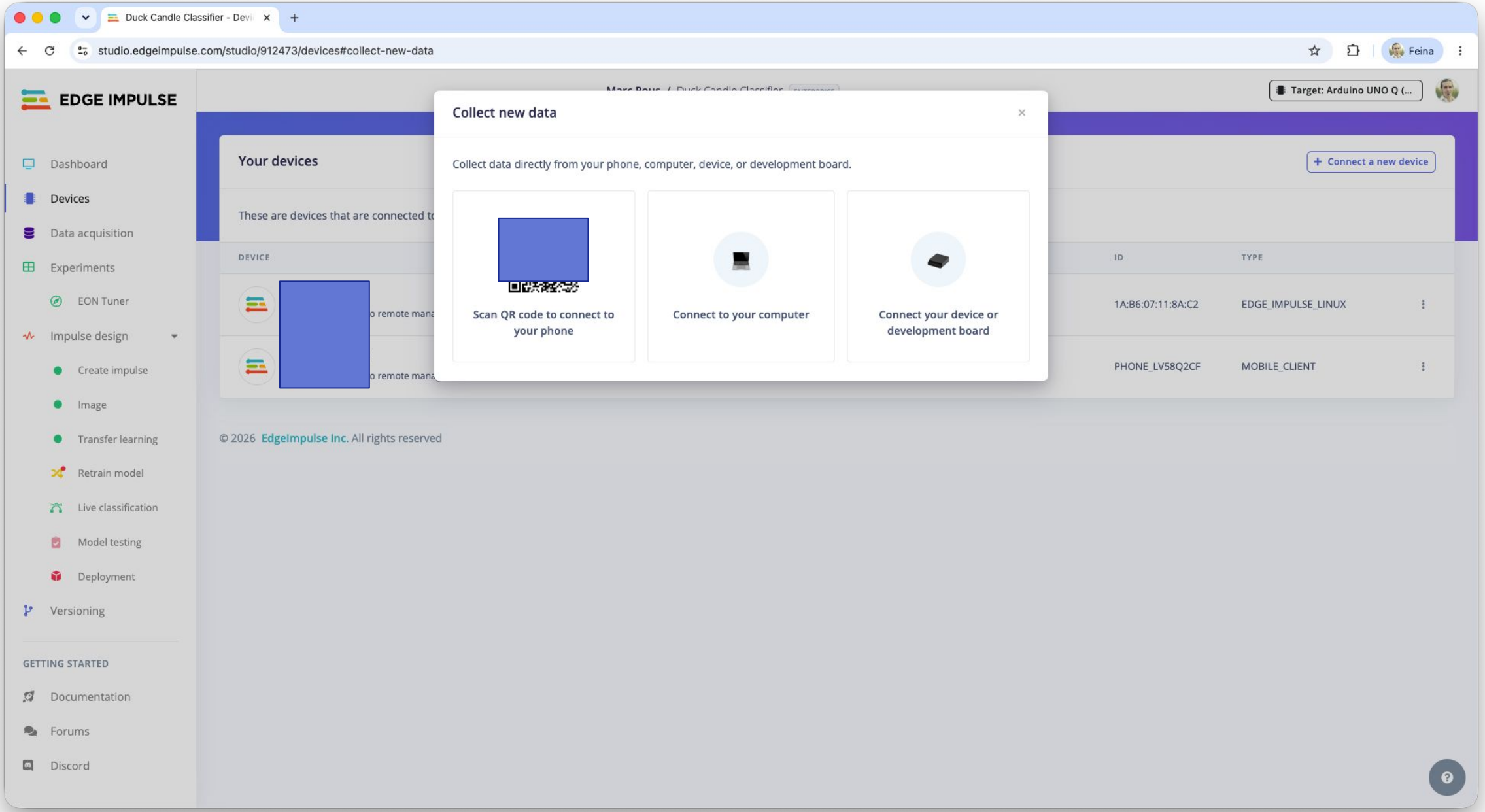

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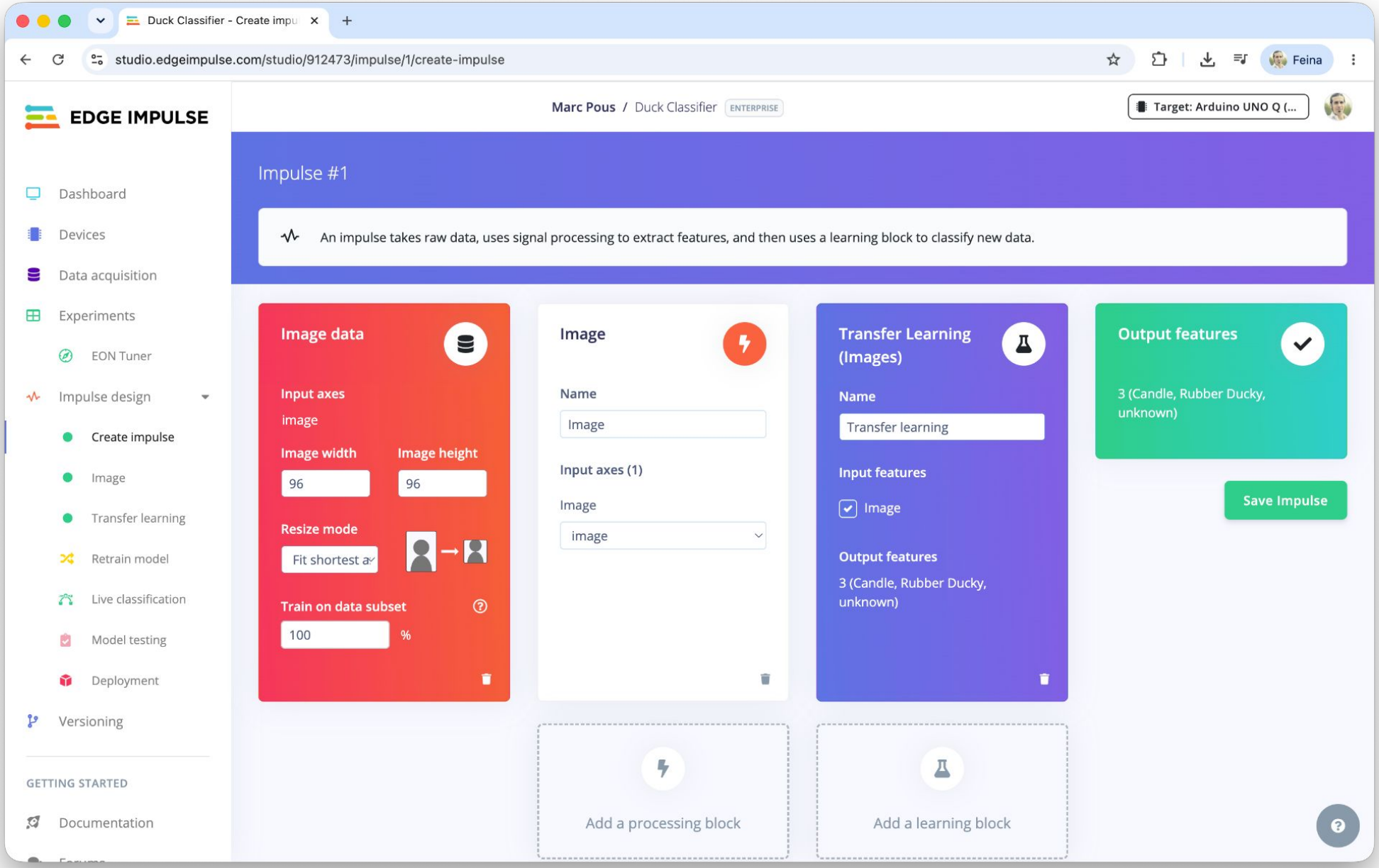

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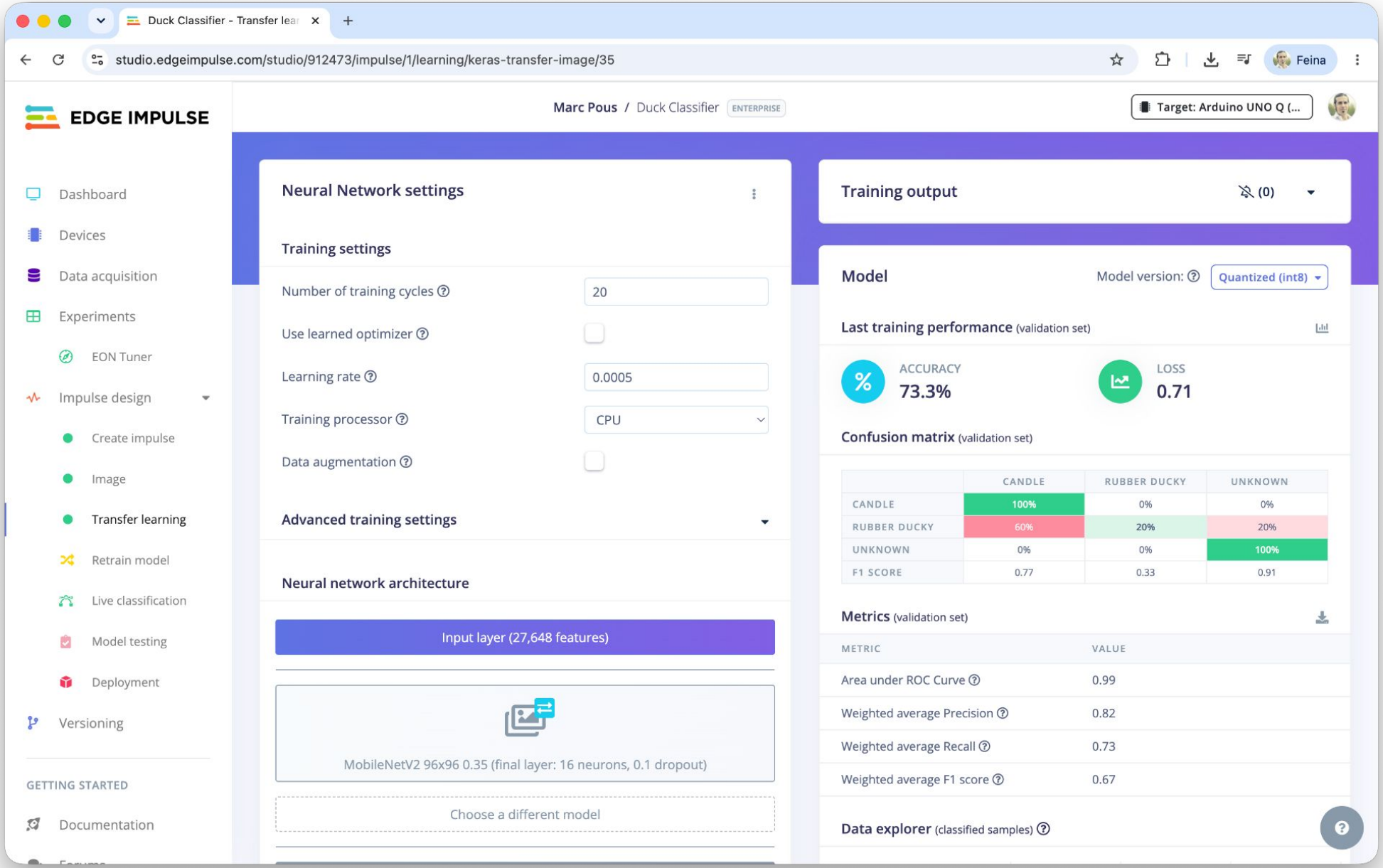

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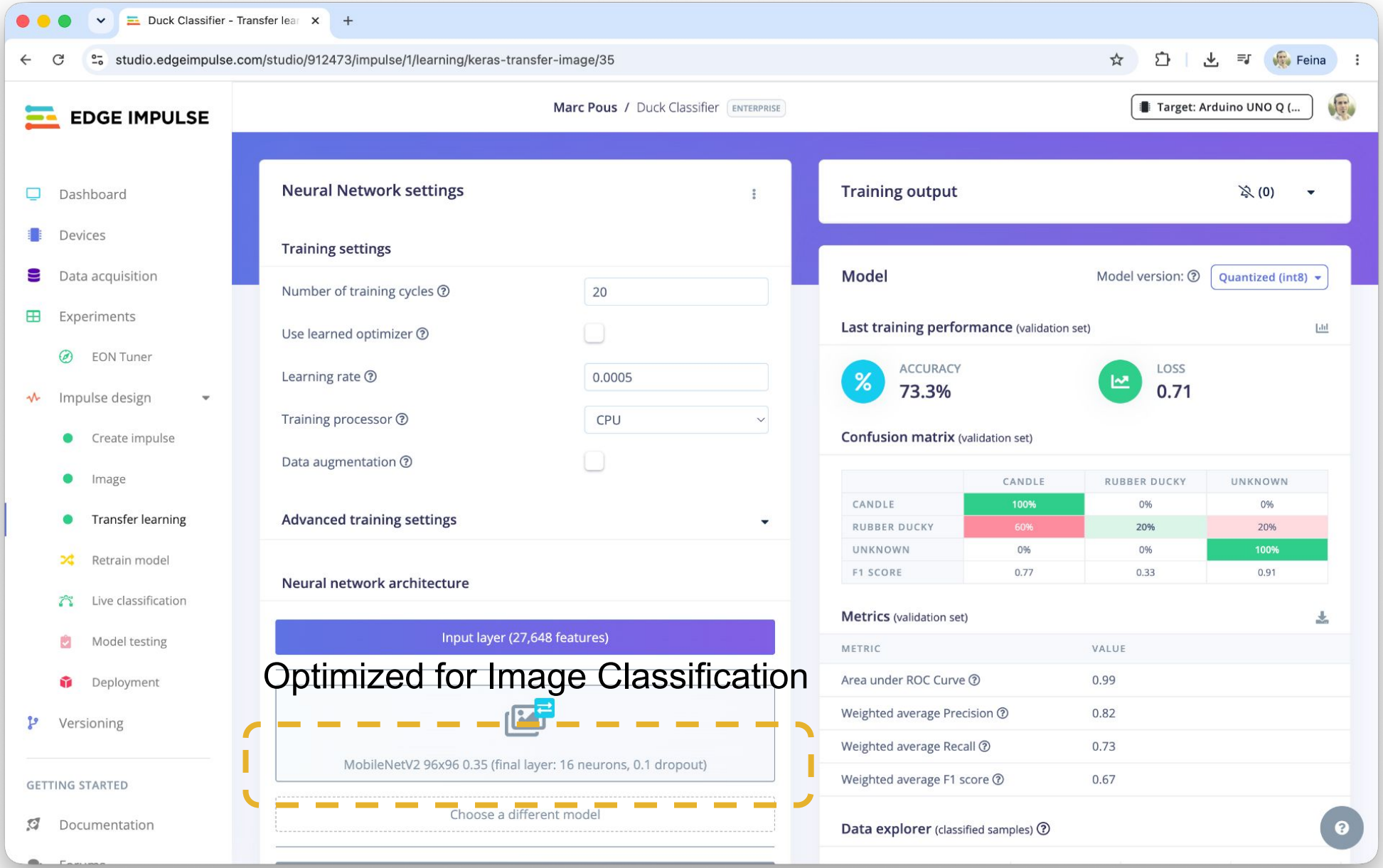

Optimized for Image Classification

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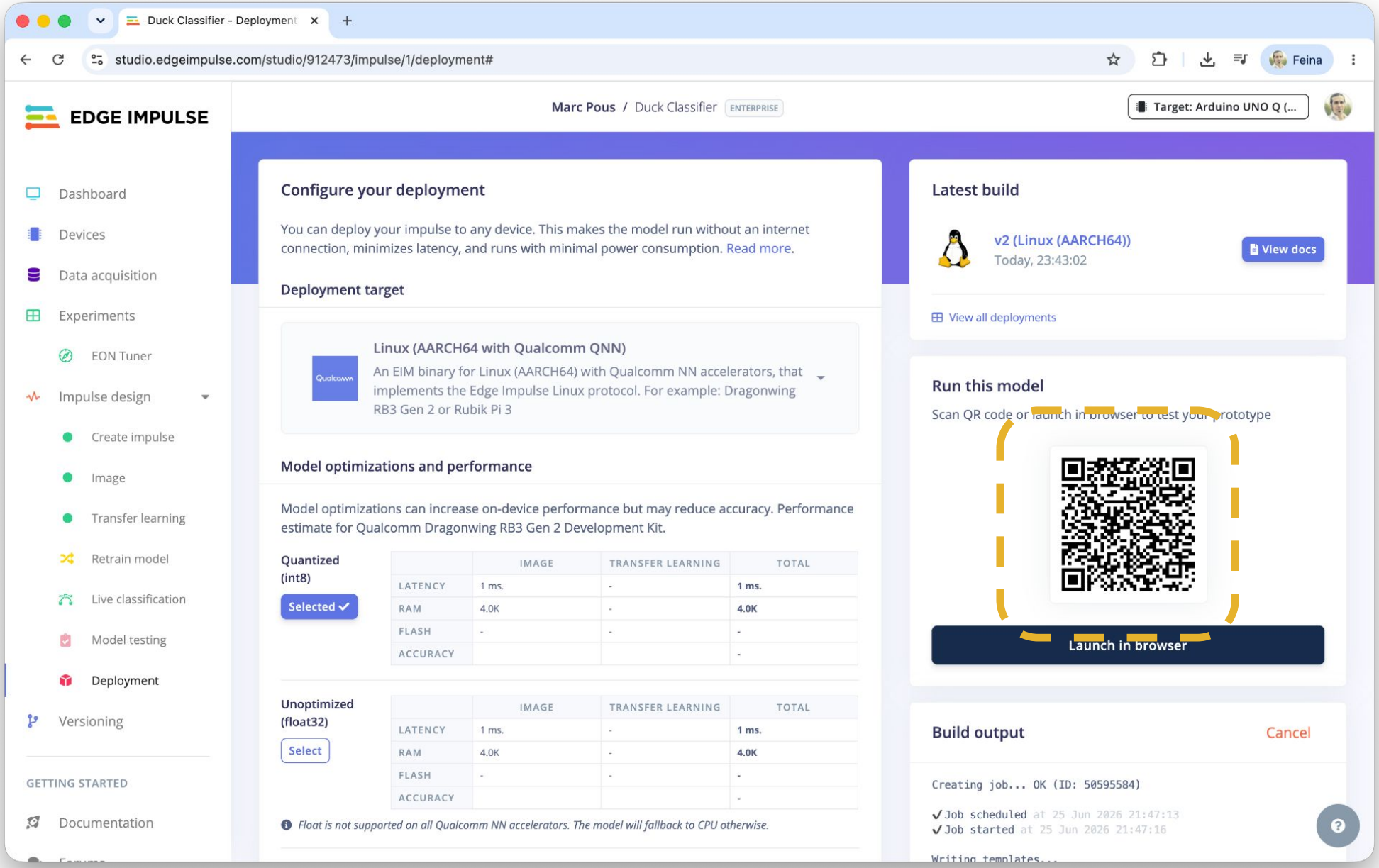

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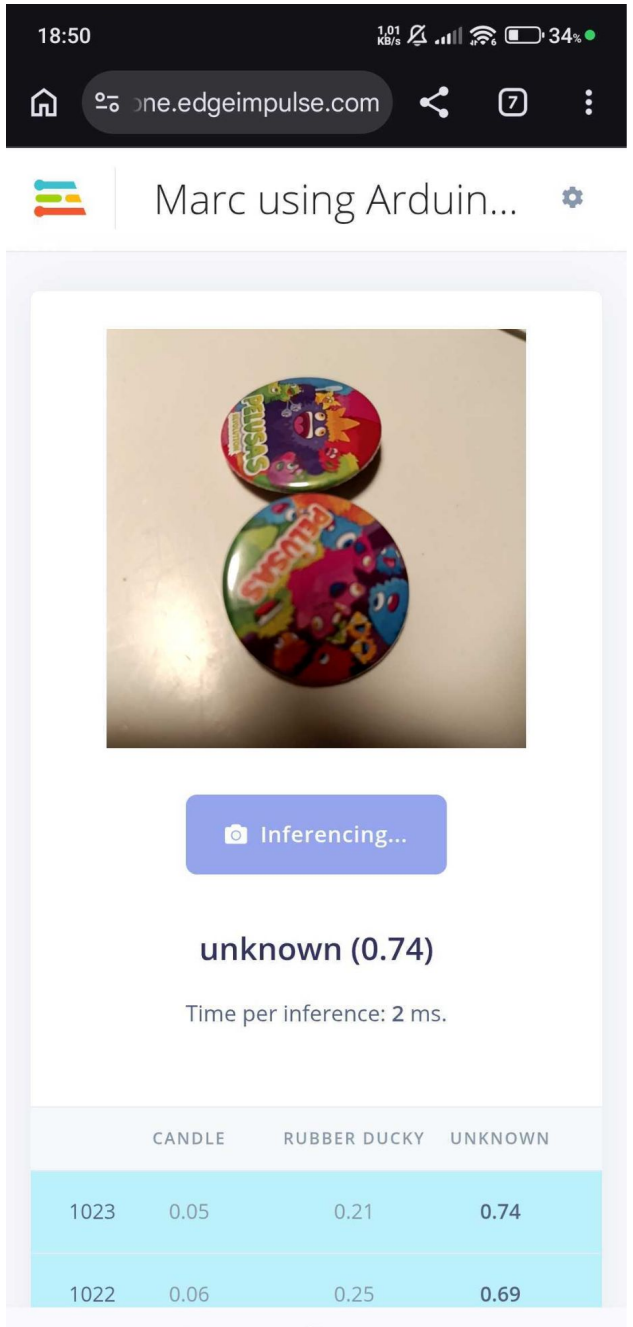

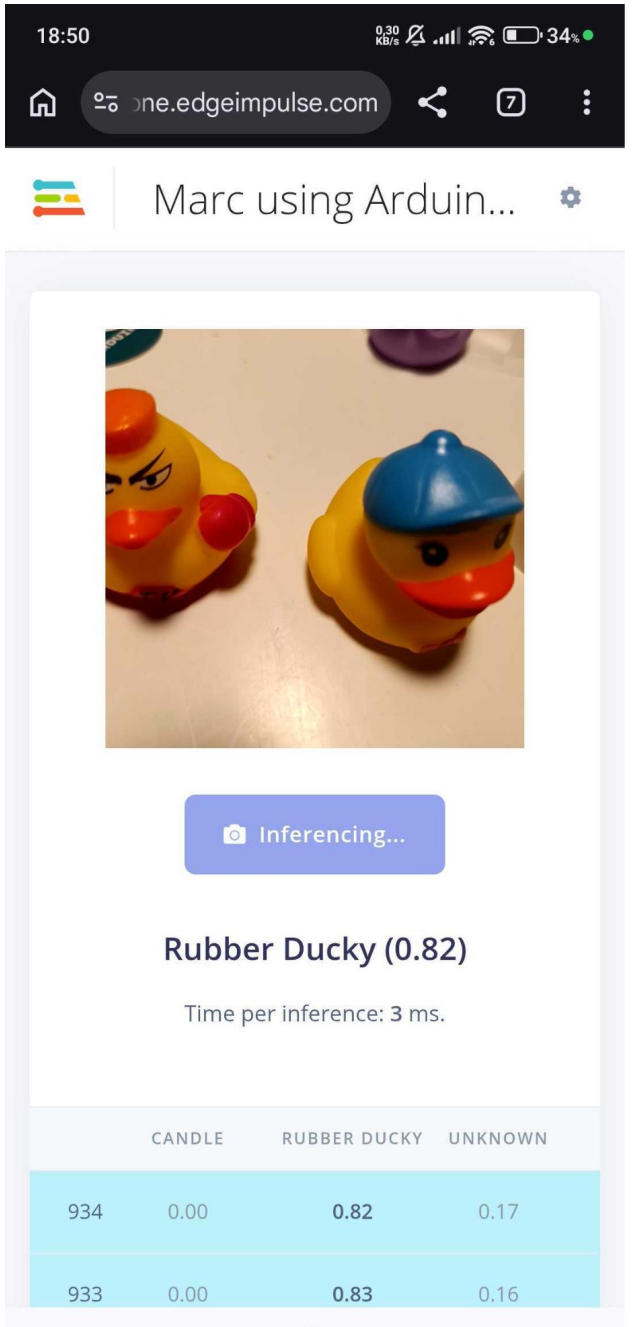

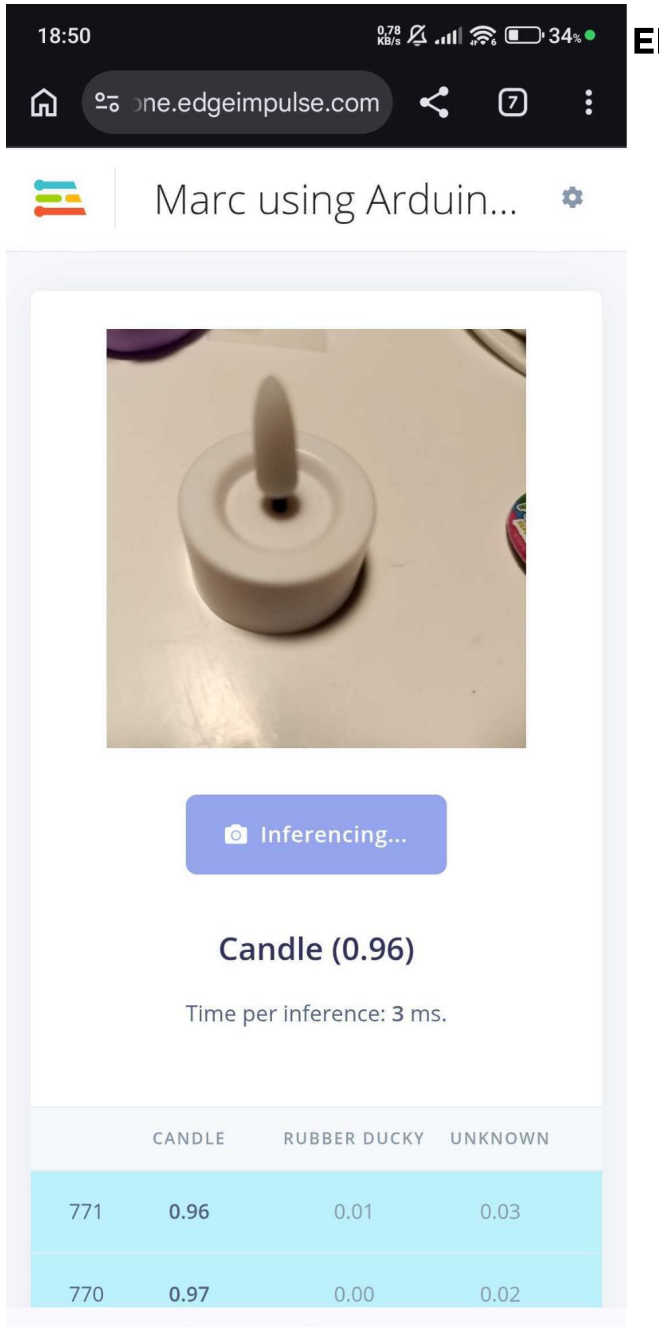

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What is a good dataset?

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What is a good dataset?

A good dataset is **representative of the real-world scenarios** your model will encounter. This includes variations in:

- Environmental conditions: Lighting, noise, weather, etc.
- Demographic factors: Different ages, genders, accents, etc.
- Edge cases: Unusual or rare situations that might occur.

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How much data is needed for a good dataset?

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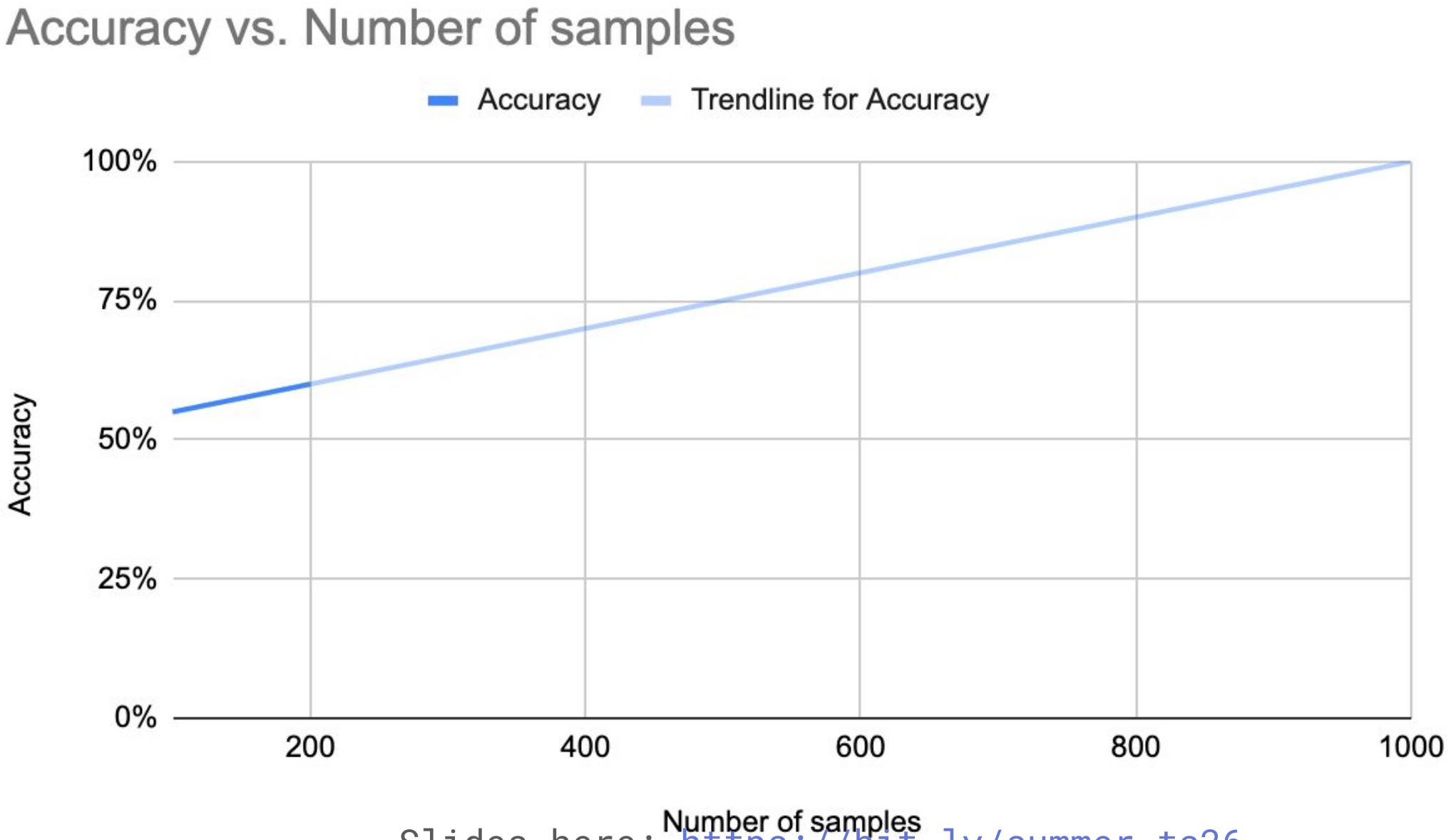

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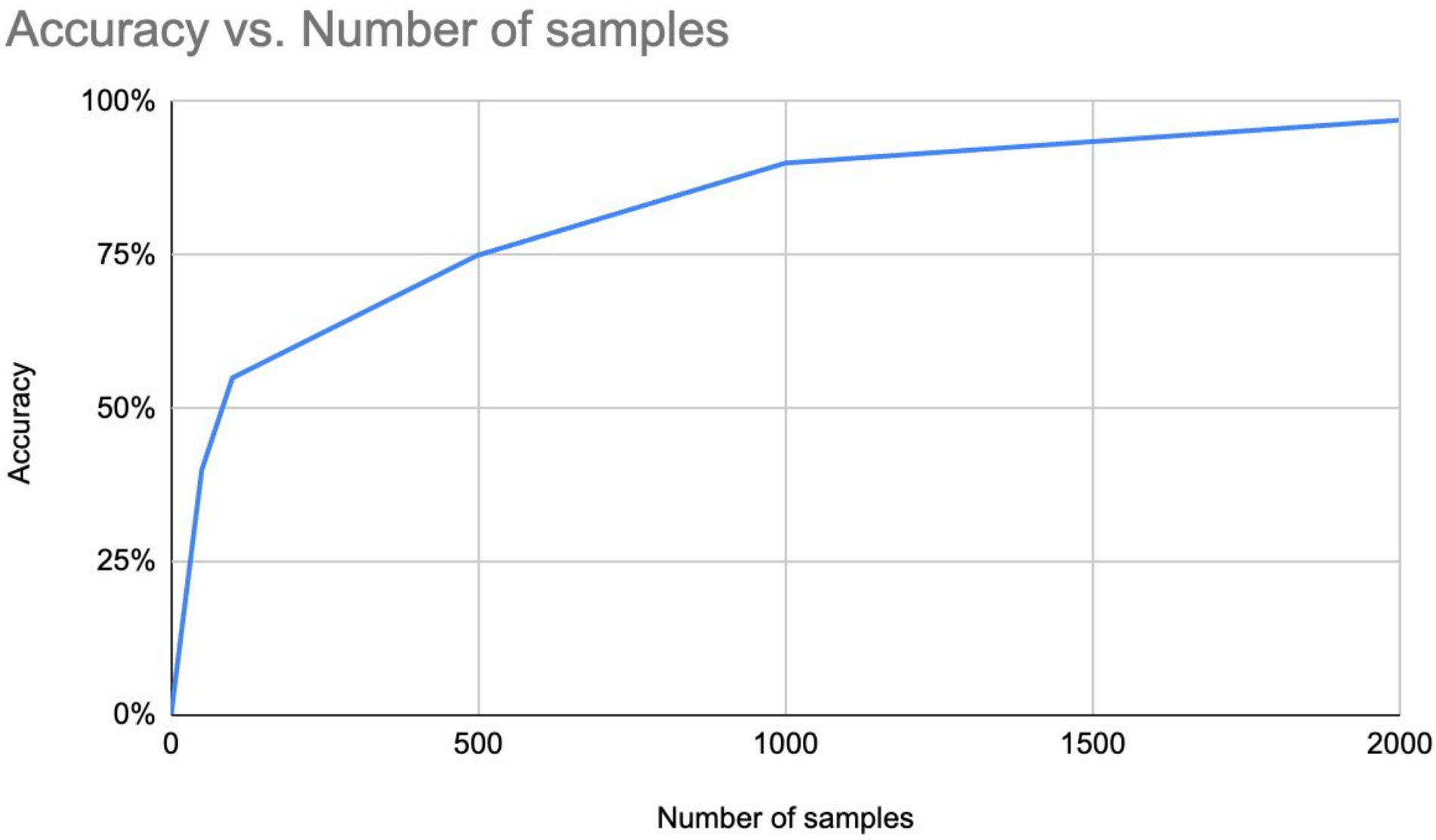

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Image processing approaches

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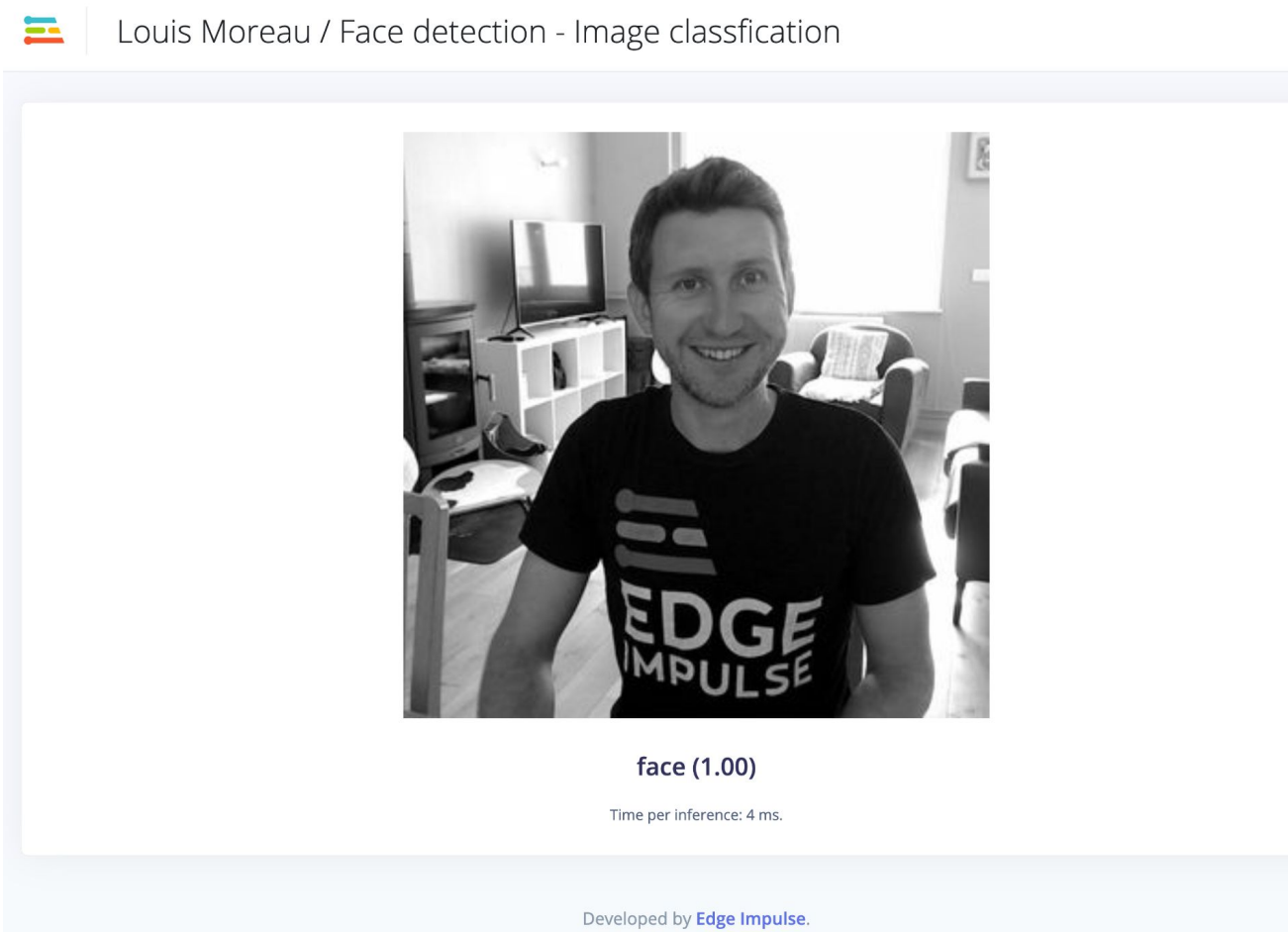
Image classification

The question the model is trying to answer is:

“Is there a face or not in the image?”



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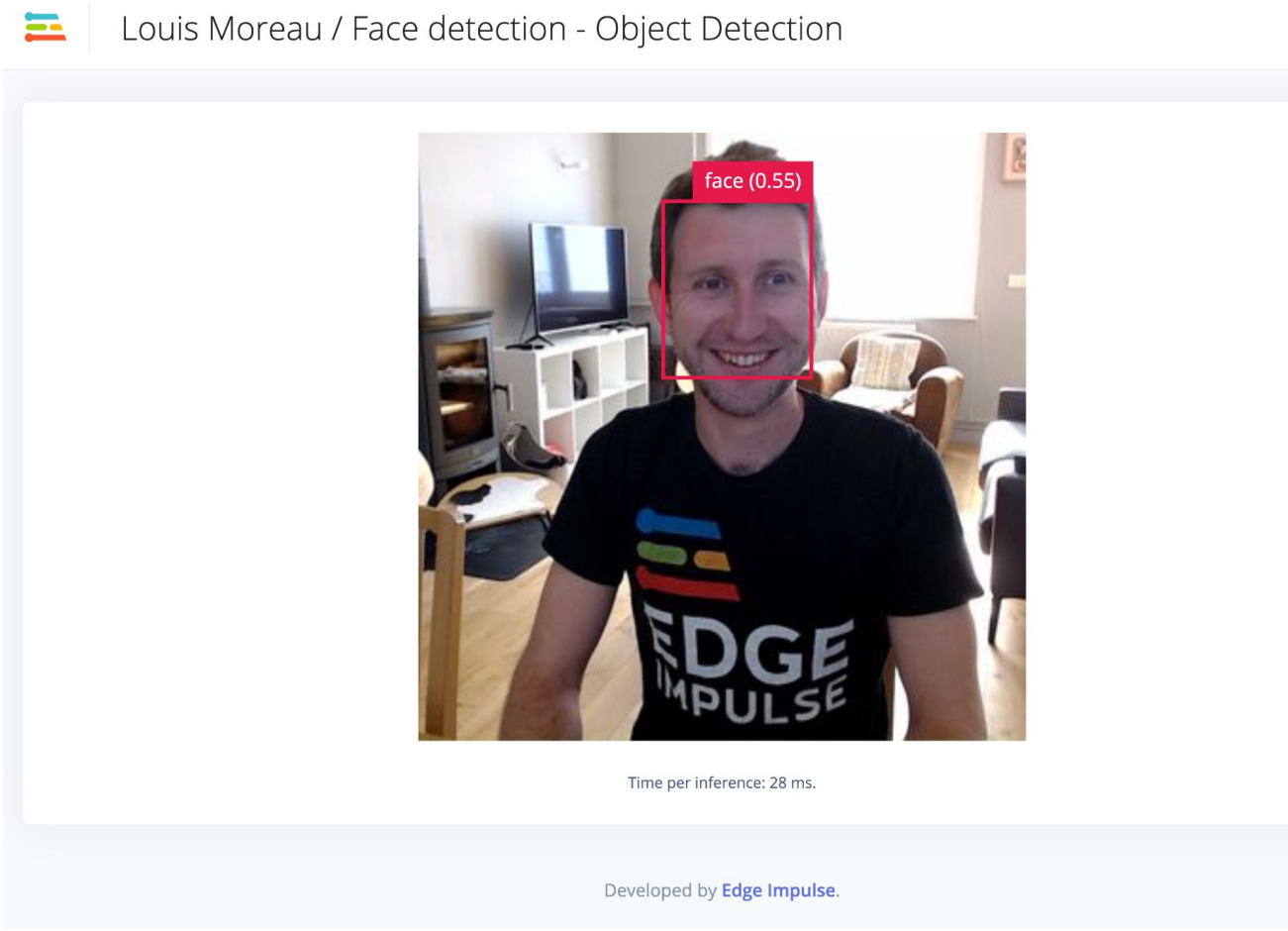
Object Detection using Bounding Boxes

The question the model is trying to answer is:

“Are there faces in the image, where **and what size are they?”**



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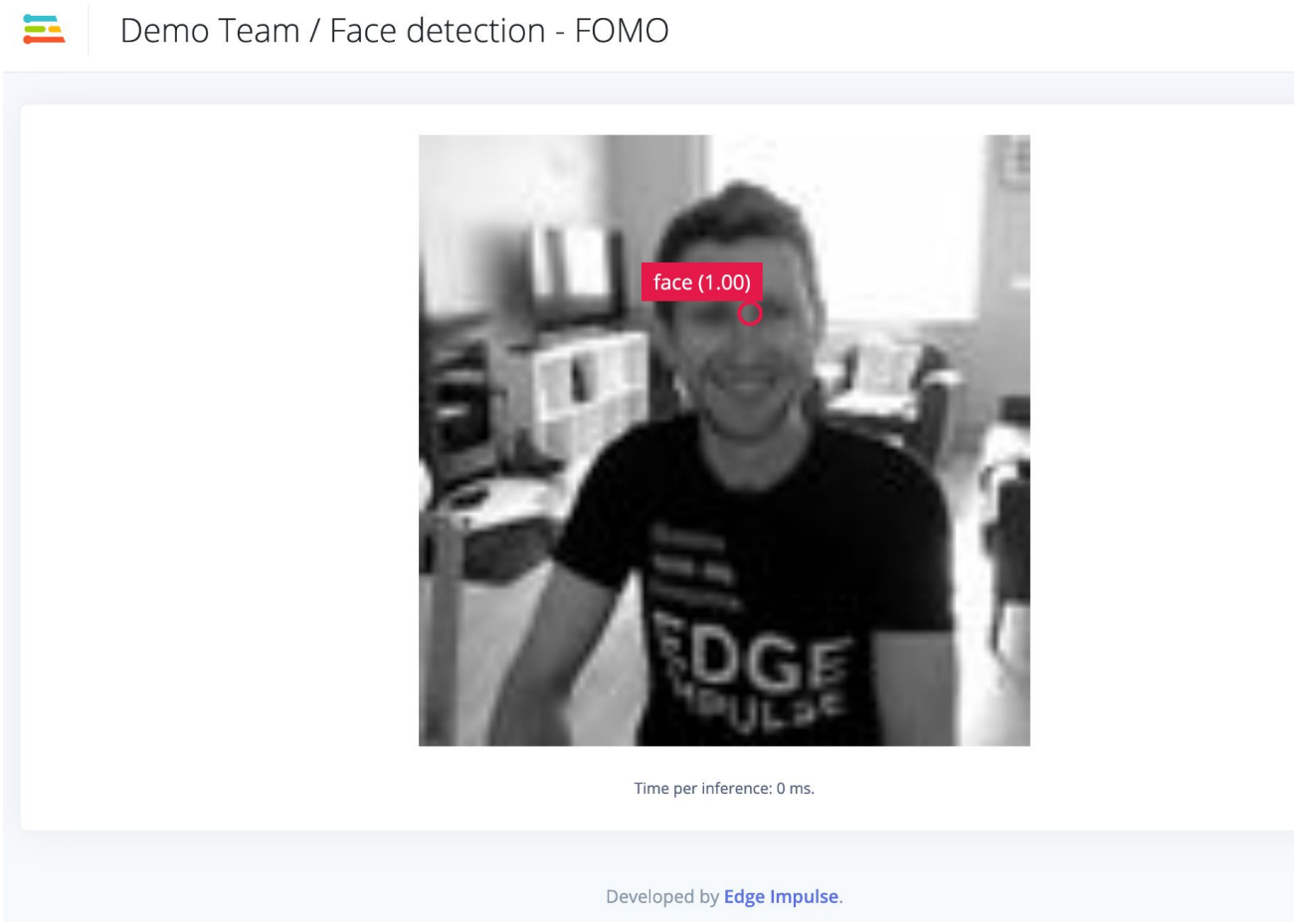
Object Detection using Centroids

The question the model is trying to answer is:

“Are there faces in the image, **where are they?”**



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Re-train an Object Detection ML model

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Duckies classification

1. Get your laptop
2. Sign-up here

https://edgeimp.com/ducks-ta26

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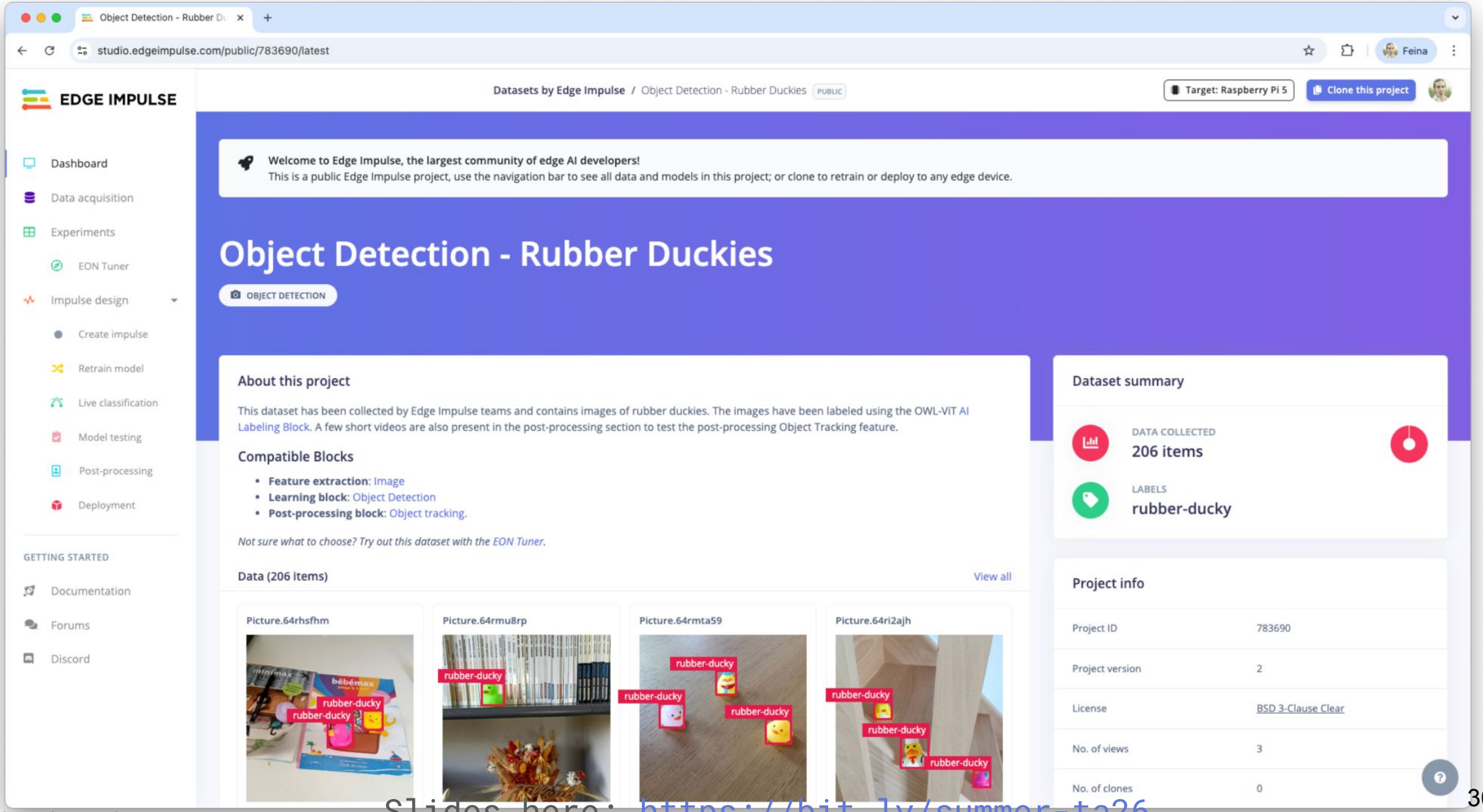

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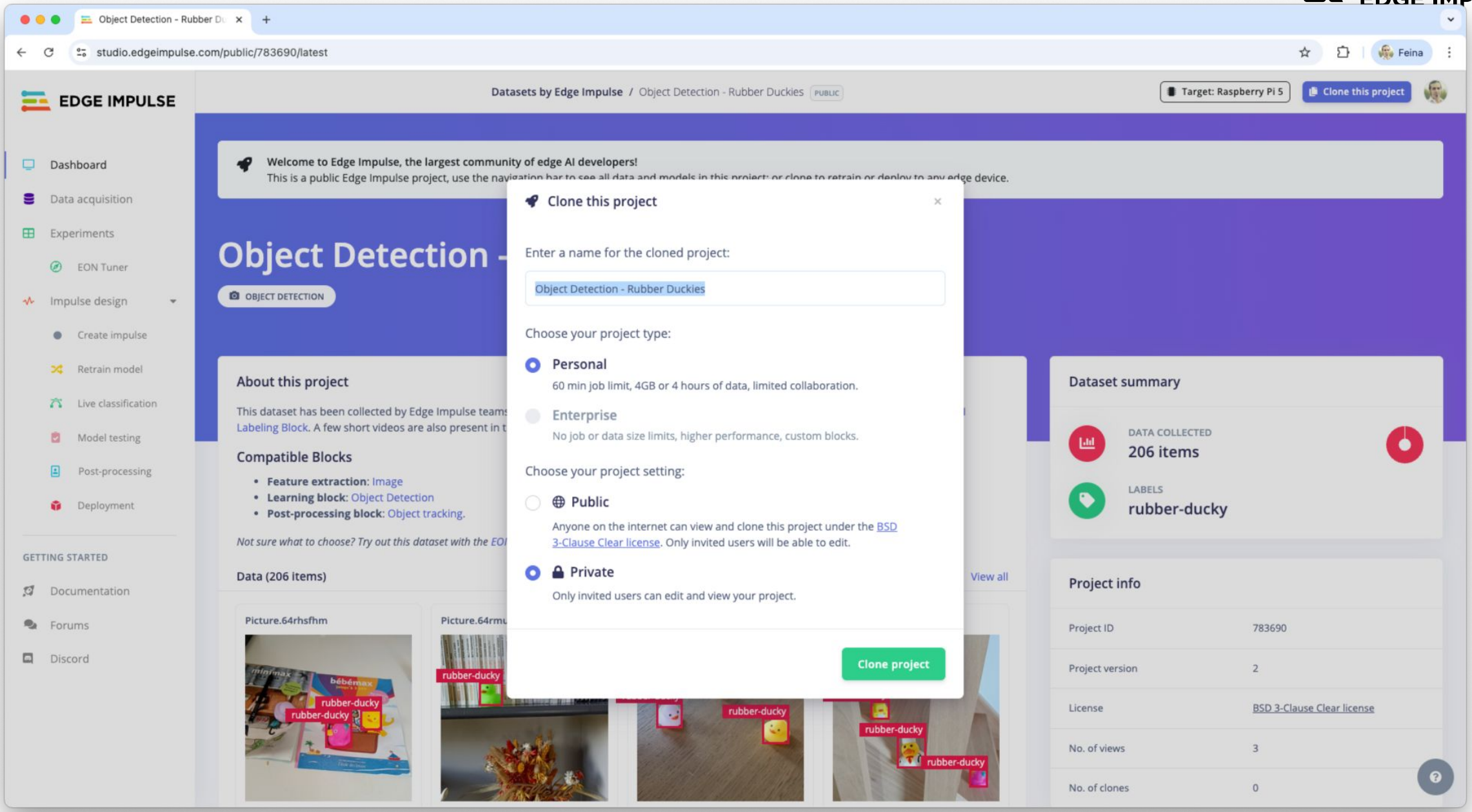

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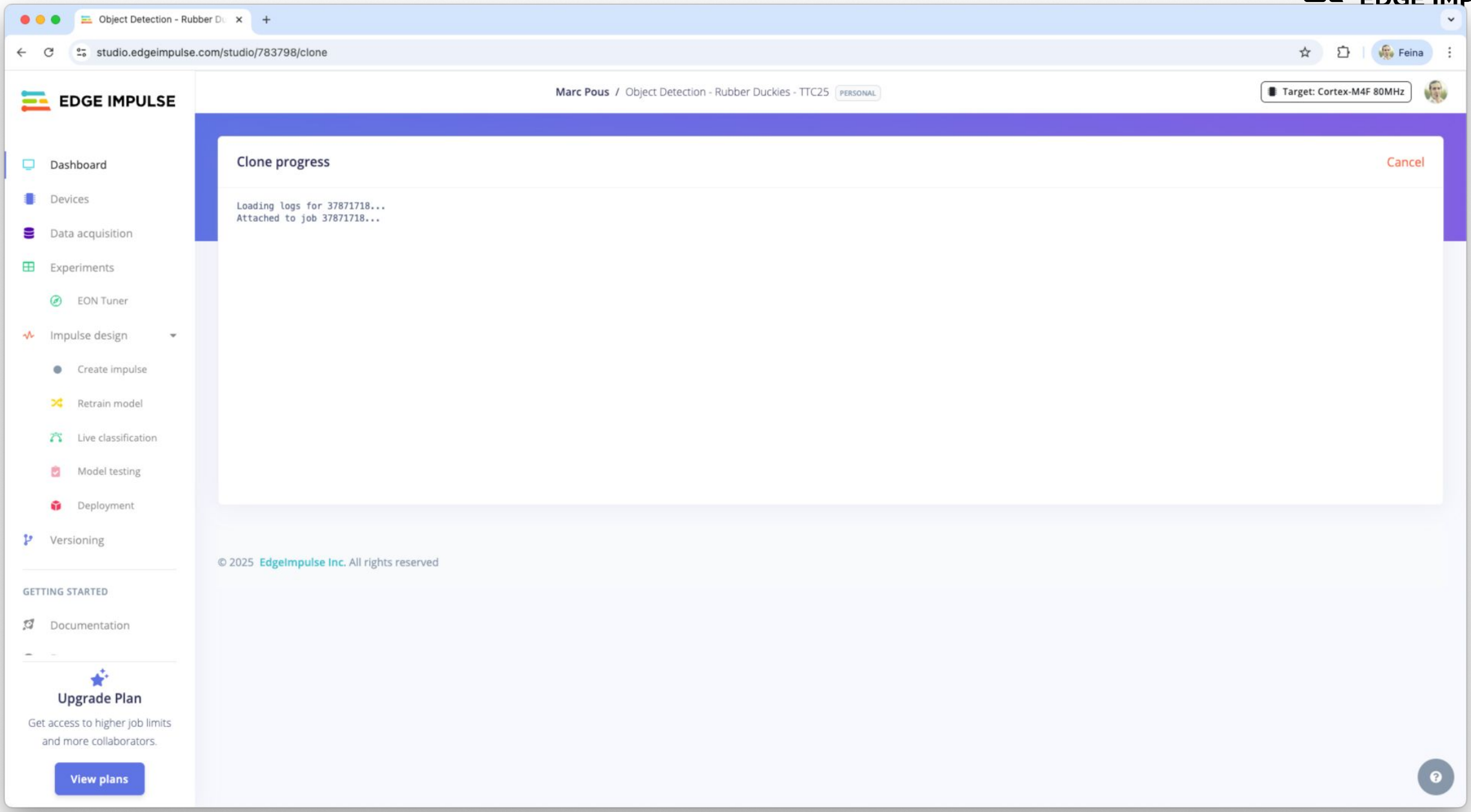

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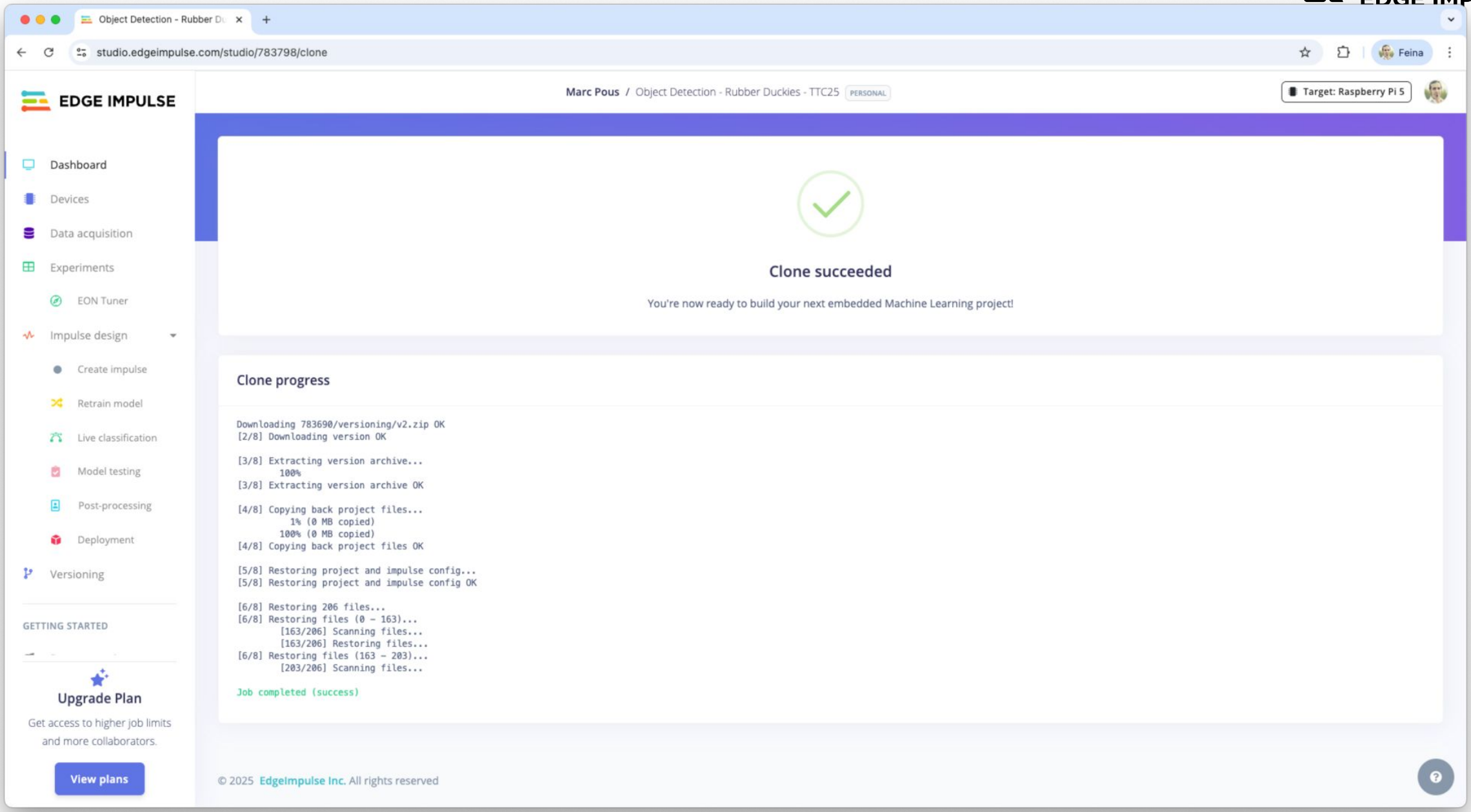

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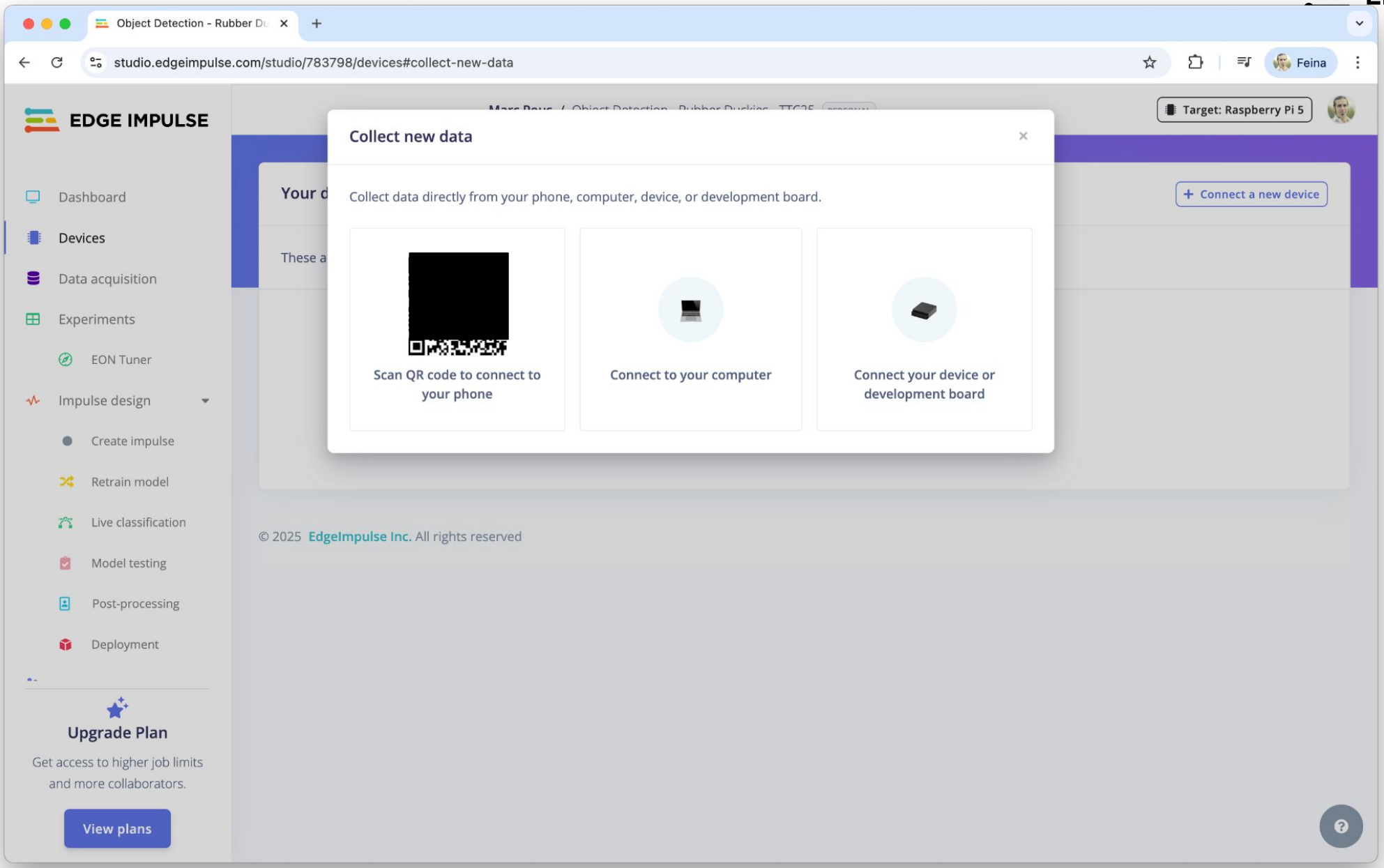

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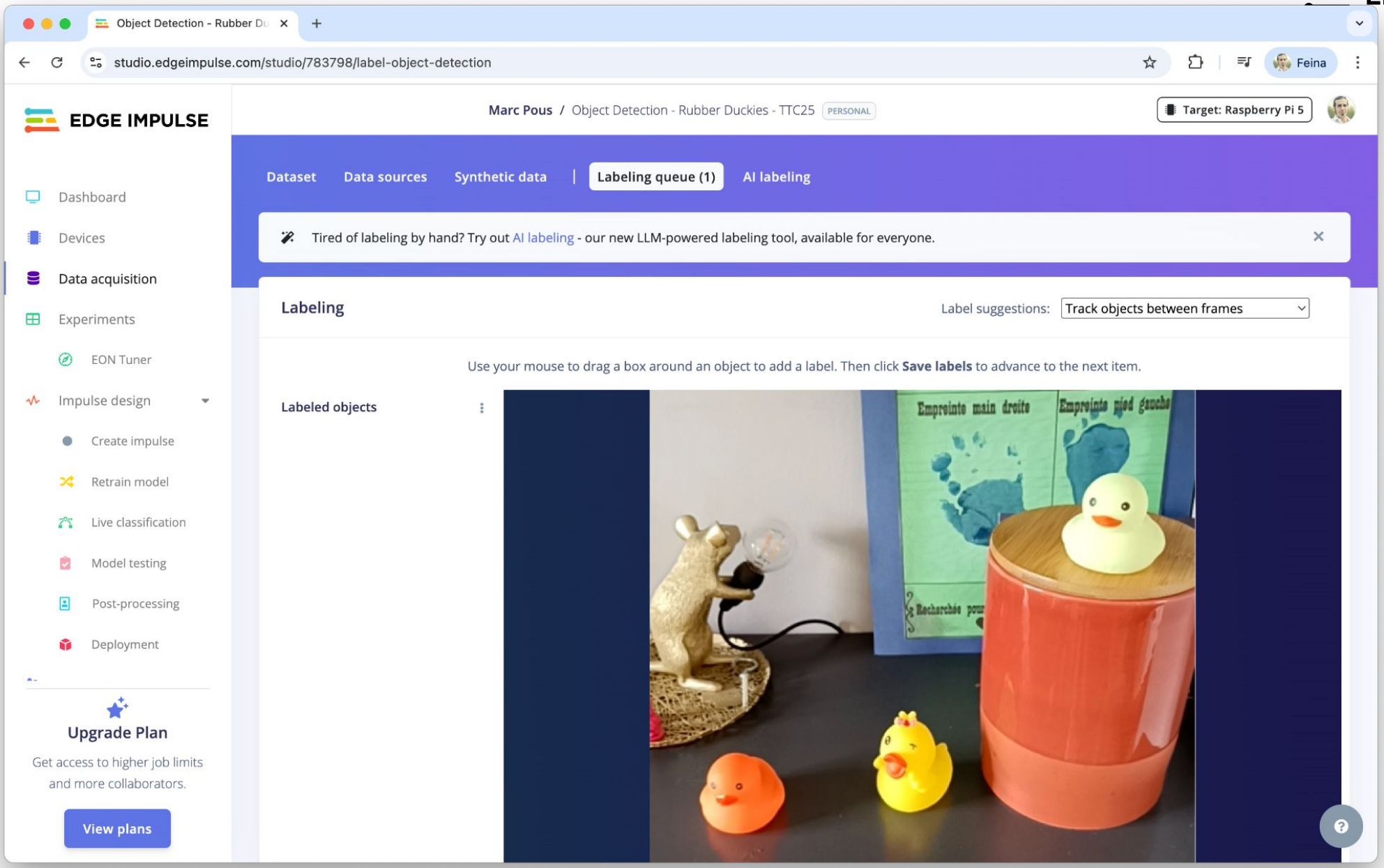

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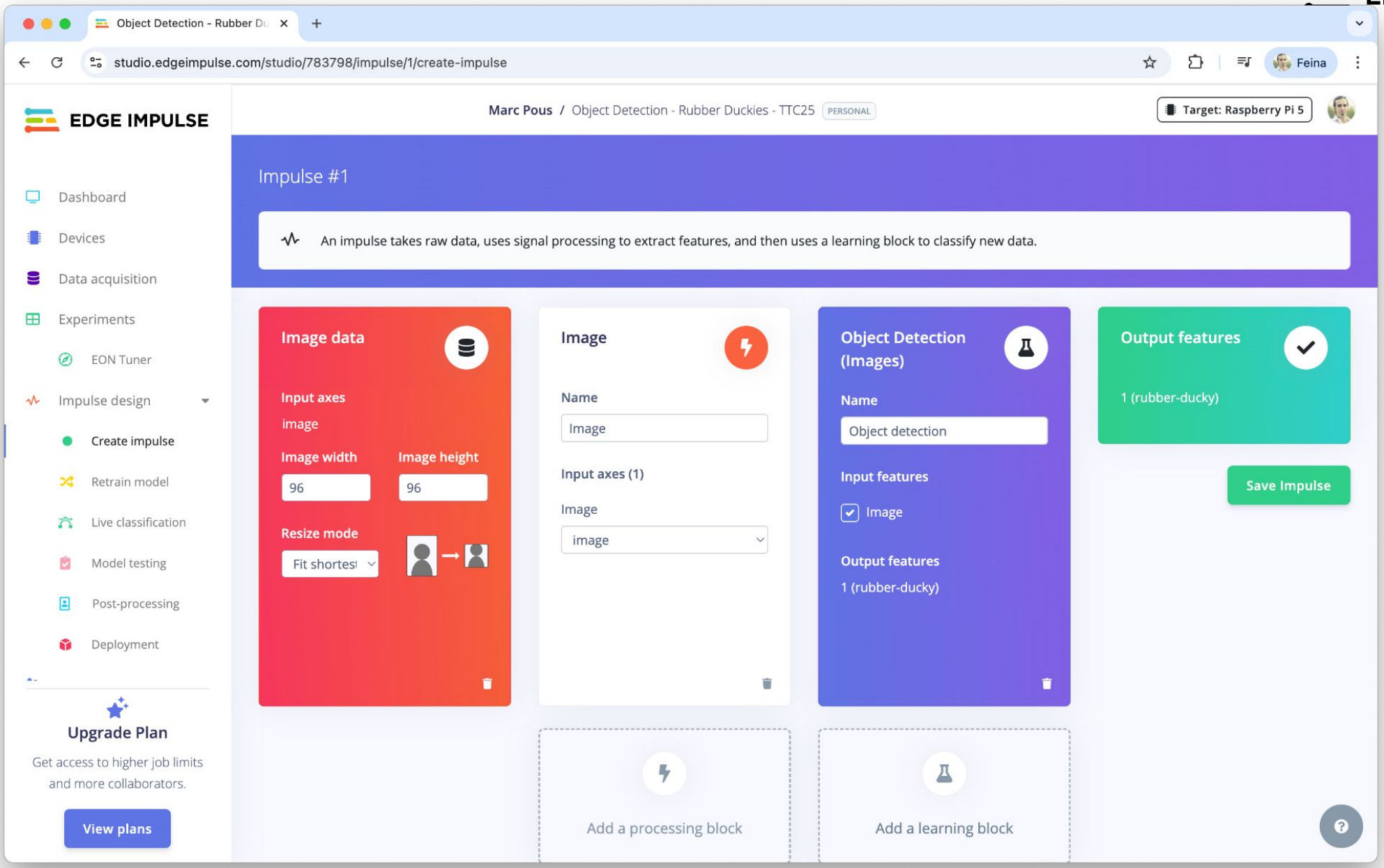

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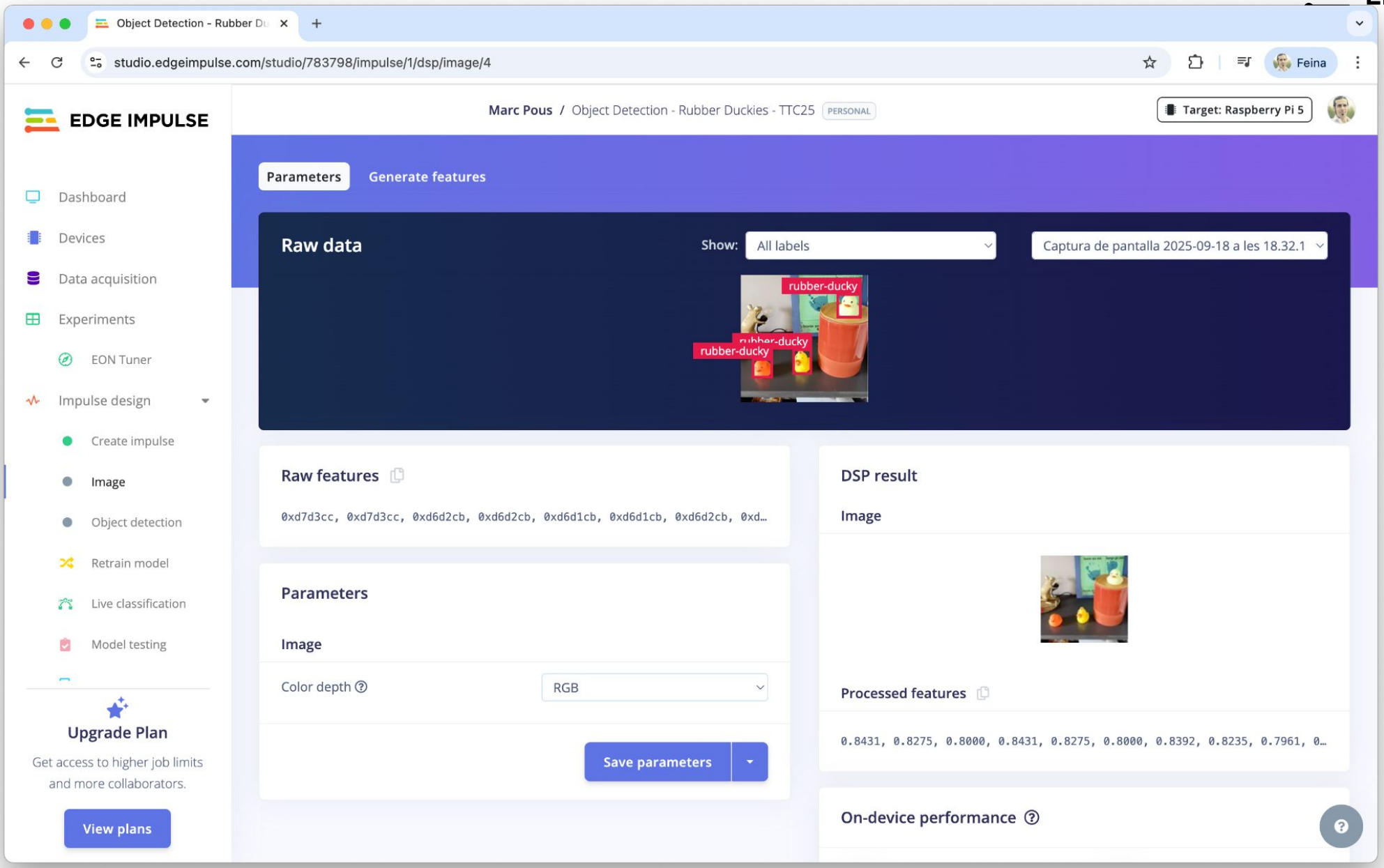

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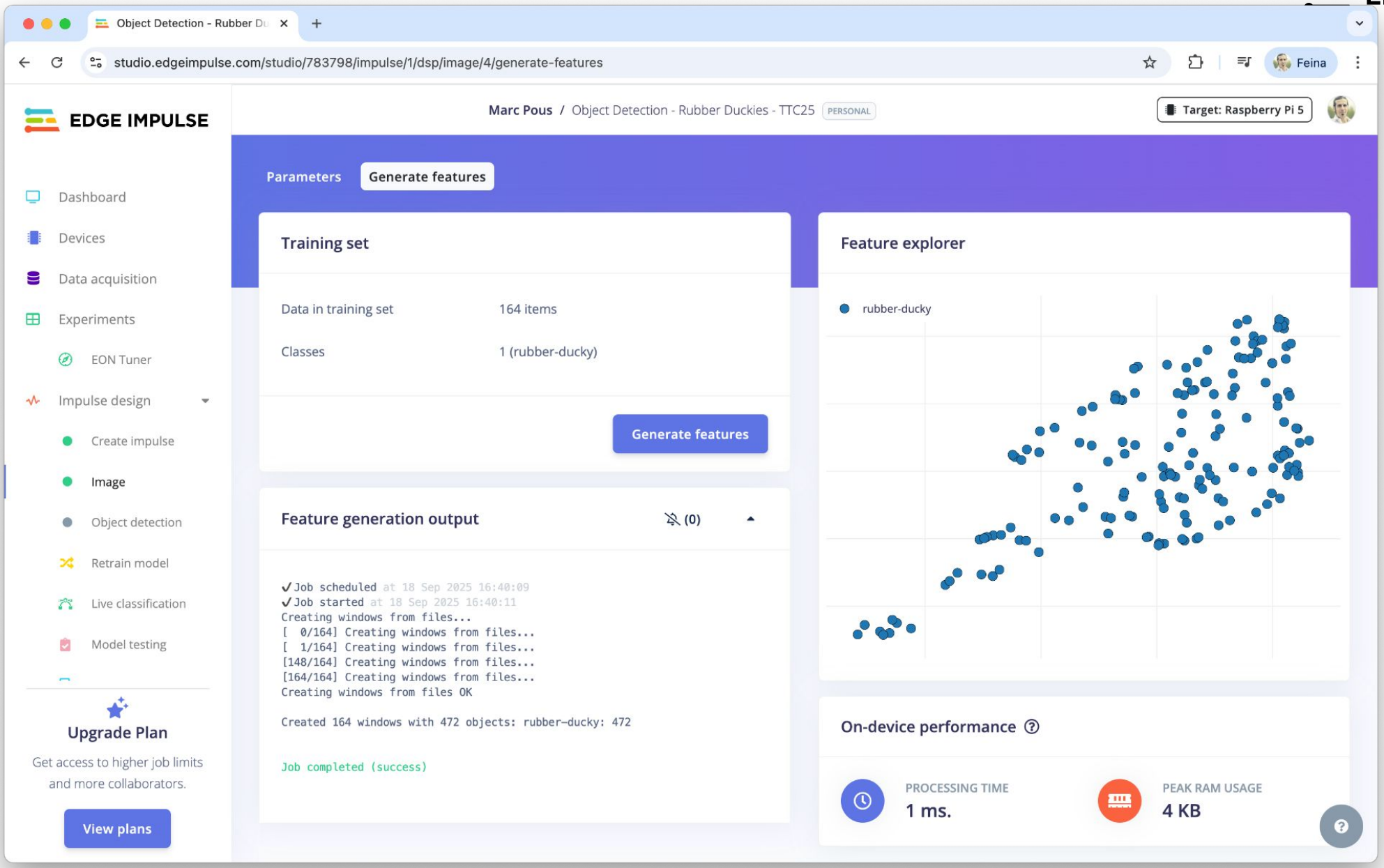

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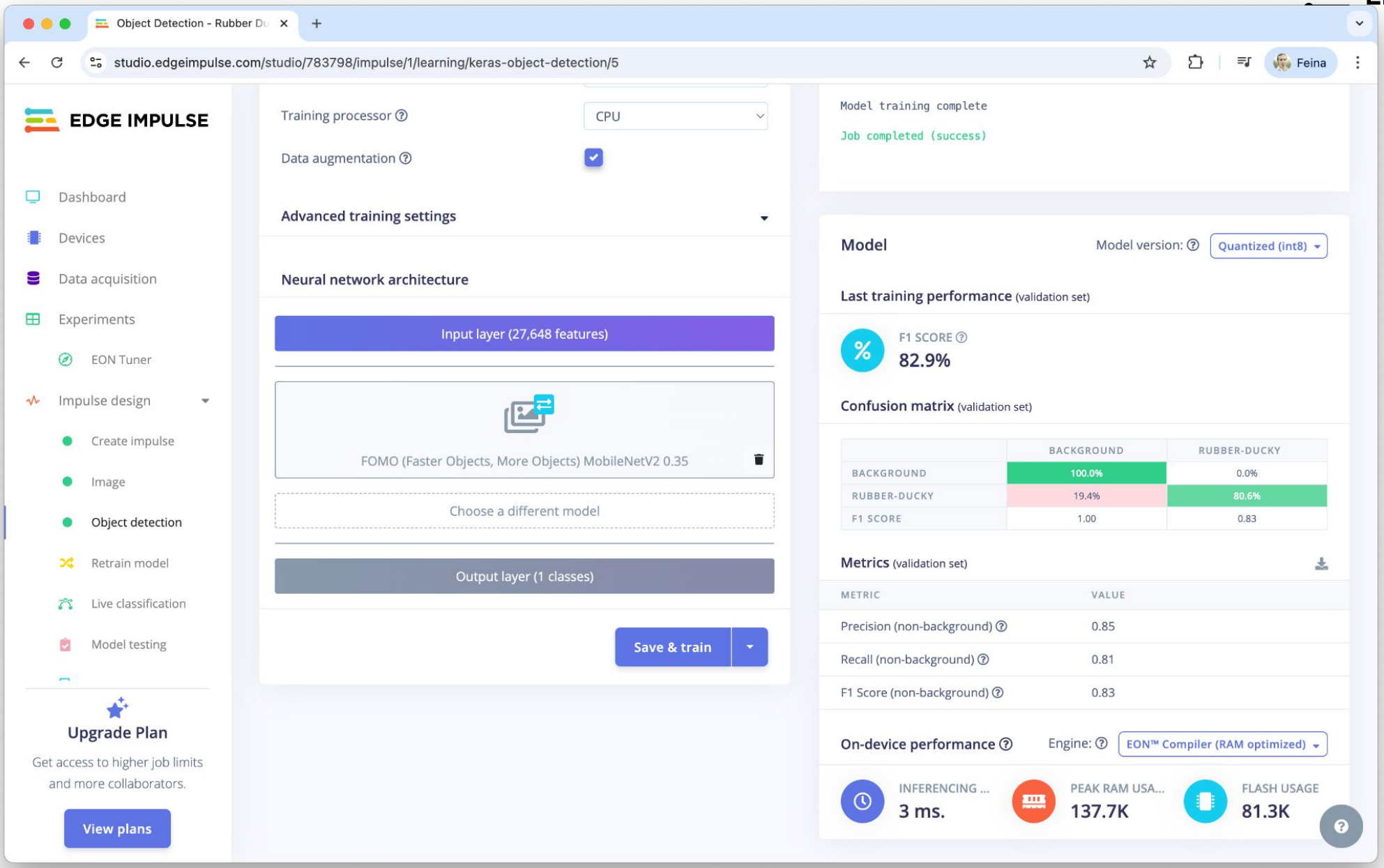

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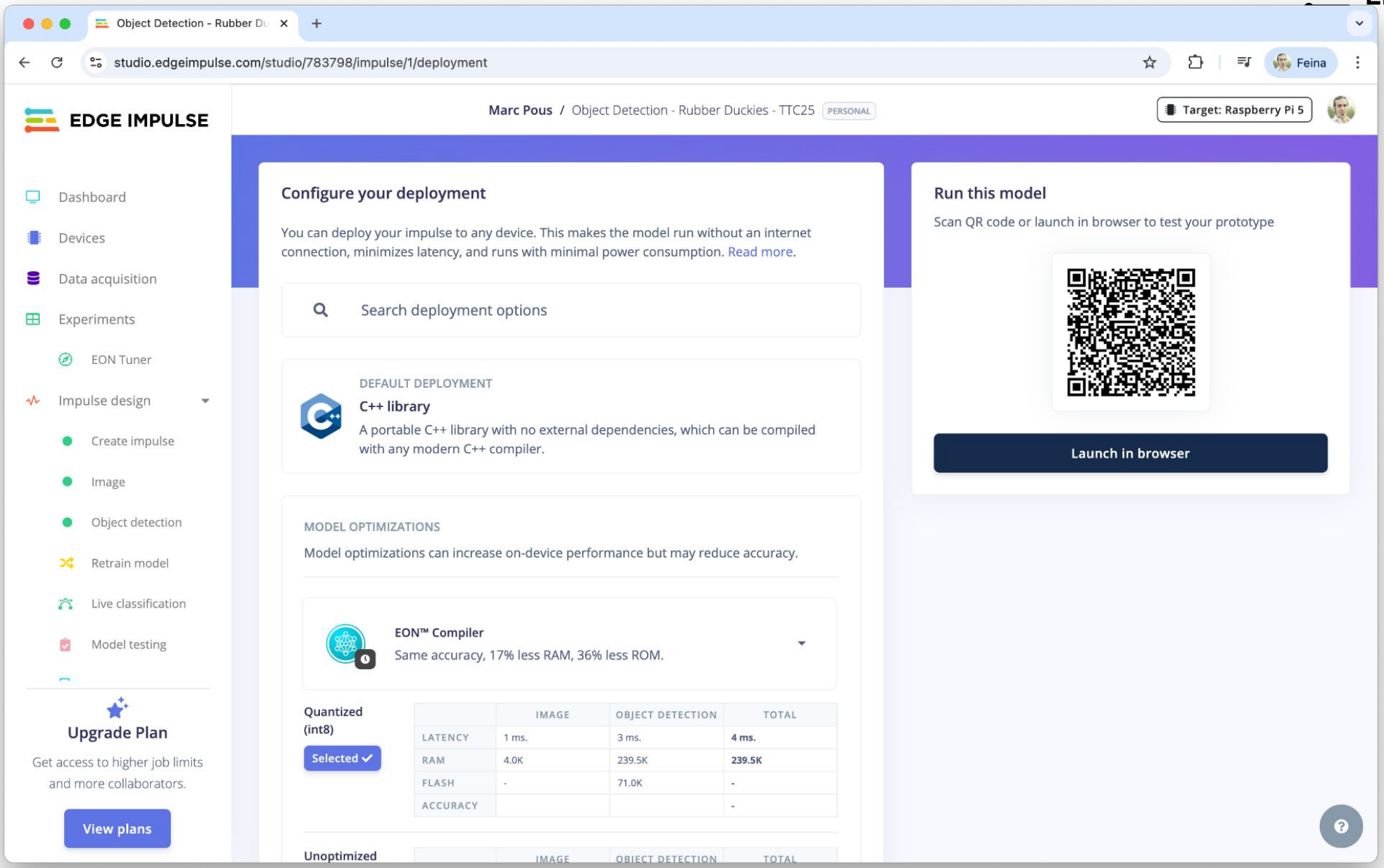

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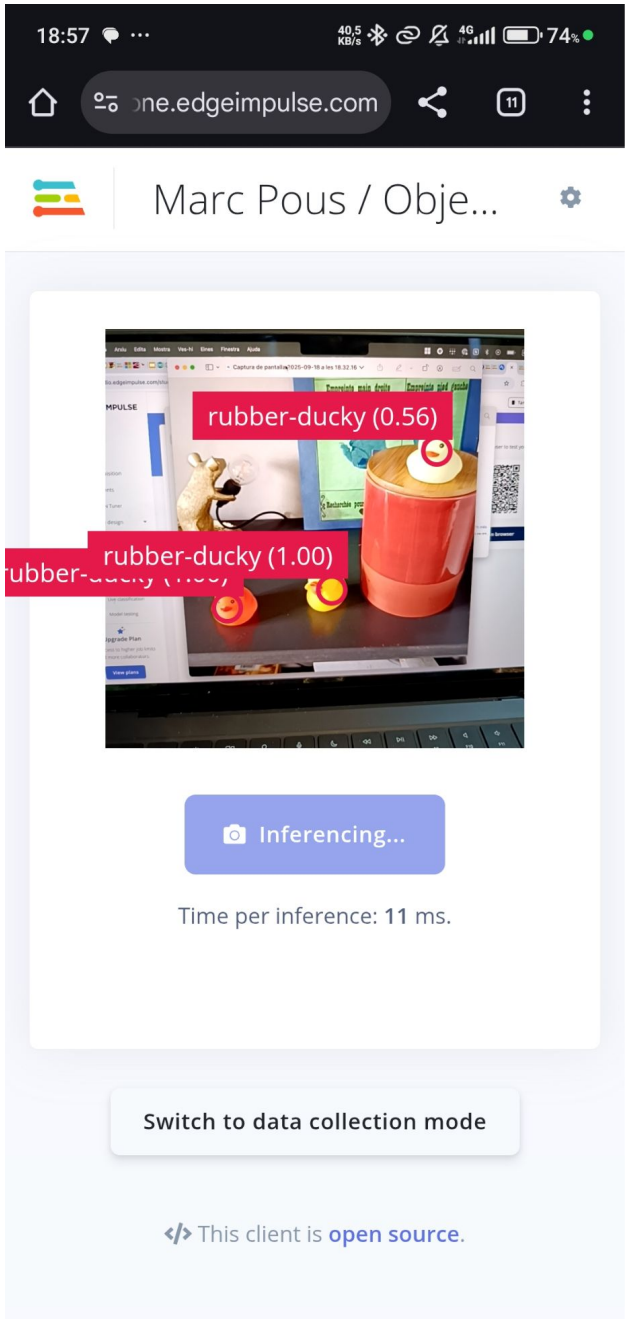

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How to deploy it into a real hardware?

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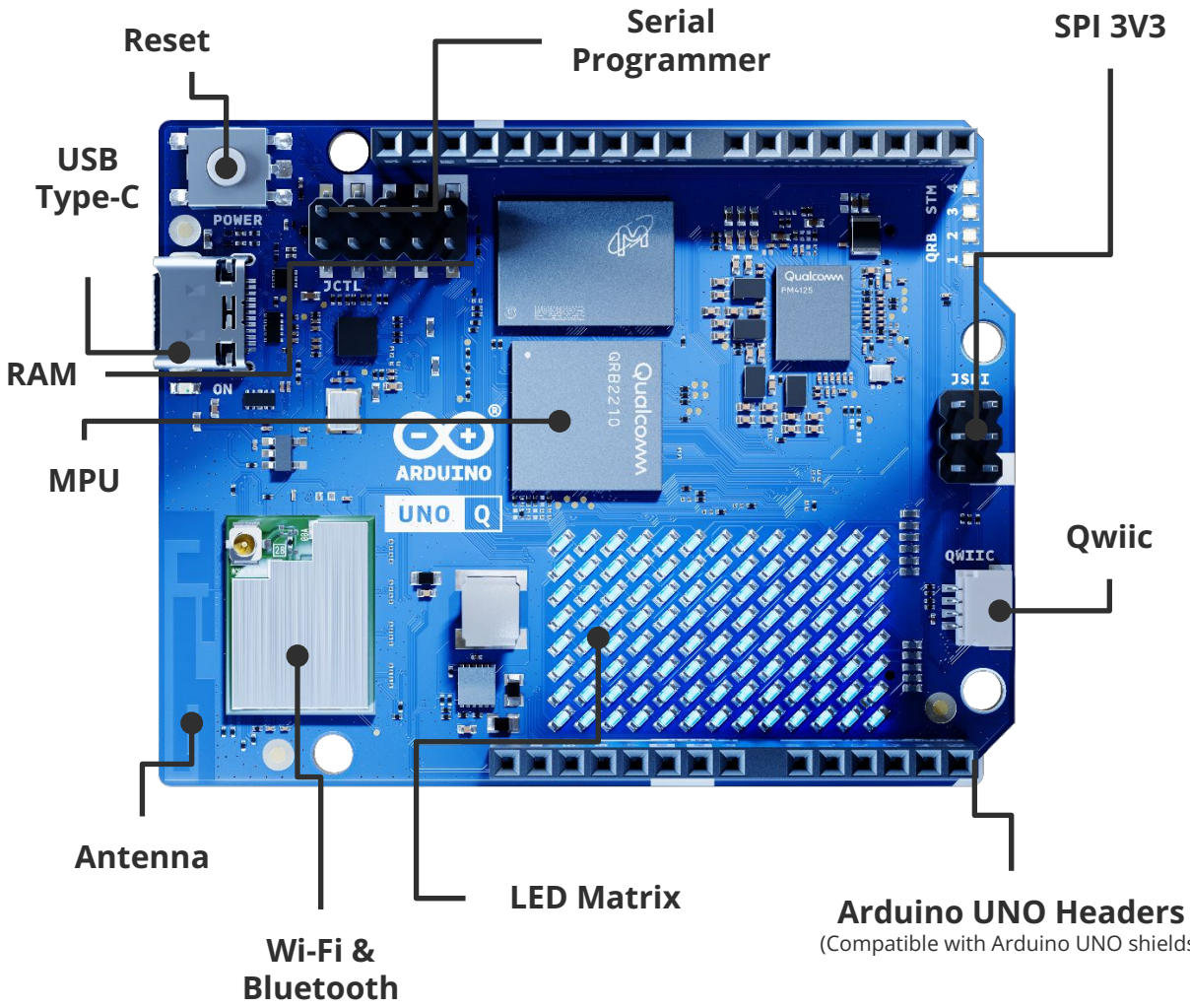
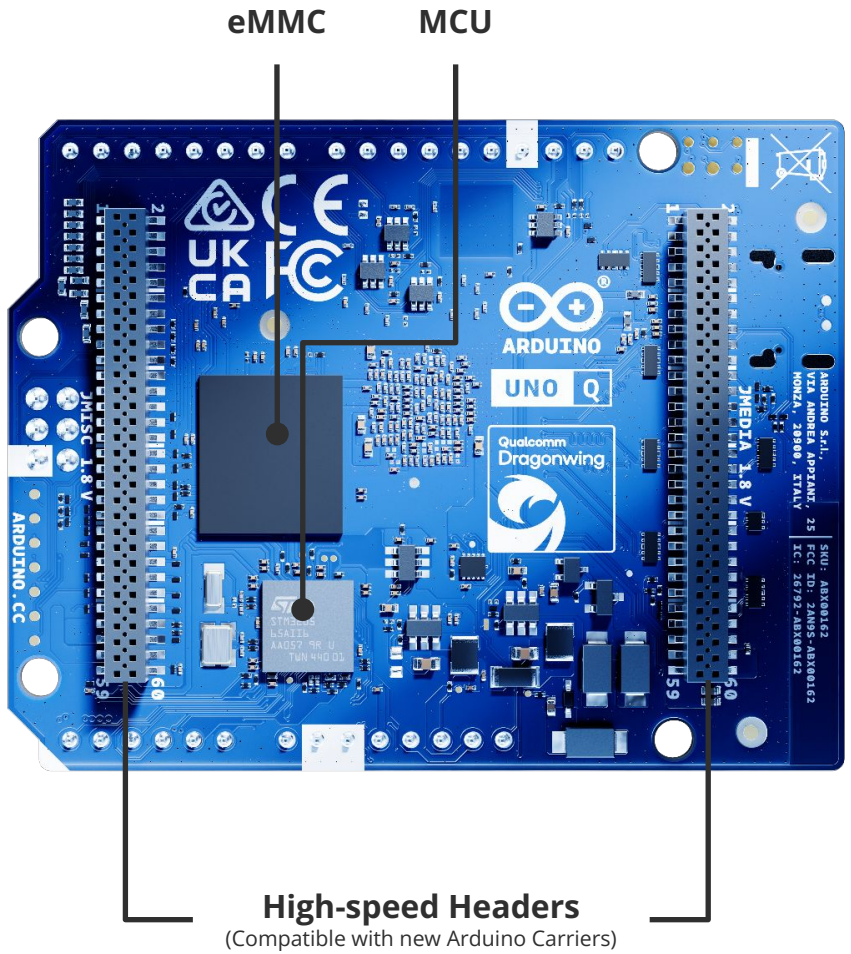
Arduino UNO Q





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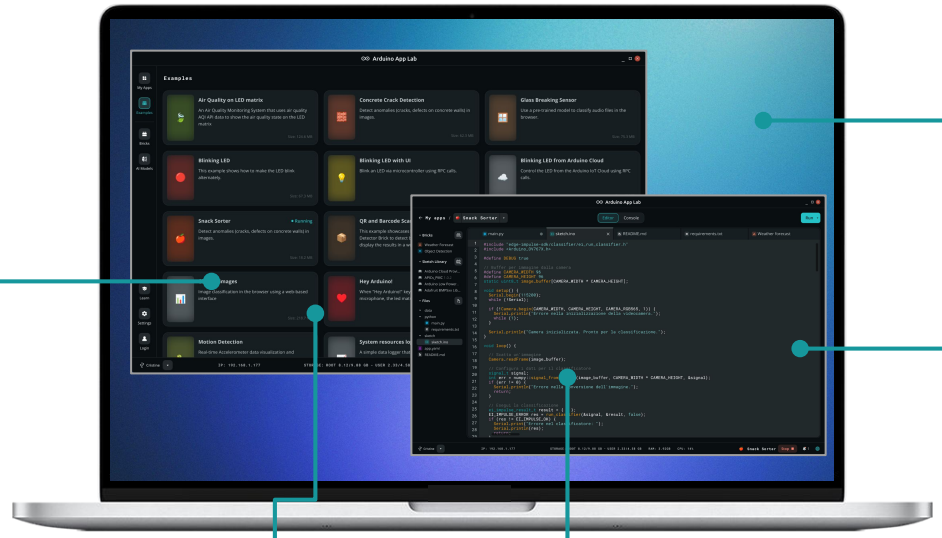
Dual-Brain Architecture: Linux Meets Arduino

UNO Q brings together **Linux** and **microcontroller worlds** through a **built-in RPC library,** so you can create powerful, responsive applications in **one seamless workflow with Arduino App Lab**.



Arduino App Lab combines Arduino sketches, Python scripts, and AI models into ready-to-run apps

Linux Debian OS with upstream support

Arduino Core and **Sketch** experience running on **Zephyr OS**

Qualcomm® Dragonwing ™ QRB2210

STM32U585

Remote Procedure Call (RPC)

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Arduino AppLab: Powerful AI Models, Ready to Deploy

Face Detection
Quick, lightweight facial detection for smart kiosks, home devices, and security systems.

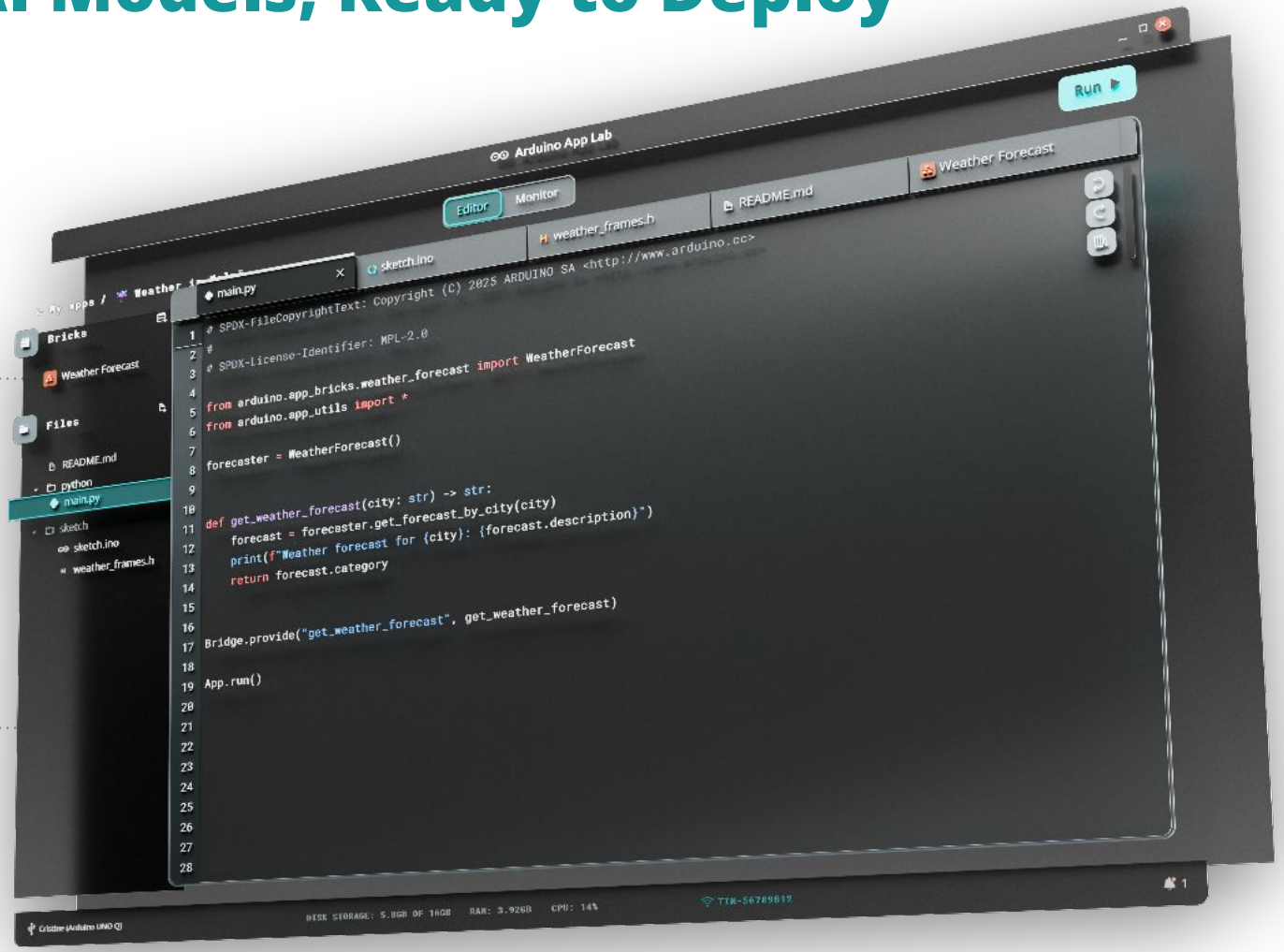
Anomaly Detection
Detect surface defects or issues instantly, ideal for industrial quality control.

Person Classification
Identify human presence in camera feeds for safety, automation, or smart access.

Image Classification
Recognize and classify custom objects

Sound Recognition
Recognize critical sounds in real time to enable smart alerts and responsive automation.

Keyword Spotting
specific keywords in real time to trigger voice-controlled actions and enhance hands-free interaction.



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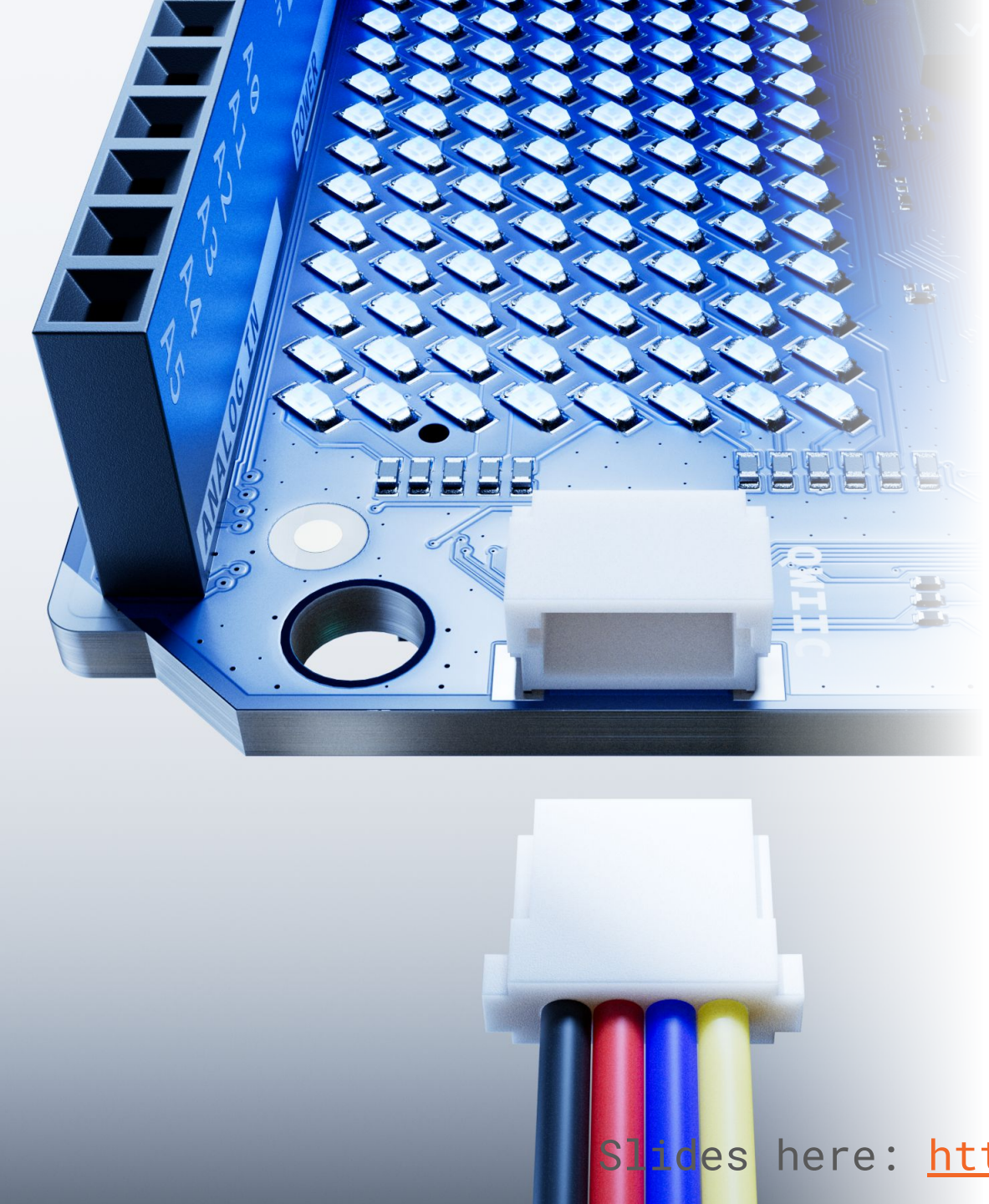

Shields, Carriers and Modulino nodes: Plug Into Possibilities

The UNO Q invites you into a world of creativity with its built-in **Qwiic** I²C connector — your link to hundreds of plug-and-play **Modulino nodes** and 3rd party **sensors, and actuators**. At the same time, combined with the iconic **Arduino UNO shield ecosystem**, the UNO Q becomes one of the most versatile platforms for building projects—no breadboards, no soldering, just seamless innovation.

The all-new **Arduino Carrier** form factor gives direct access to high-speed GPIOs linked to the Qualcomm Dragonwing™ chip, for advanced applications such as cameras, display, audio, etc...

No soldering needed

Focus on building your next idea, little wiring and no soldering will help you to prototype faster.

Polarized connector

No worries about wrong connections, there is only a way to connect the board and the modules

Chain-able MODULINOS

Connect multiple Qwiic compatible modules one after the other and expand the possibilities of your next project

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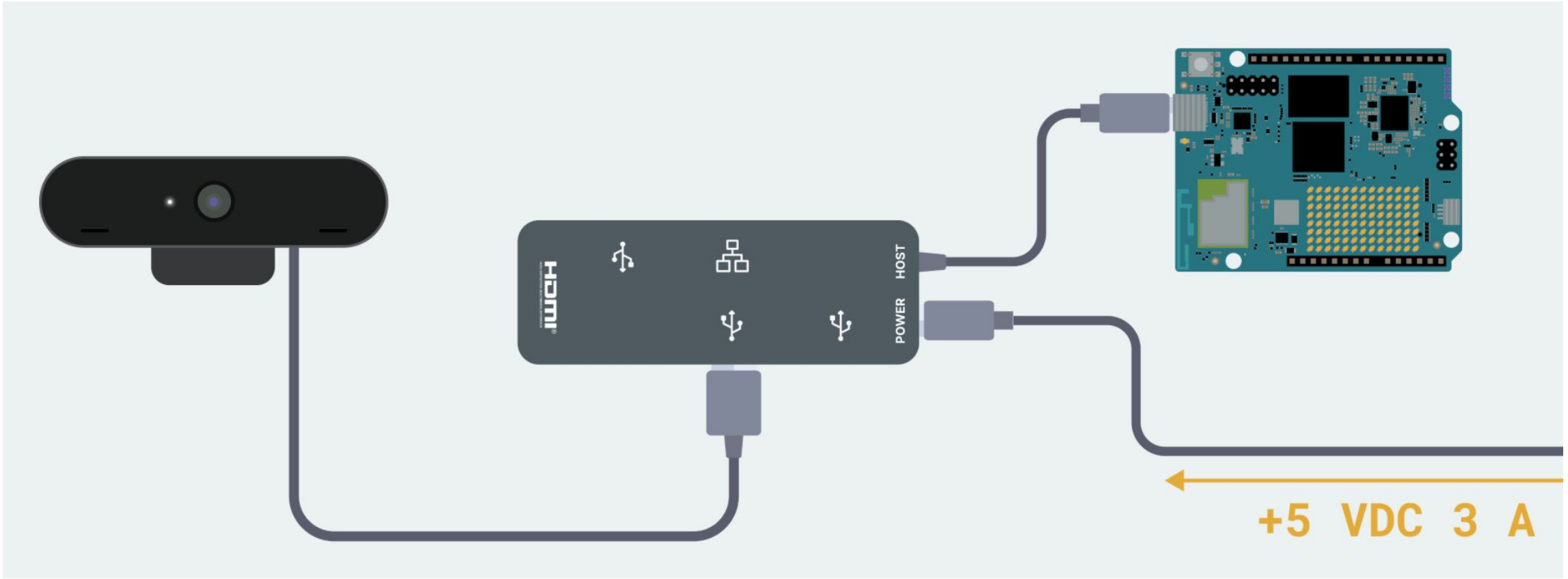
Cameras unlock vision AI



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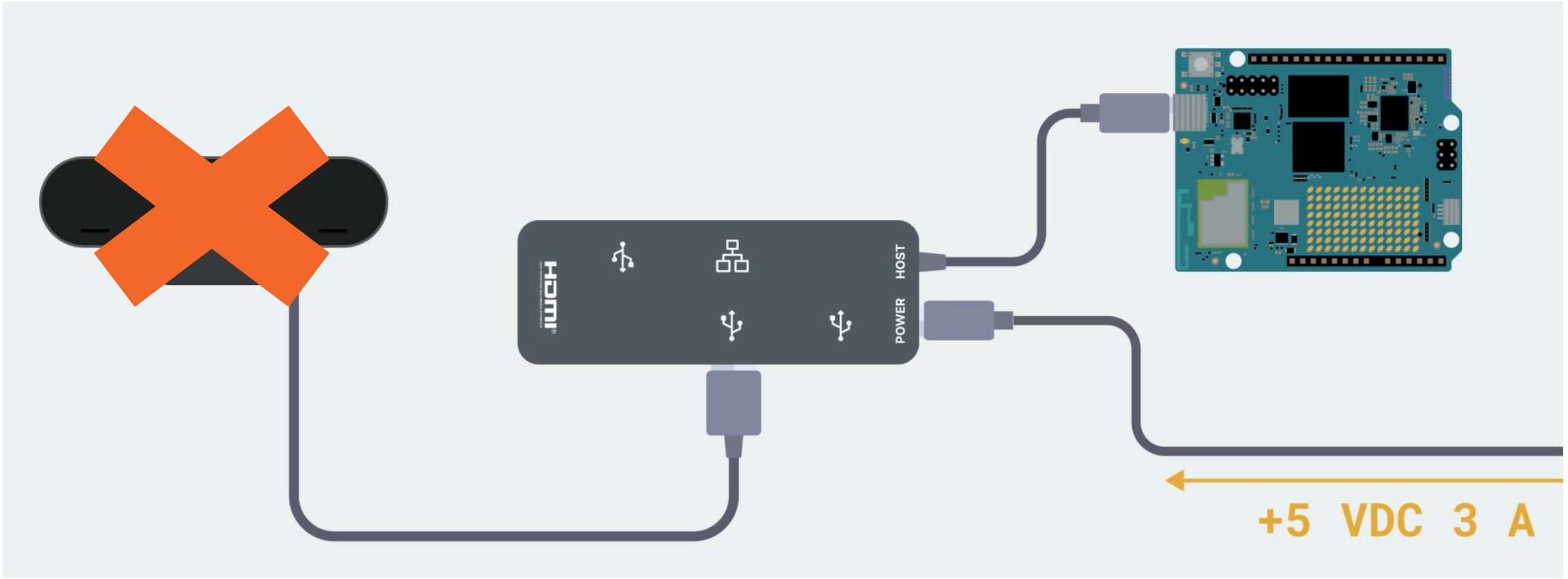
However, today...



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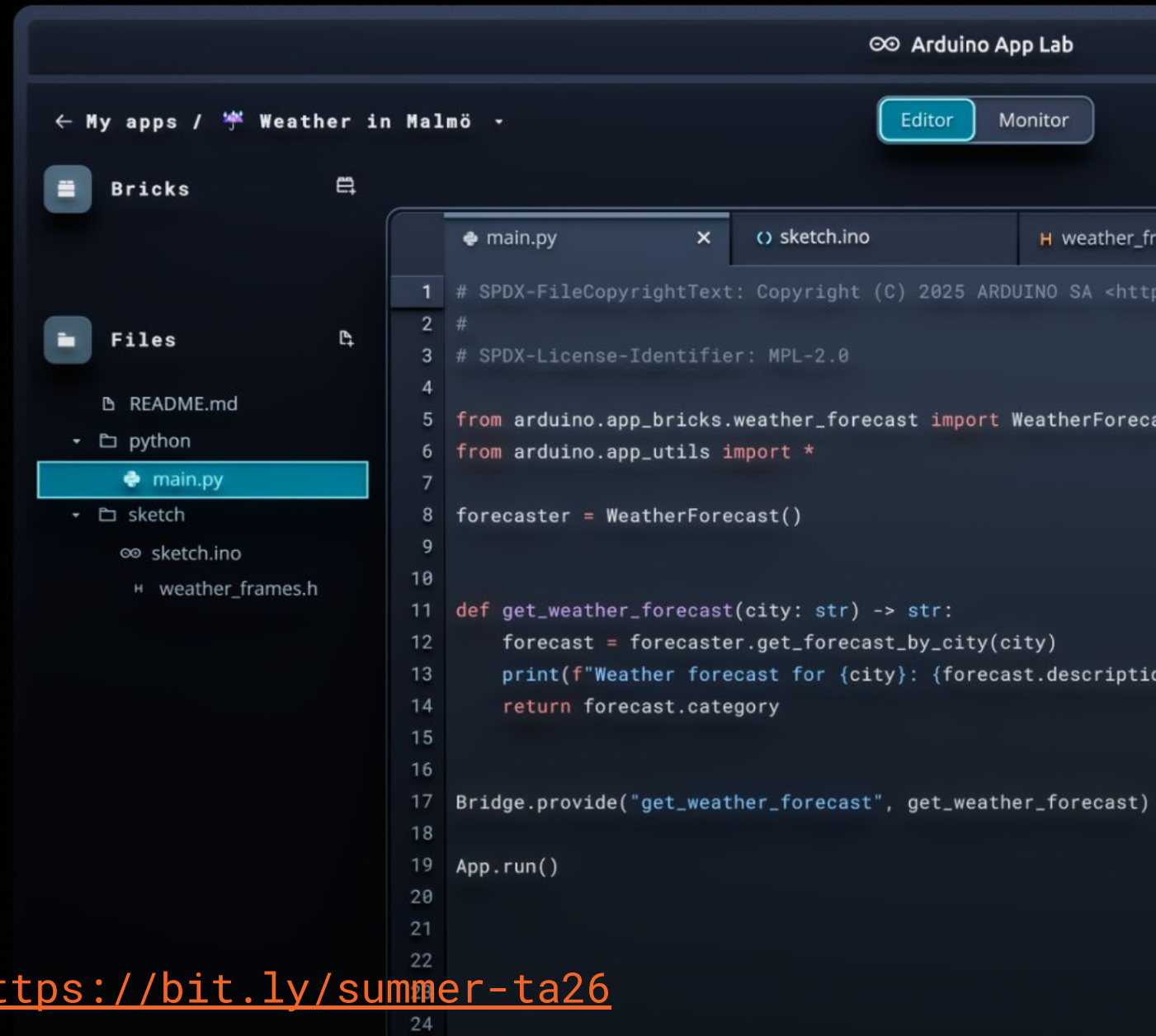
Introducing
Arduino App Lab

Download Arduino App Lab using:

https://bit.ly/arduino-applab



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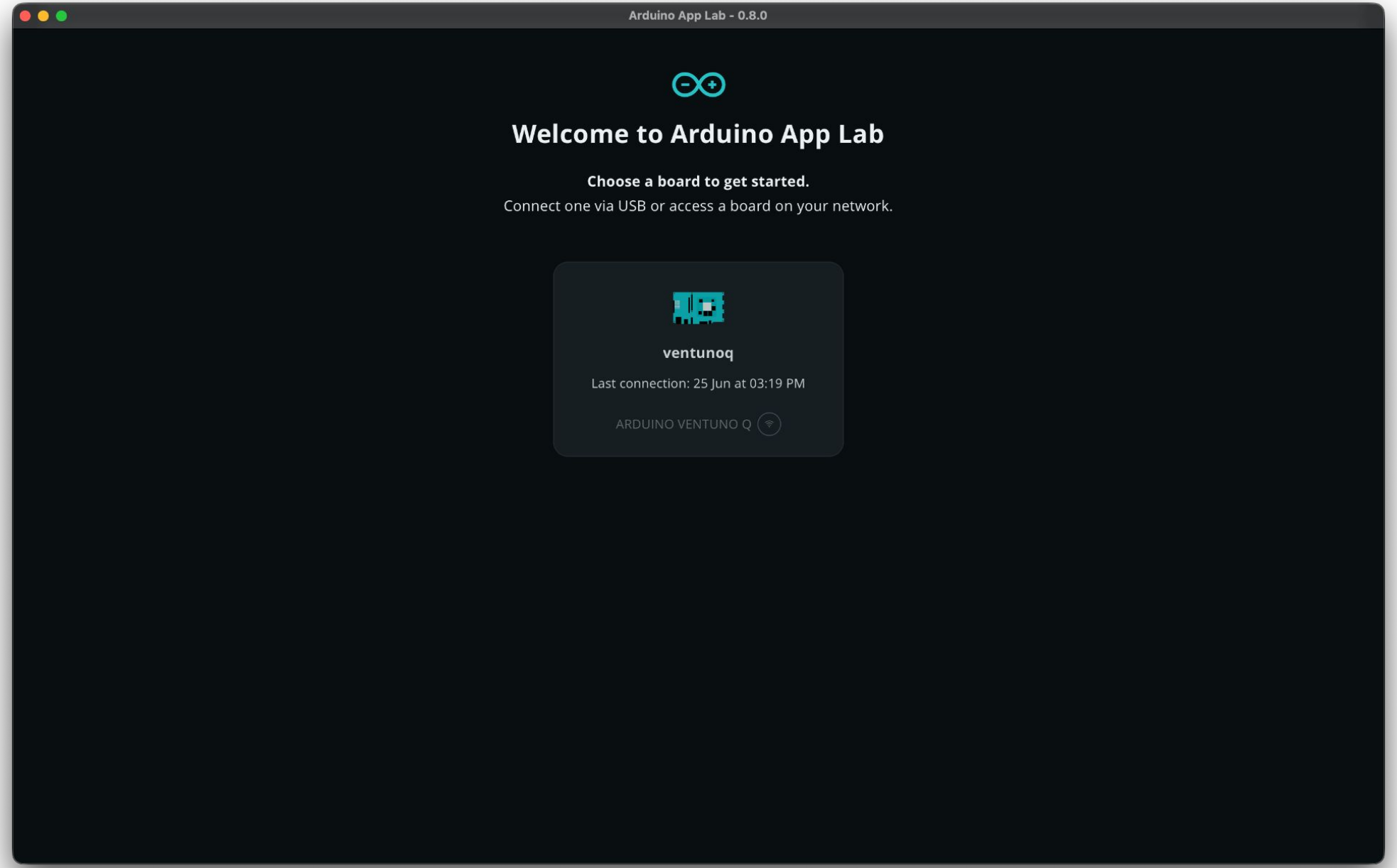

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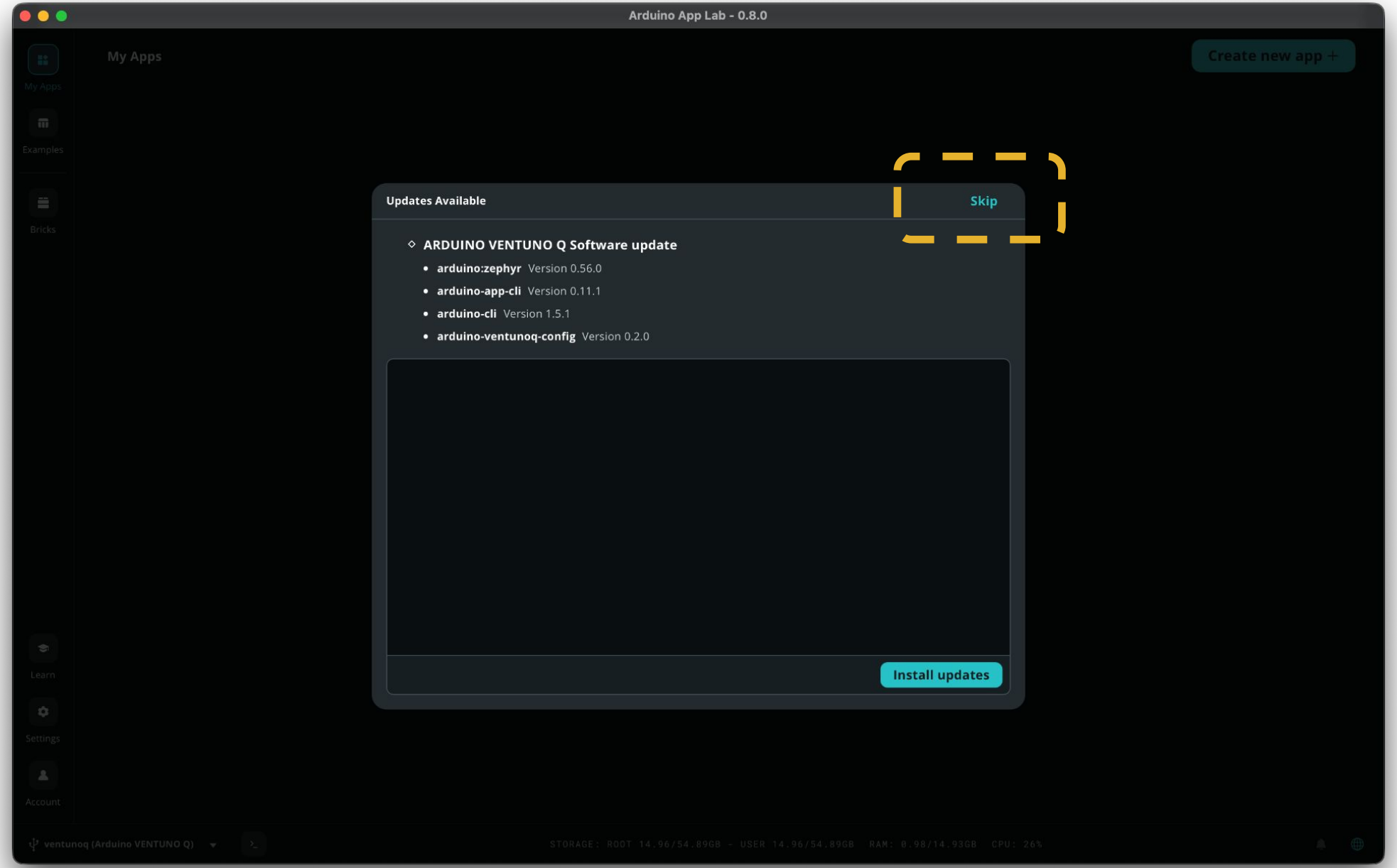

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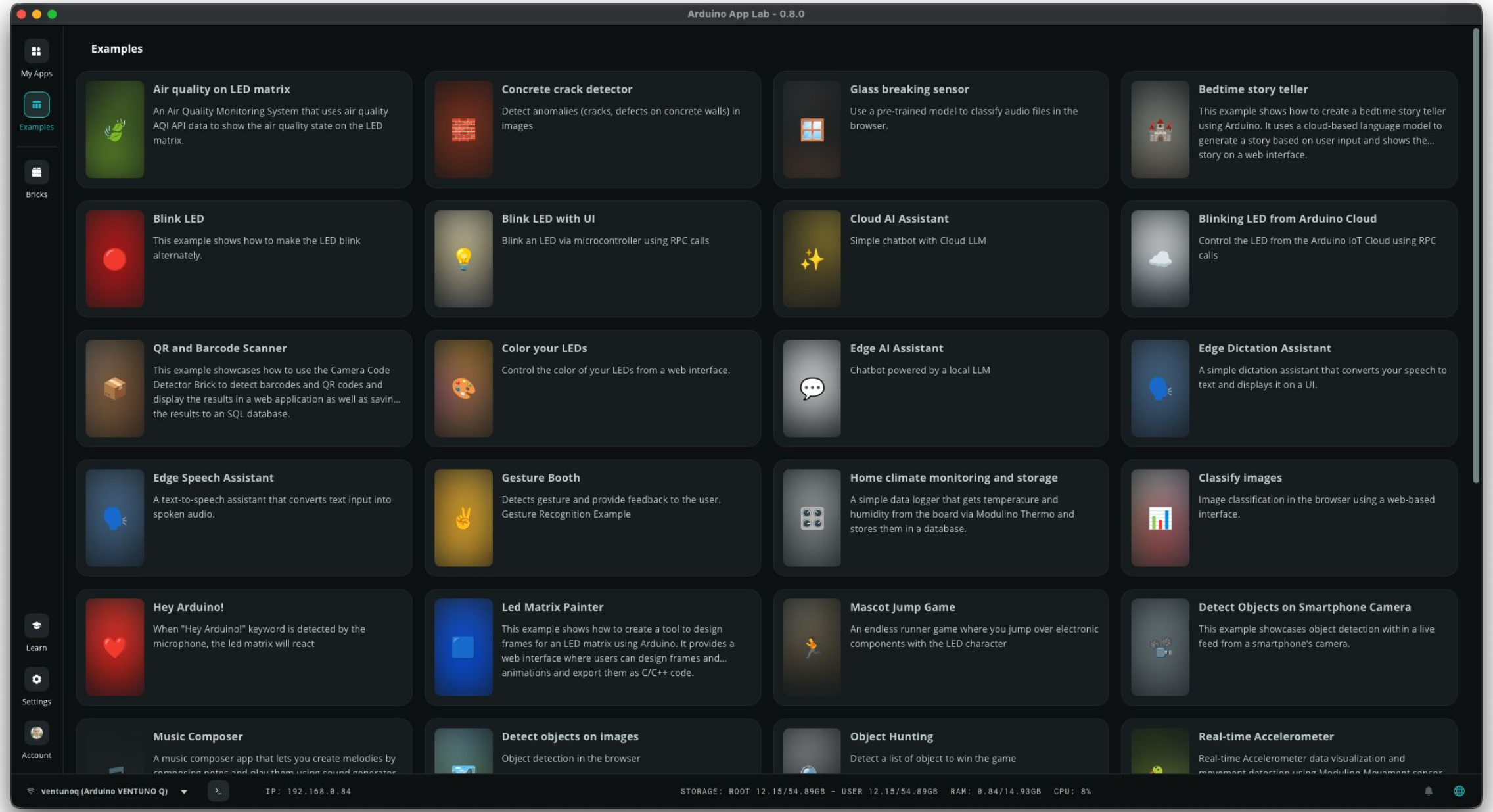

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https://github.com/edgeimpulse/example-arduino-app-lab-object-detection-using-flask

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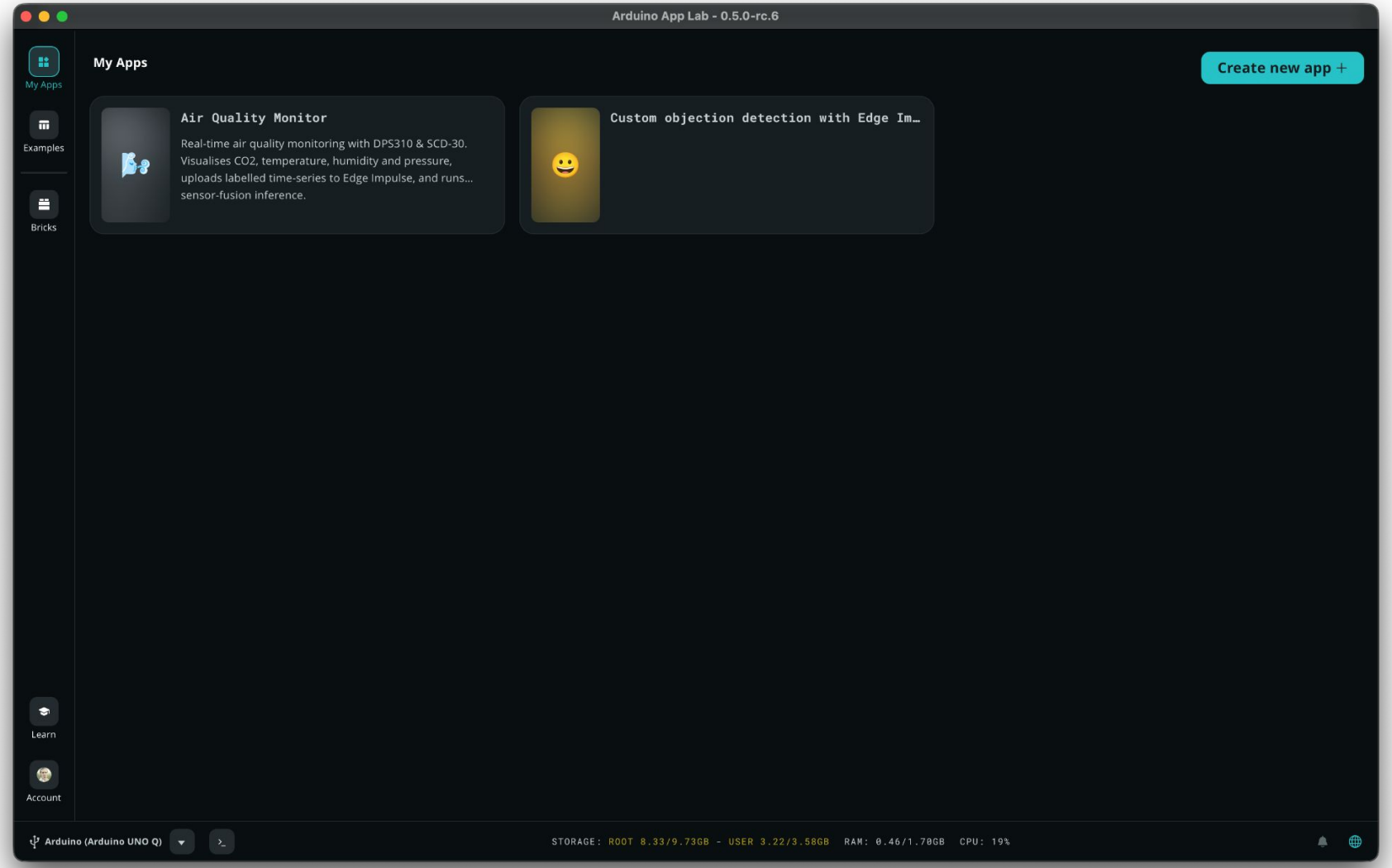

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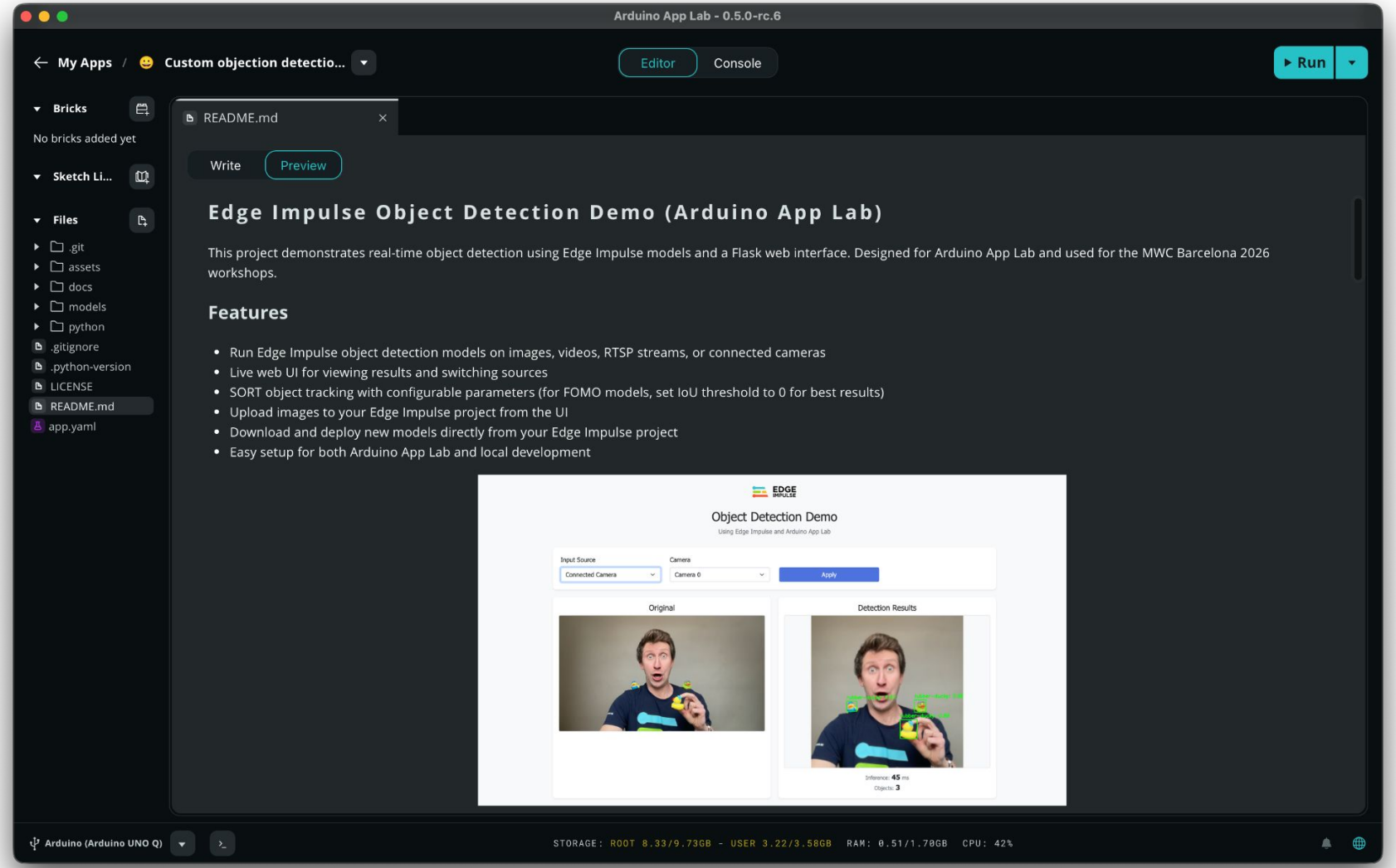

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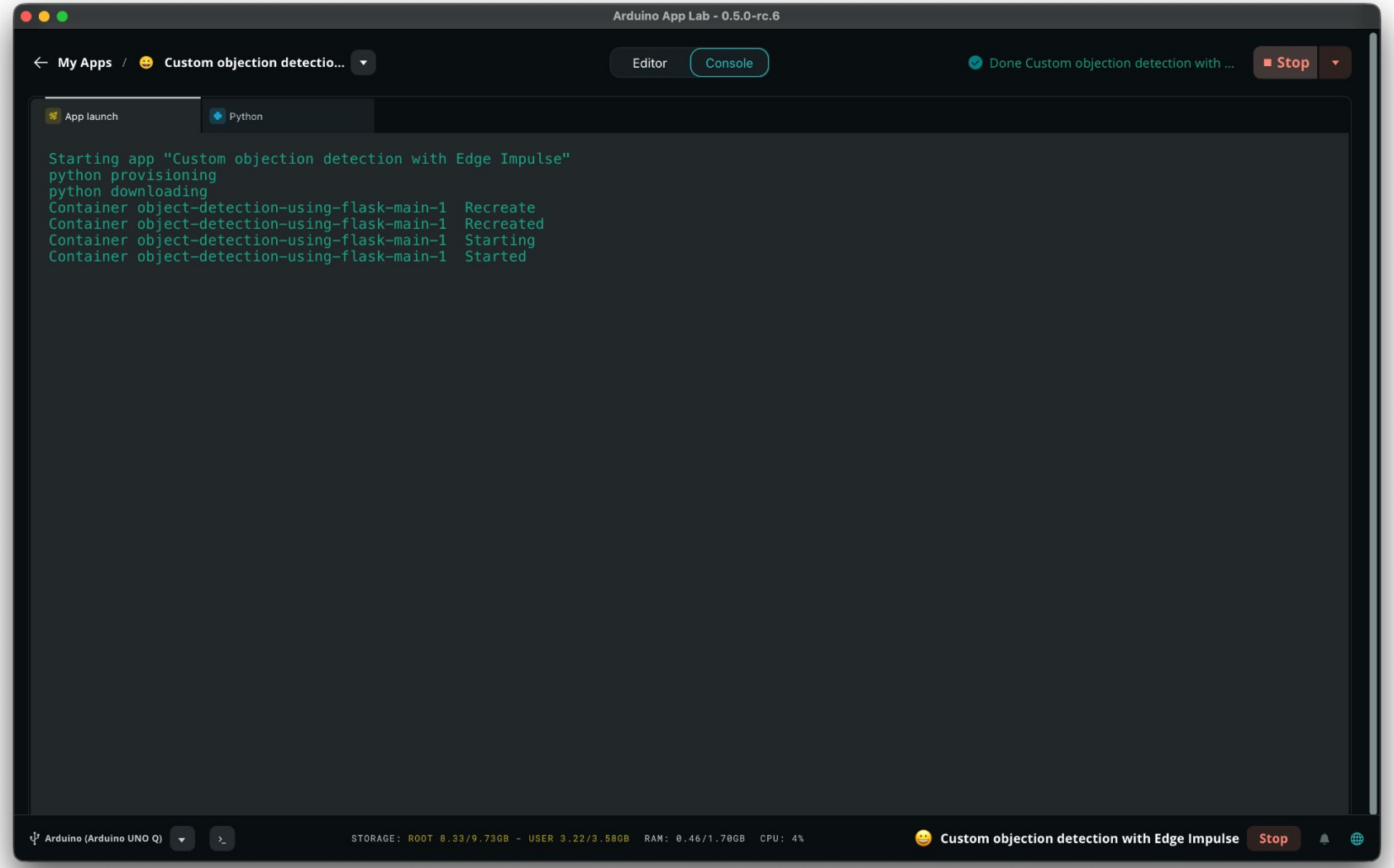

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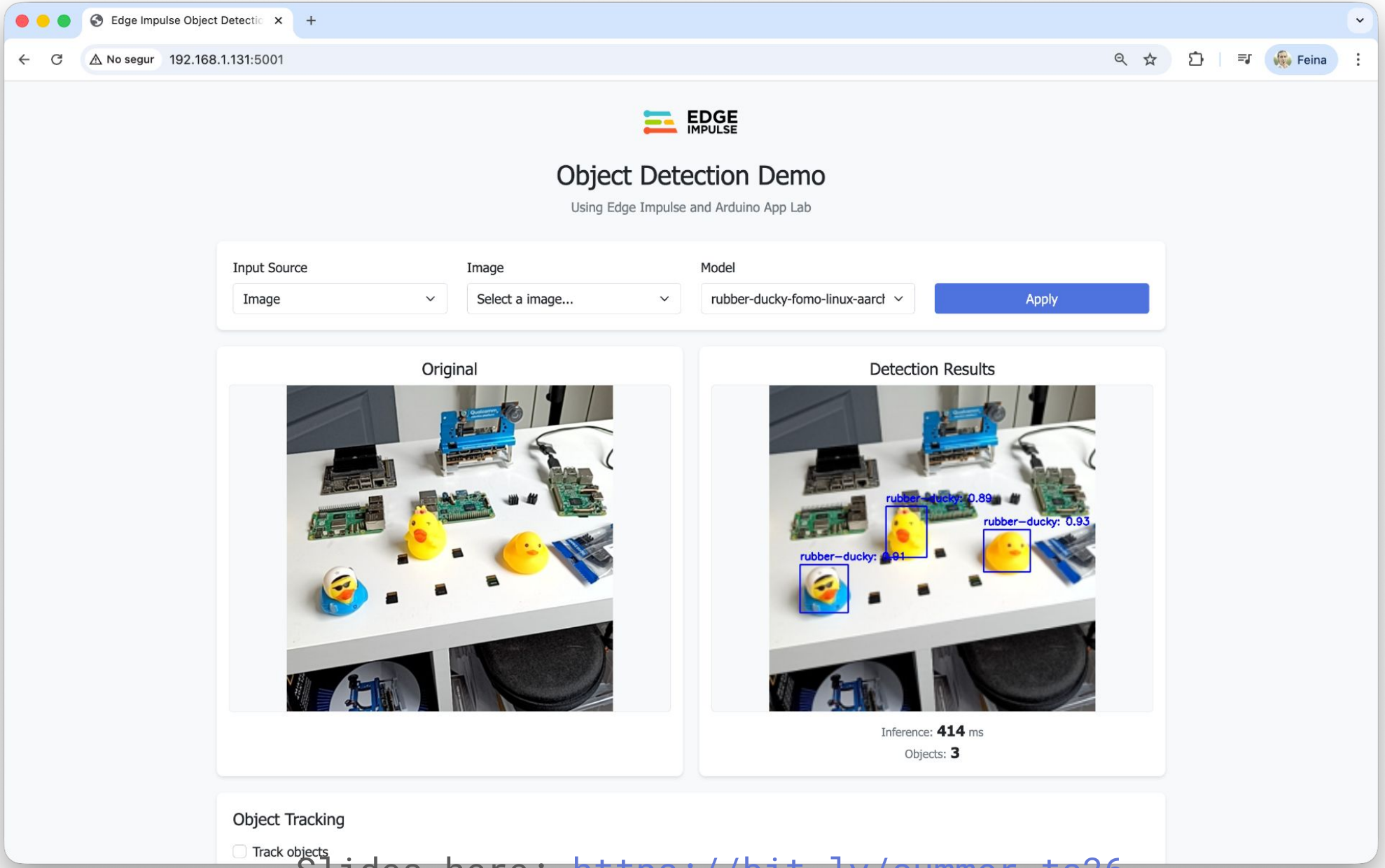

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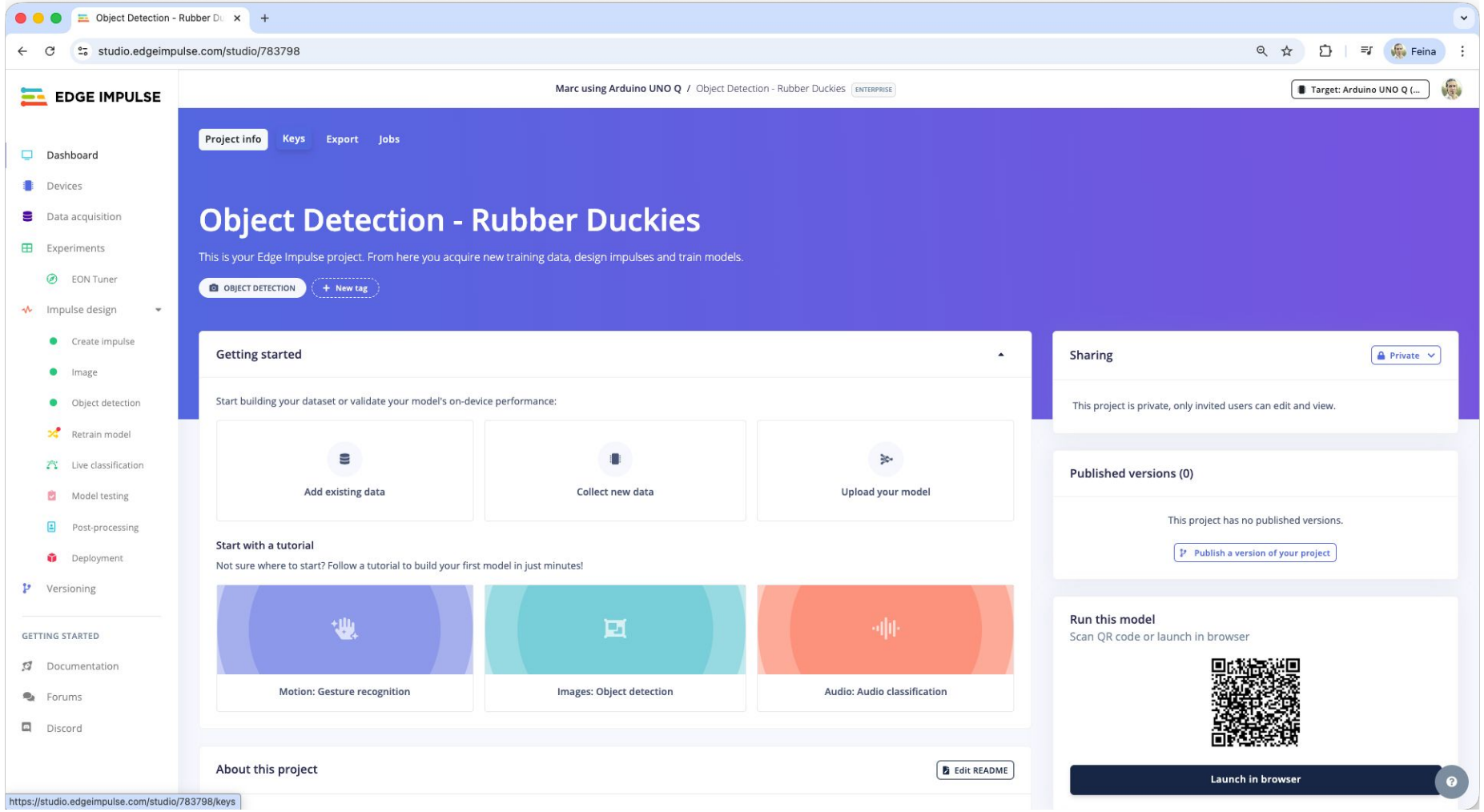
Let’s use our own Rubber Duckies project



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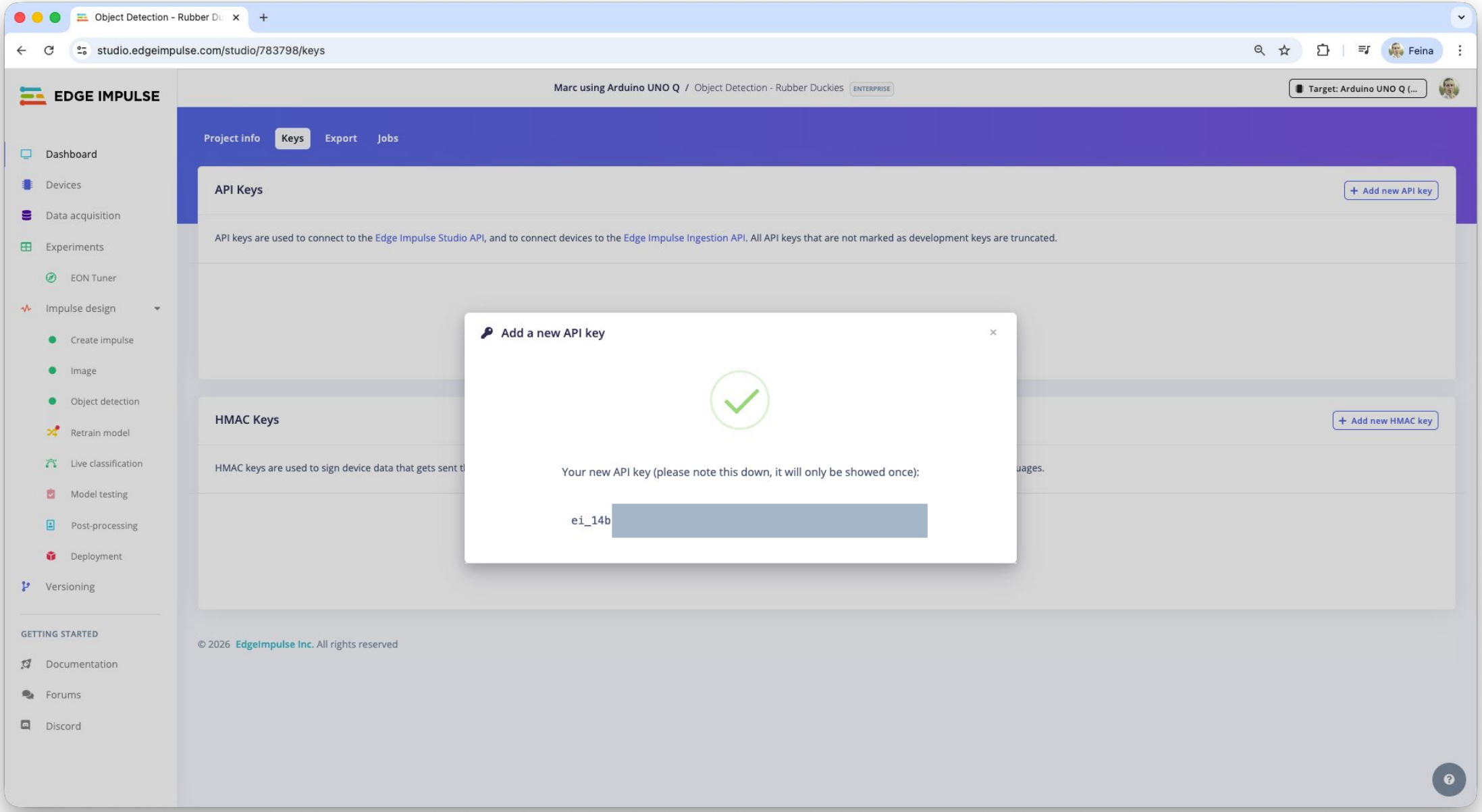

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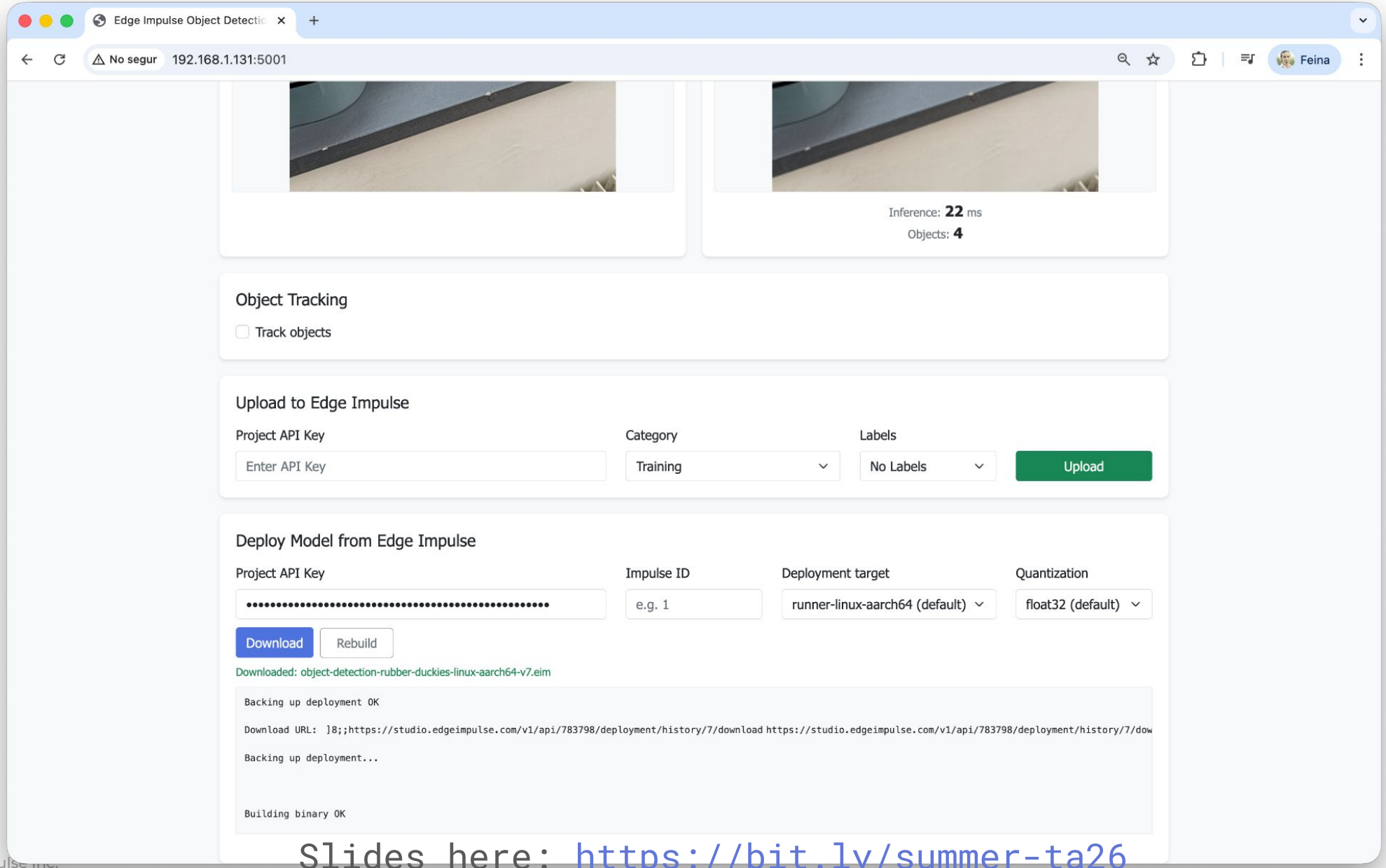

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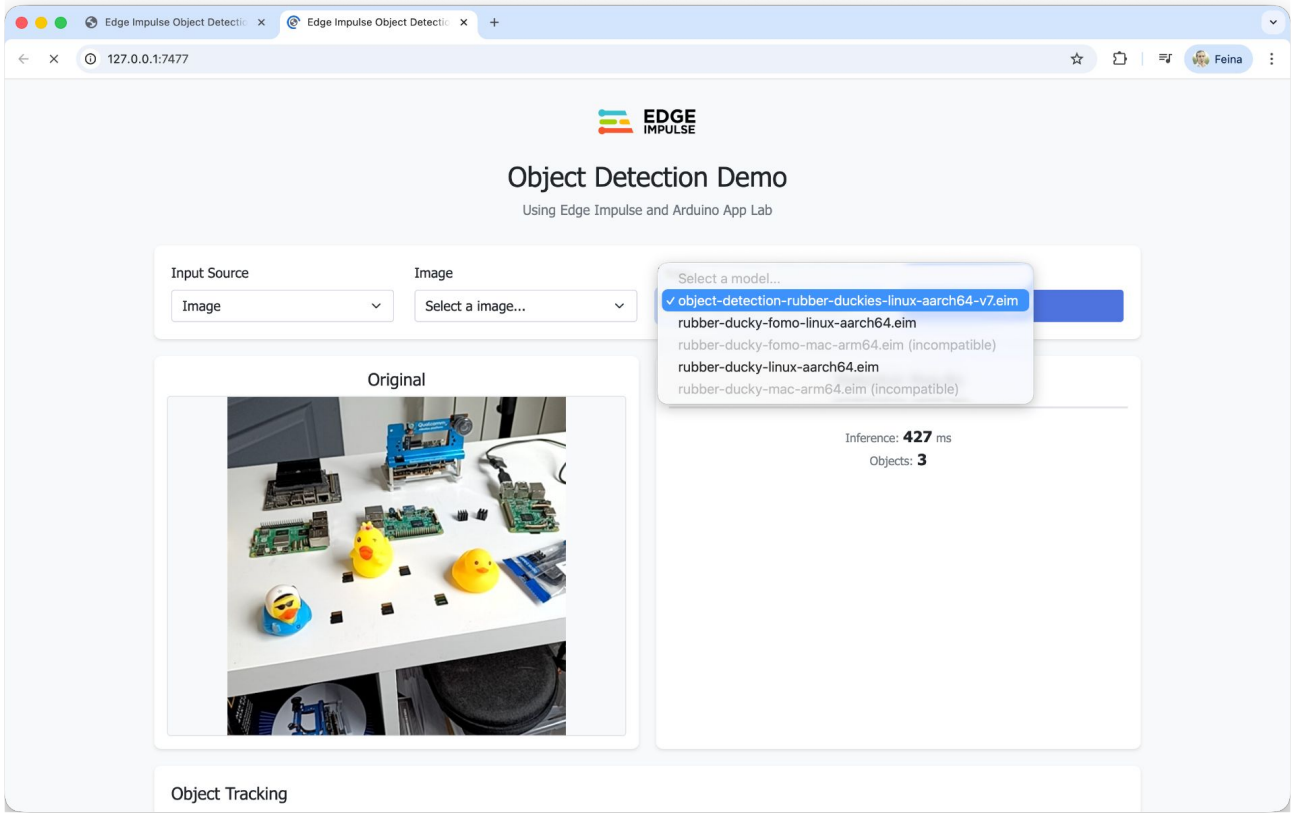
```
arduino@Arduino:~/ArduinoApps/object-detection-using-flask$ ls -la models/
total 128700
drwxrwxr-x 2 arduino arduino     4096 Feb 25 15:02 .
drwxrwxr-x 8 arduino arduino     4096 Feb 25 14:38 ..
-rwxr-xr-x 1 arduino arduino 14177624 Feb 25 15:02 object-detection-rubber-duckies-linux-aarch64-v7.eim
-rwxrwxr-x 1 arduino arduino 14802960 Feb 24 15:02 rubber-ducky-fomo-linux-aarch64.eim
-rwxrwxr-x 1 arduino arduino  6366352 Feb 24 15:02 rubber-ducky-fomo-mac-arm64.eim
-rwxrwxr-x 1 arduino arduino 51962504 Feb 24 15:02 rubber-ducky-linux-aarch64.eim
-rwxrwxr-x 1 arduino arduino 44447296 Feb 24 15:02 rubber-ducky-mac-arm64.eim
```



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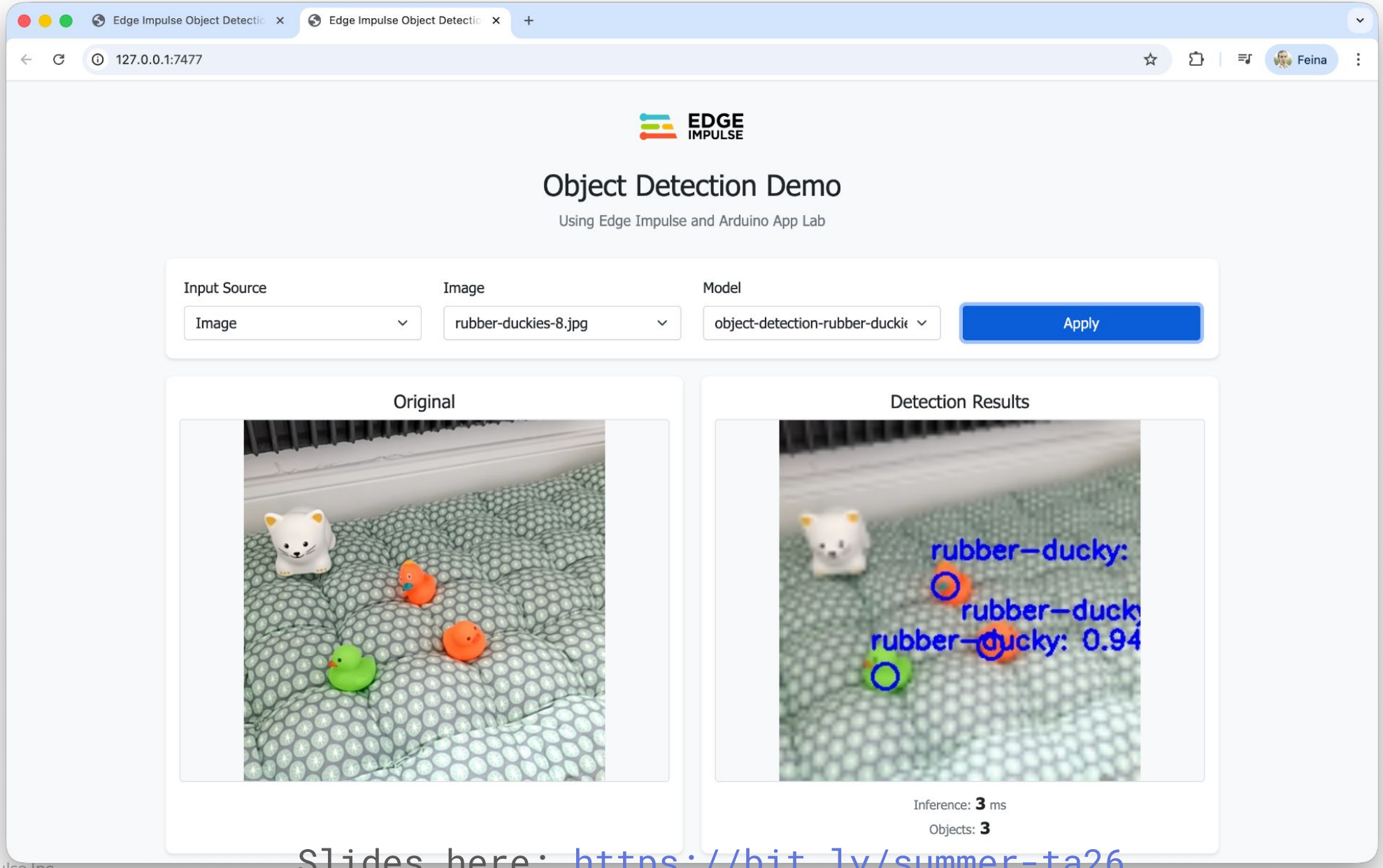

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Deploy AI models with Arduino AppLab Bricks

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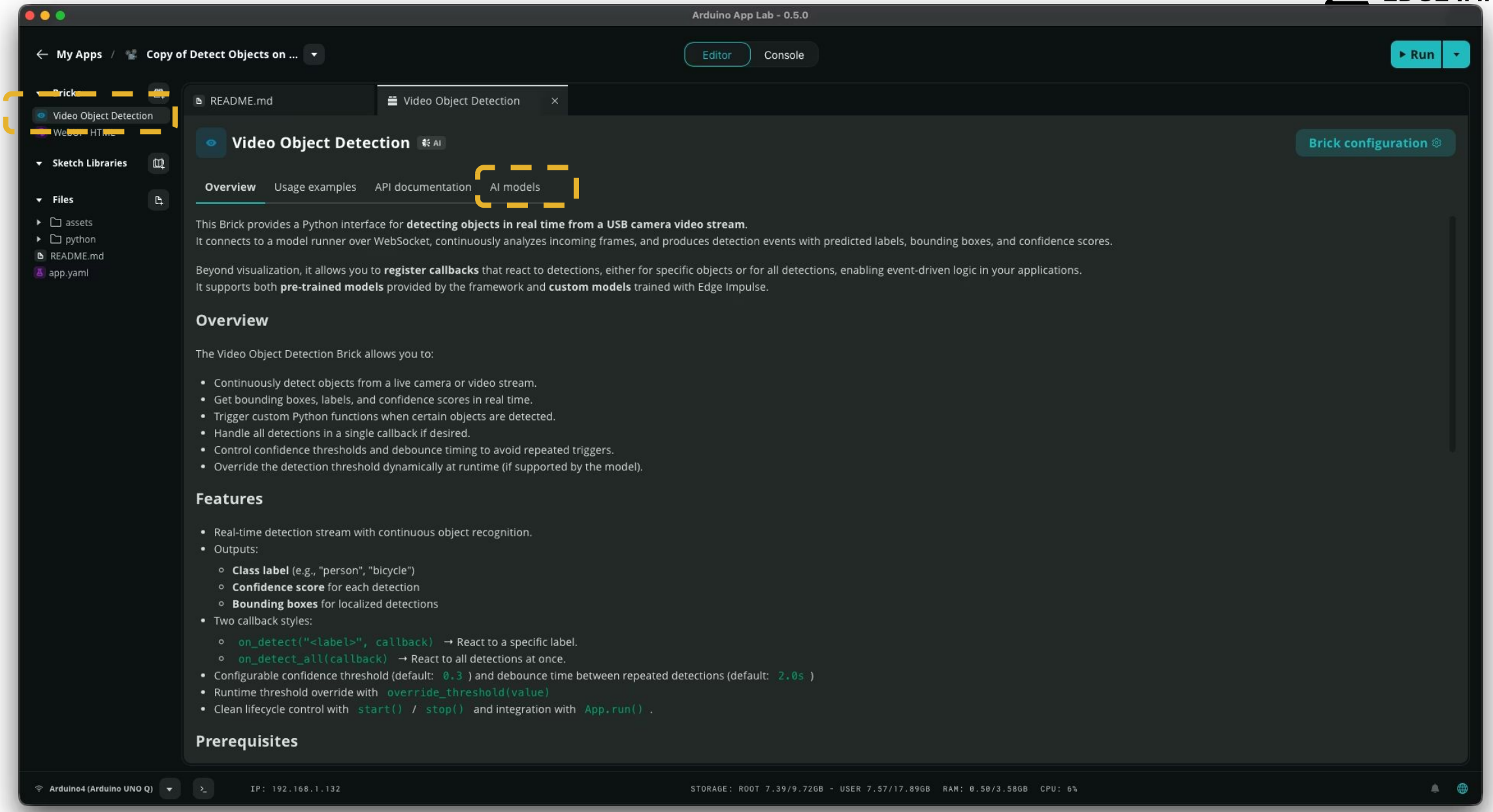

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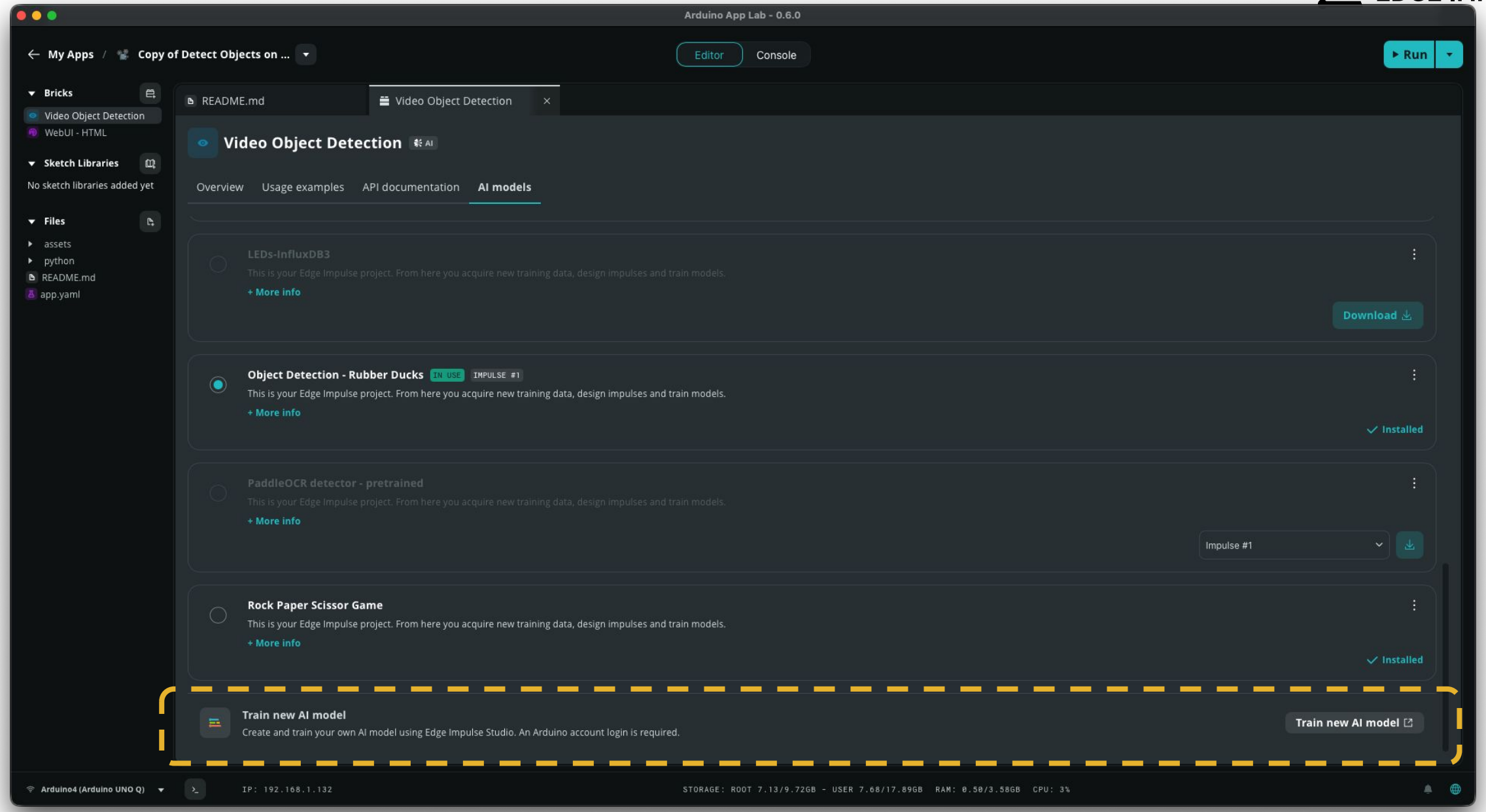

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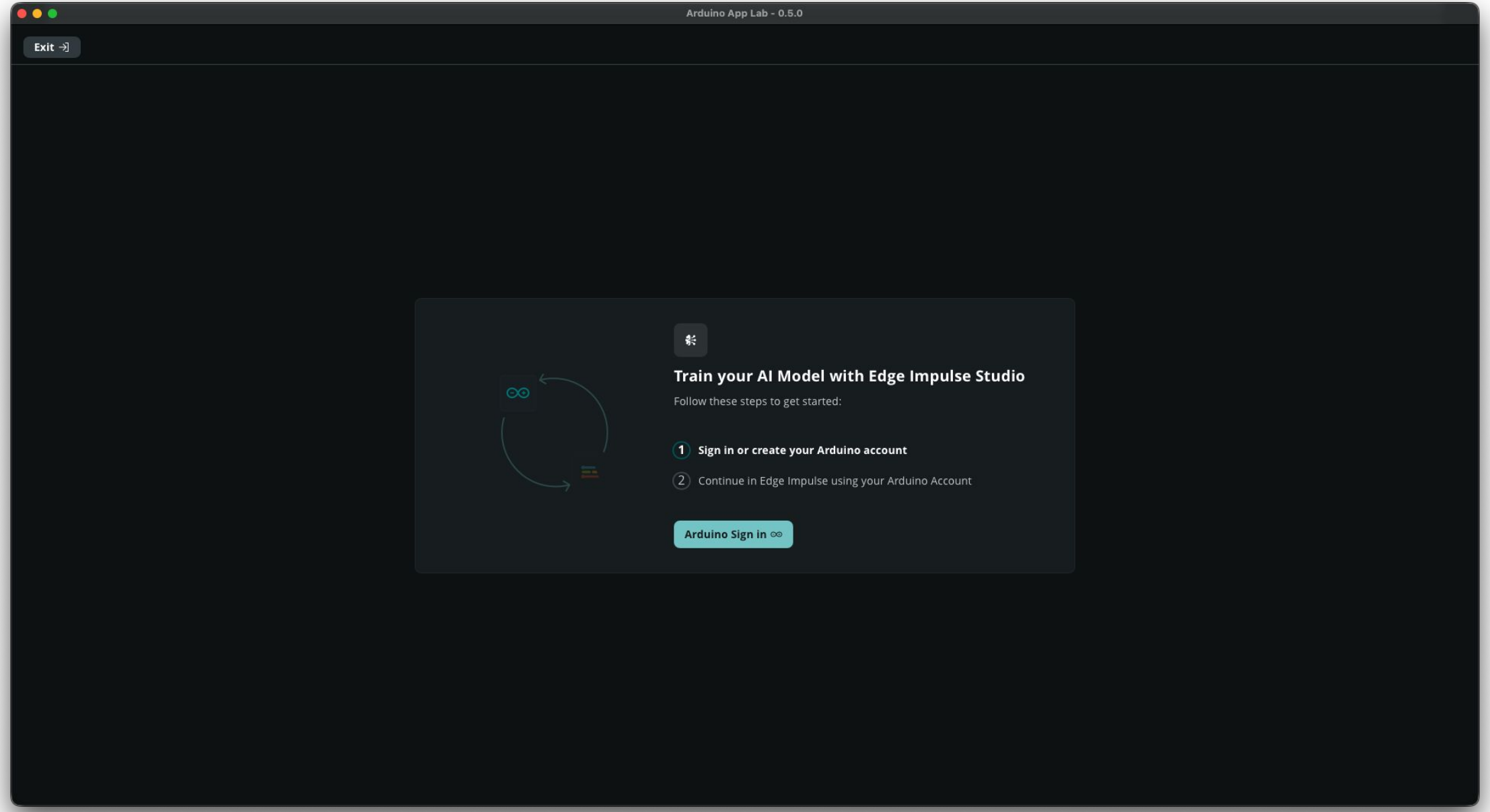

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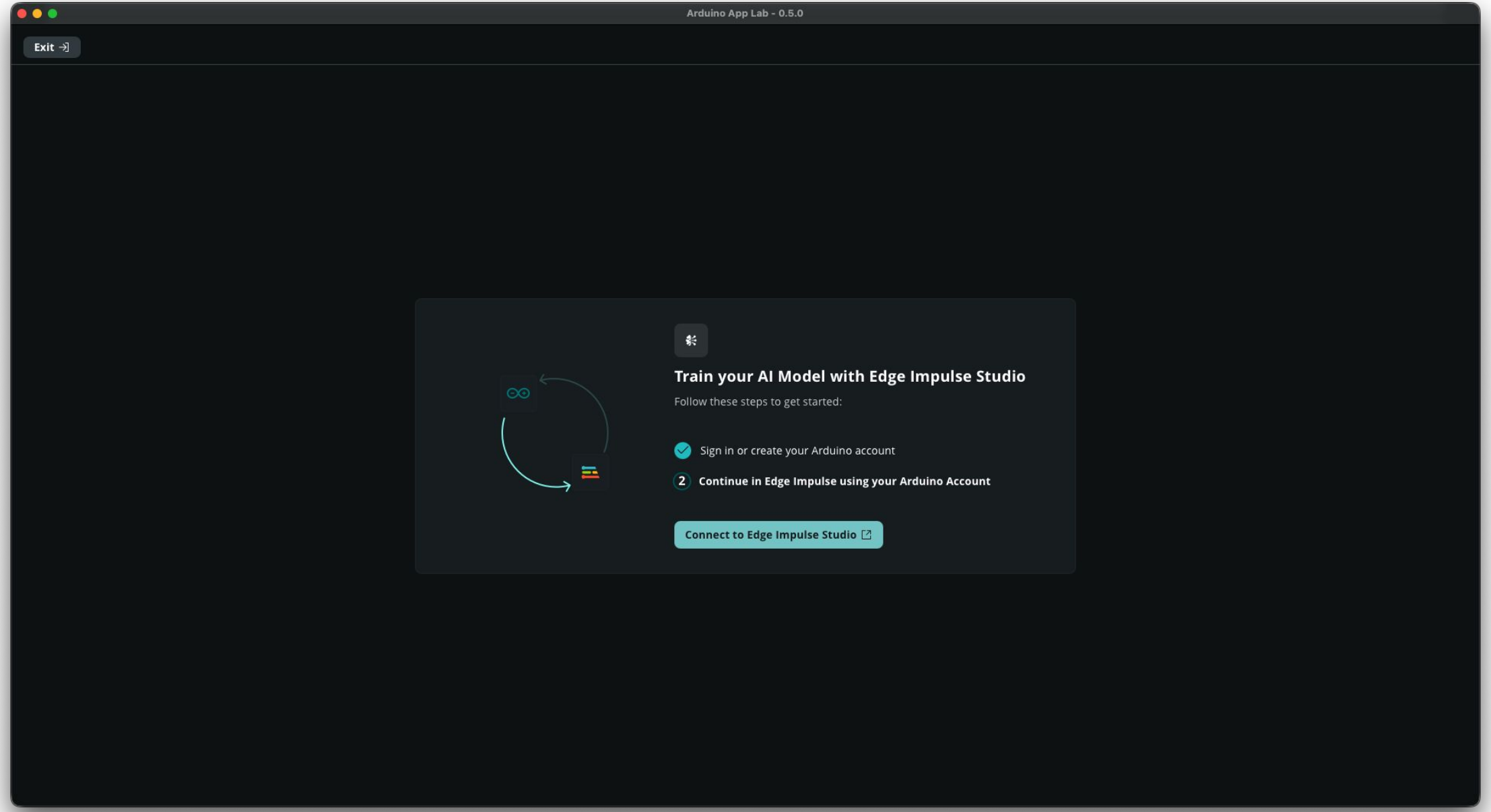

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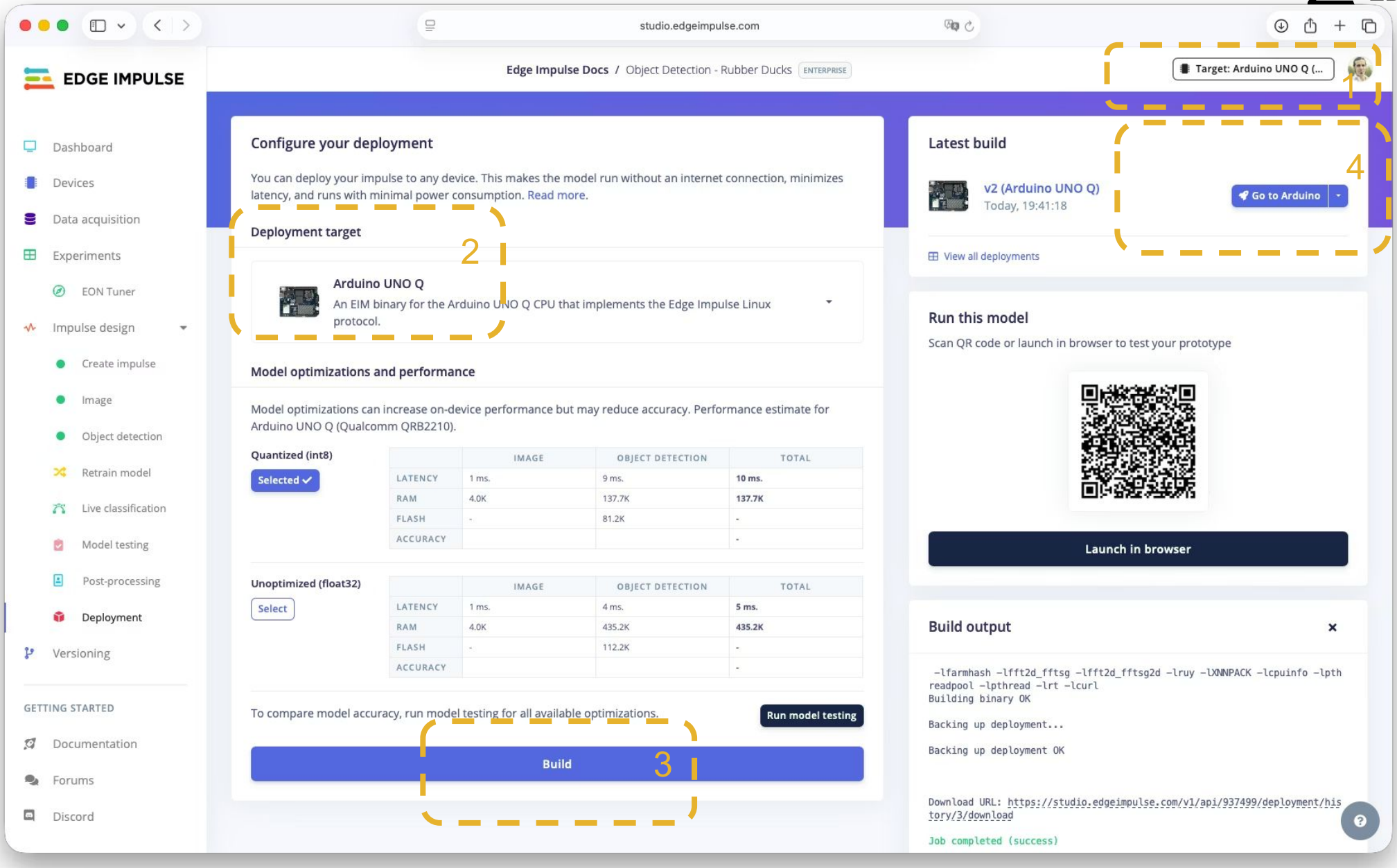

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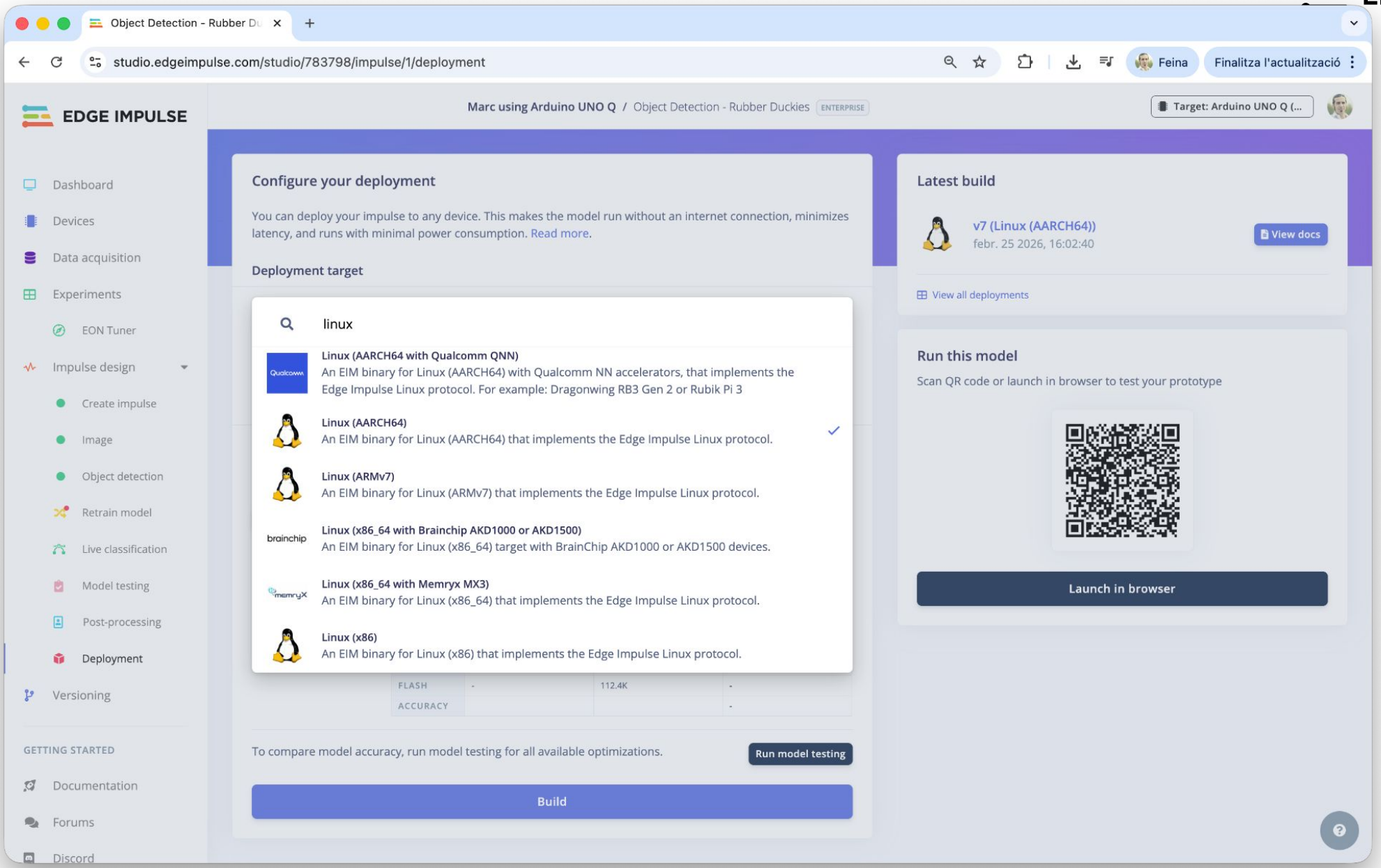

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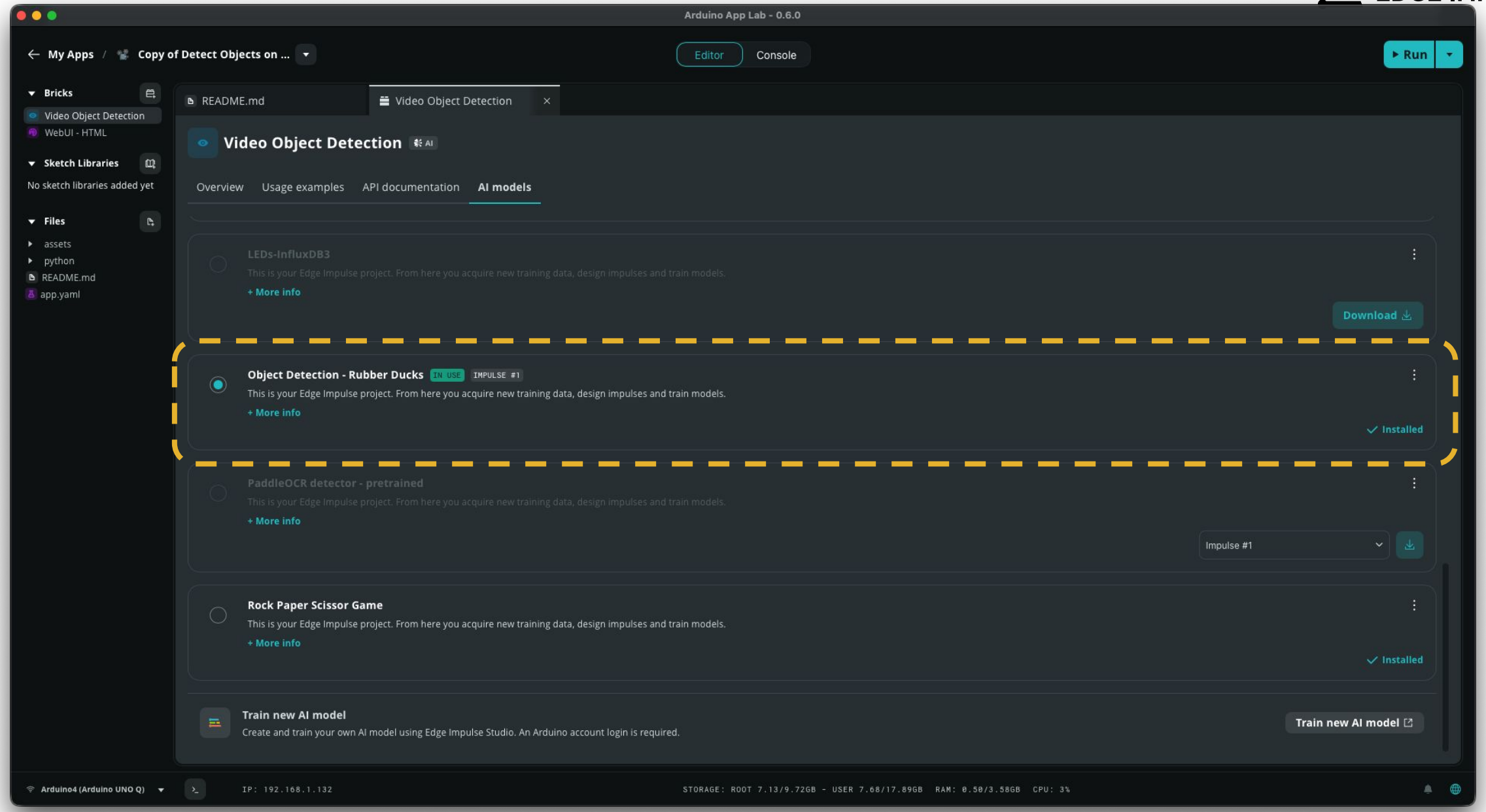

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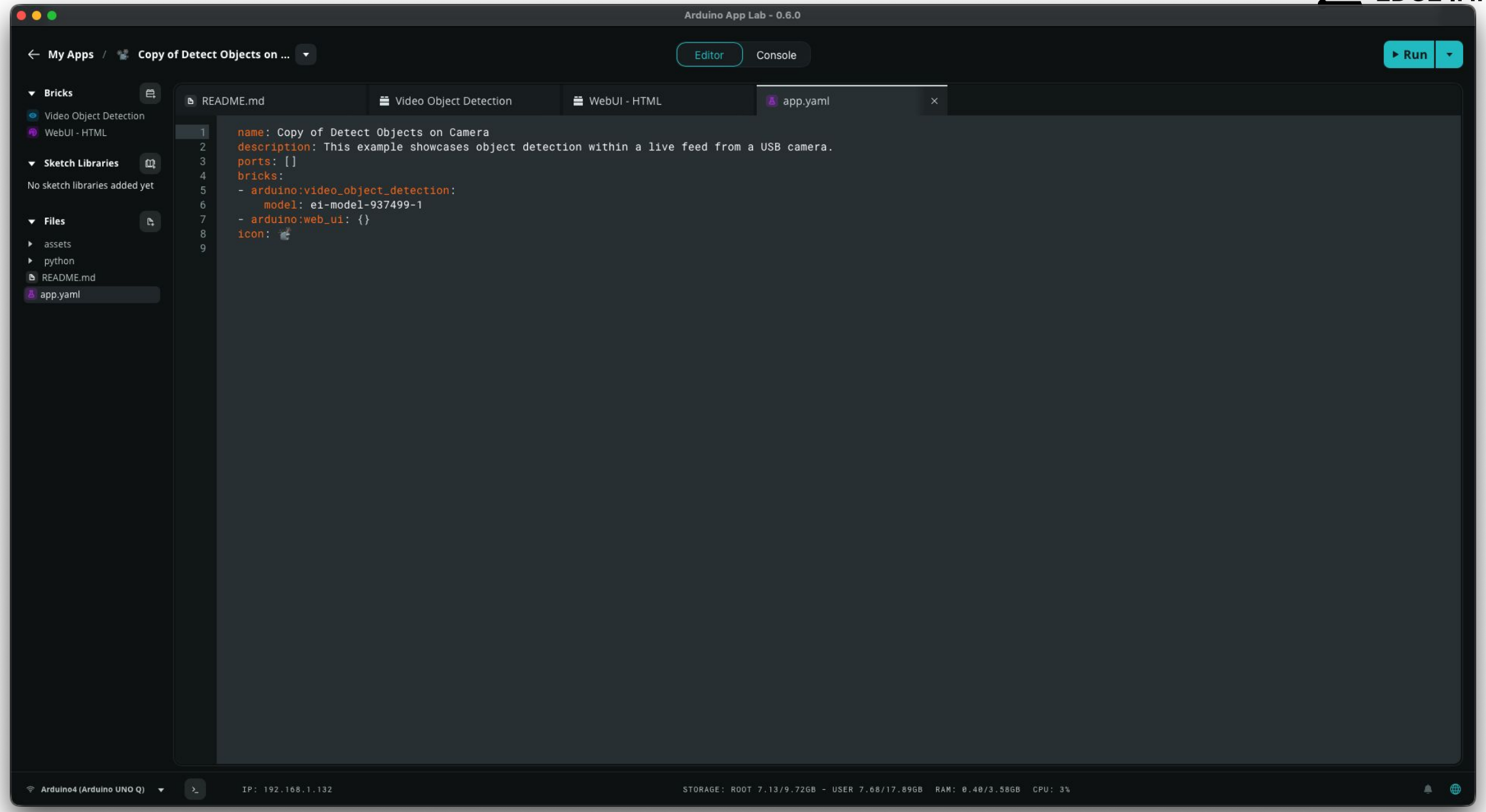

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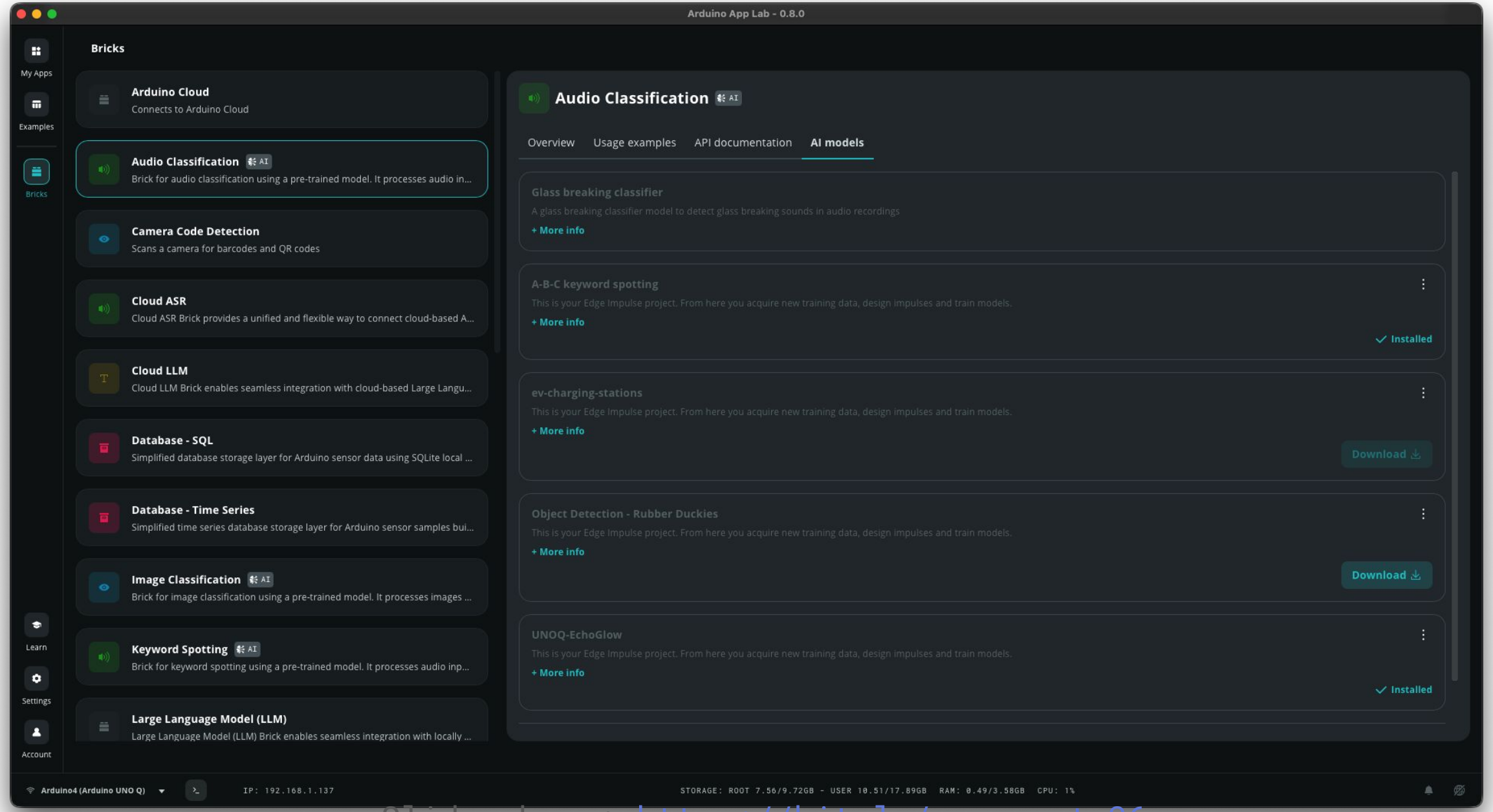

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Visual Anomaly Detection

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Machine Learning

Supervised learning

Task-driven

- Regression
- Classification
- Object detection

Unsupervised learning

Data-driven

- Clustering
- Segmentation
- Anomaly detection

Reinforcement learning

Learn from experience

- Robotics
- Games
- Recommender systems

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Anomaly detection today

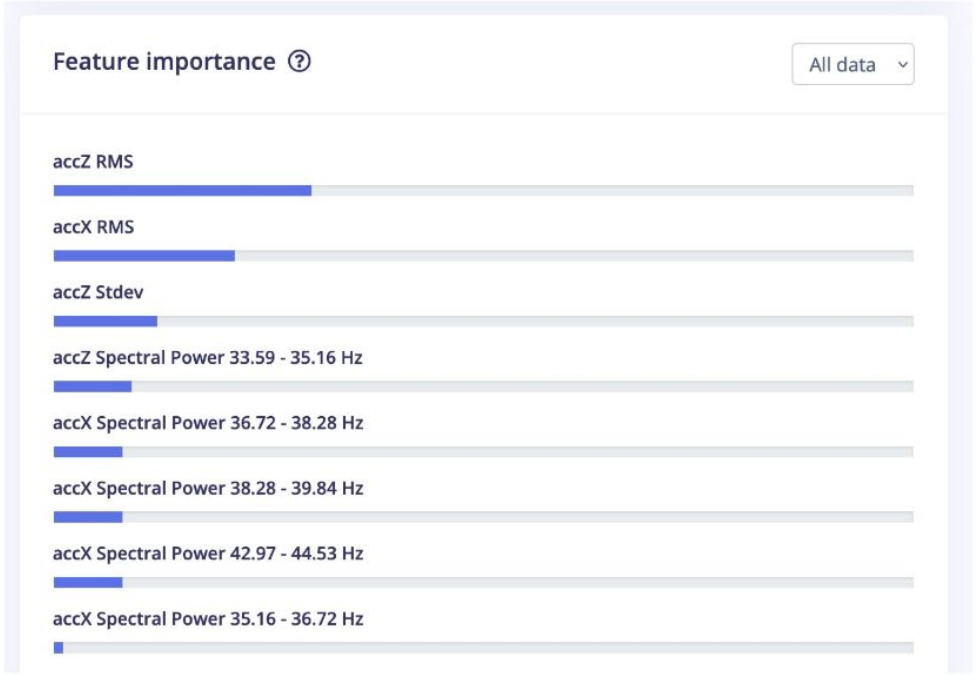
Input

↓

DSP Features

↓

Clustering algorithm



Great for basic sensor data
for which you can reason about features

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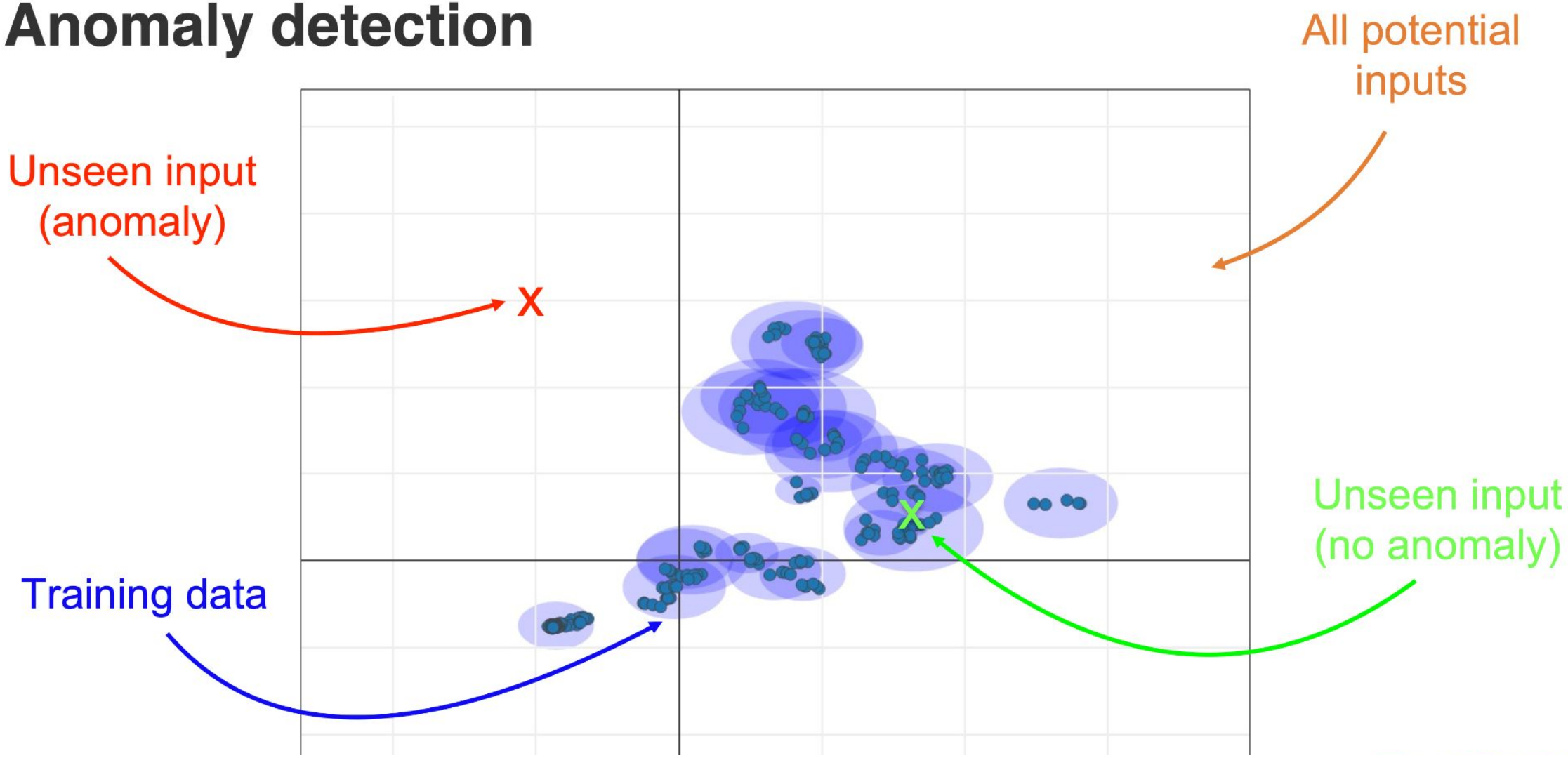
Anomaly detection



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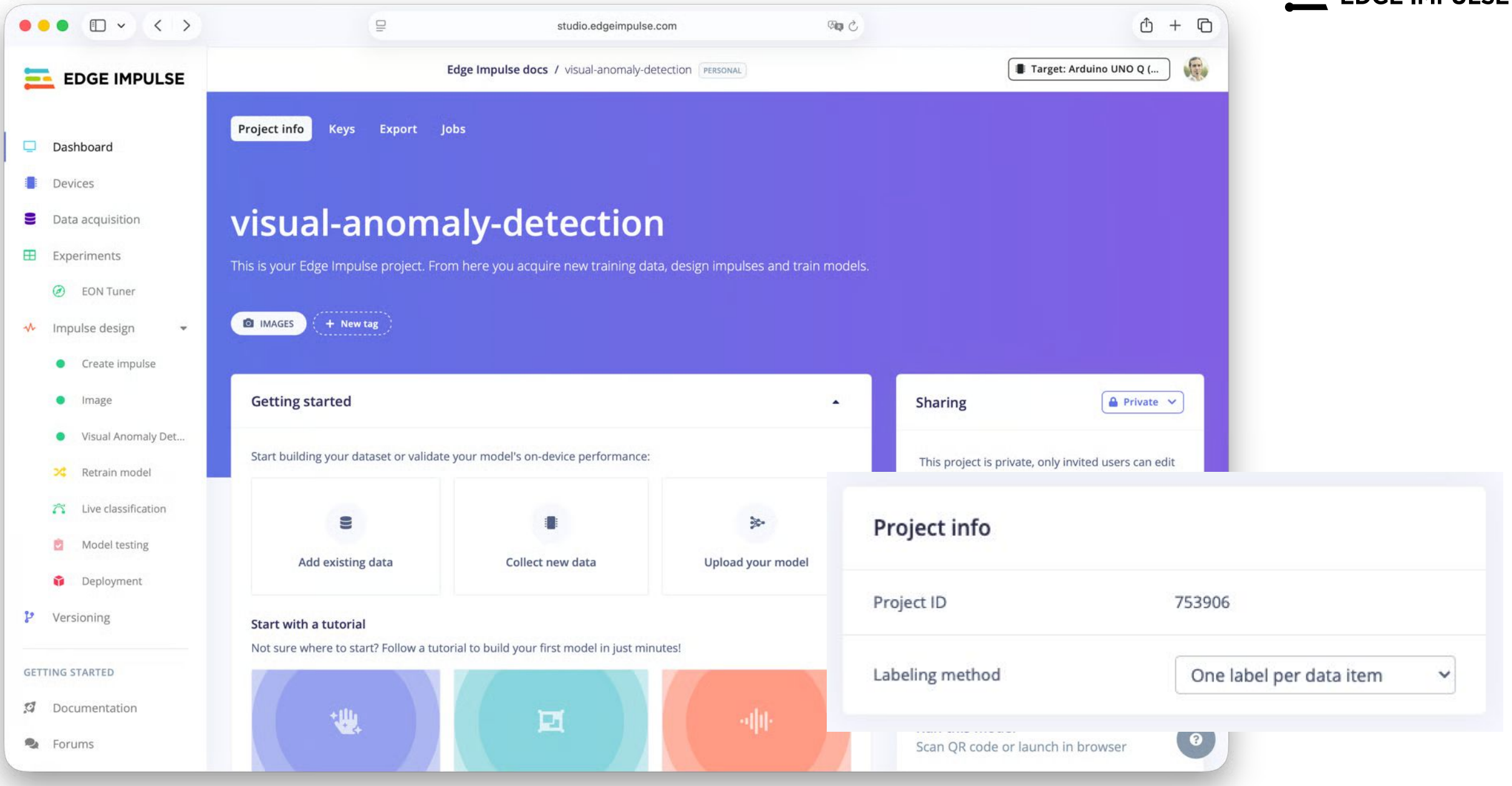

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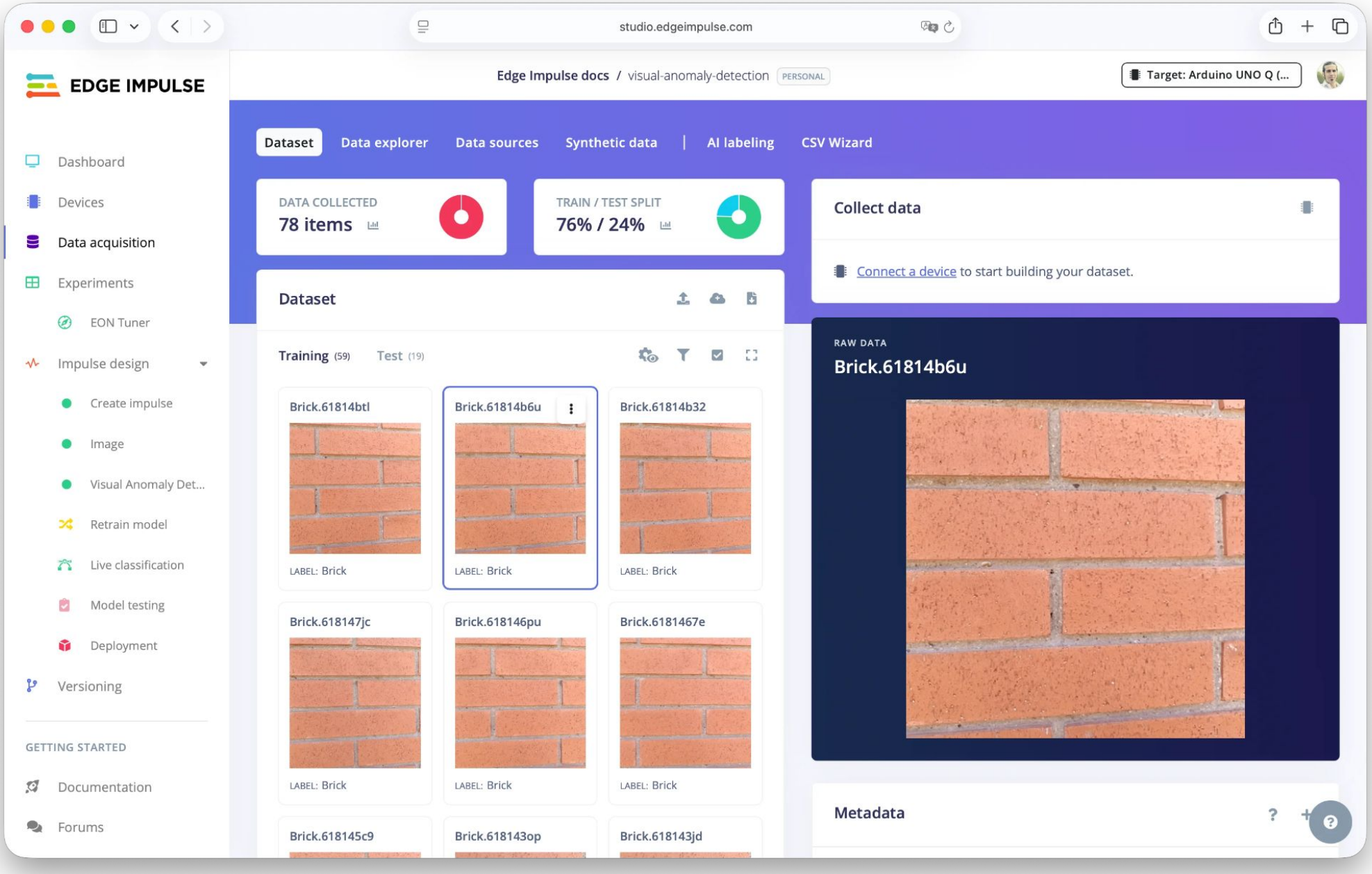

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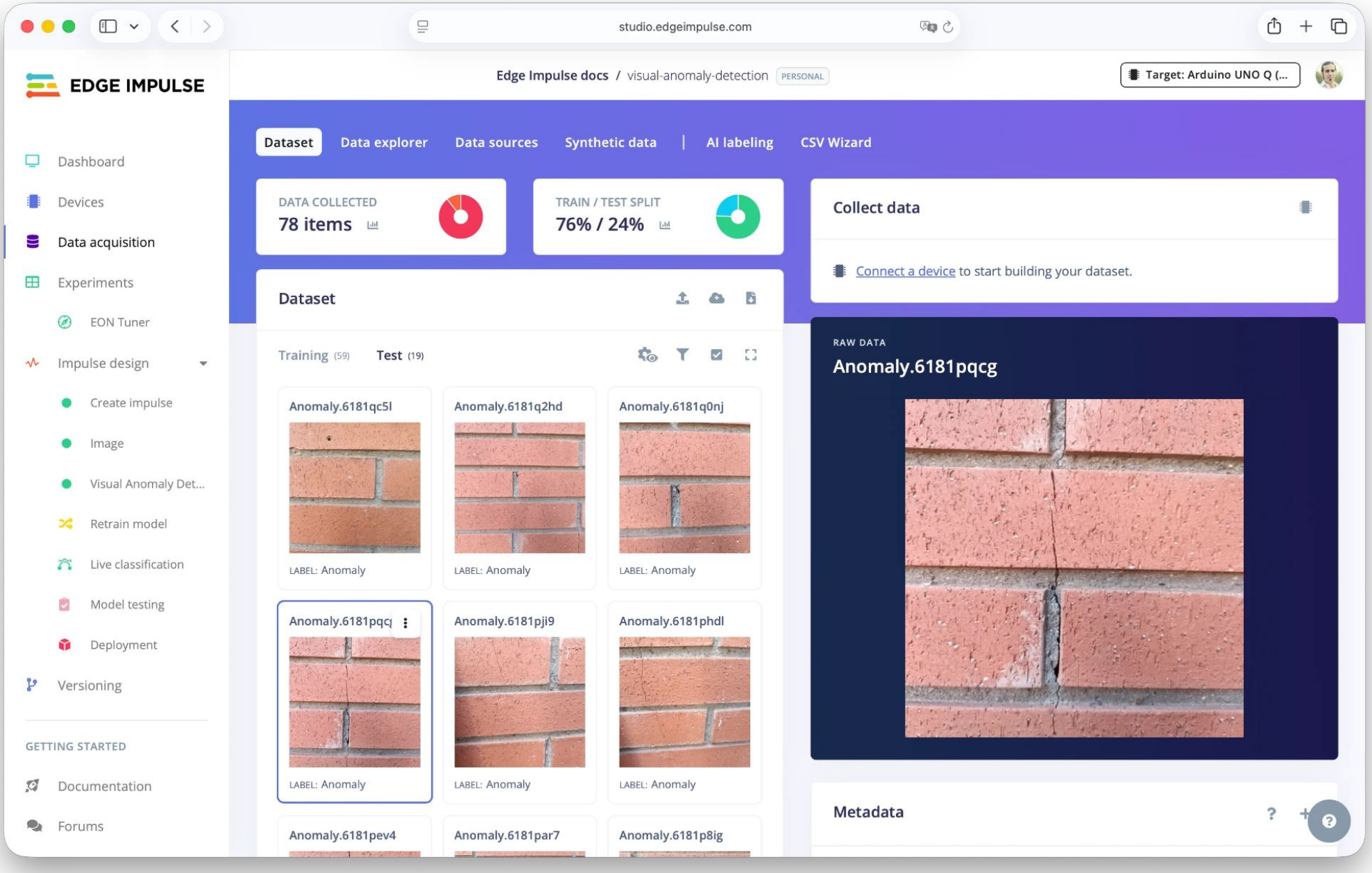

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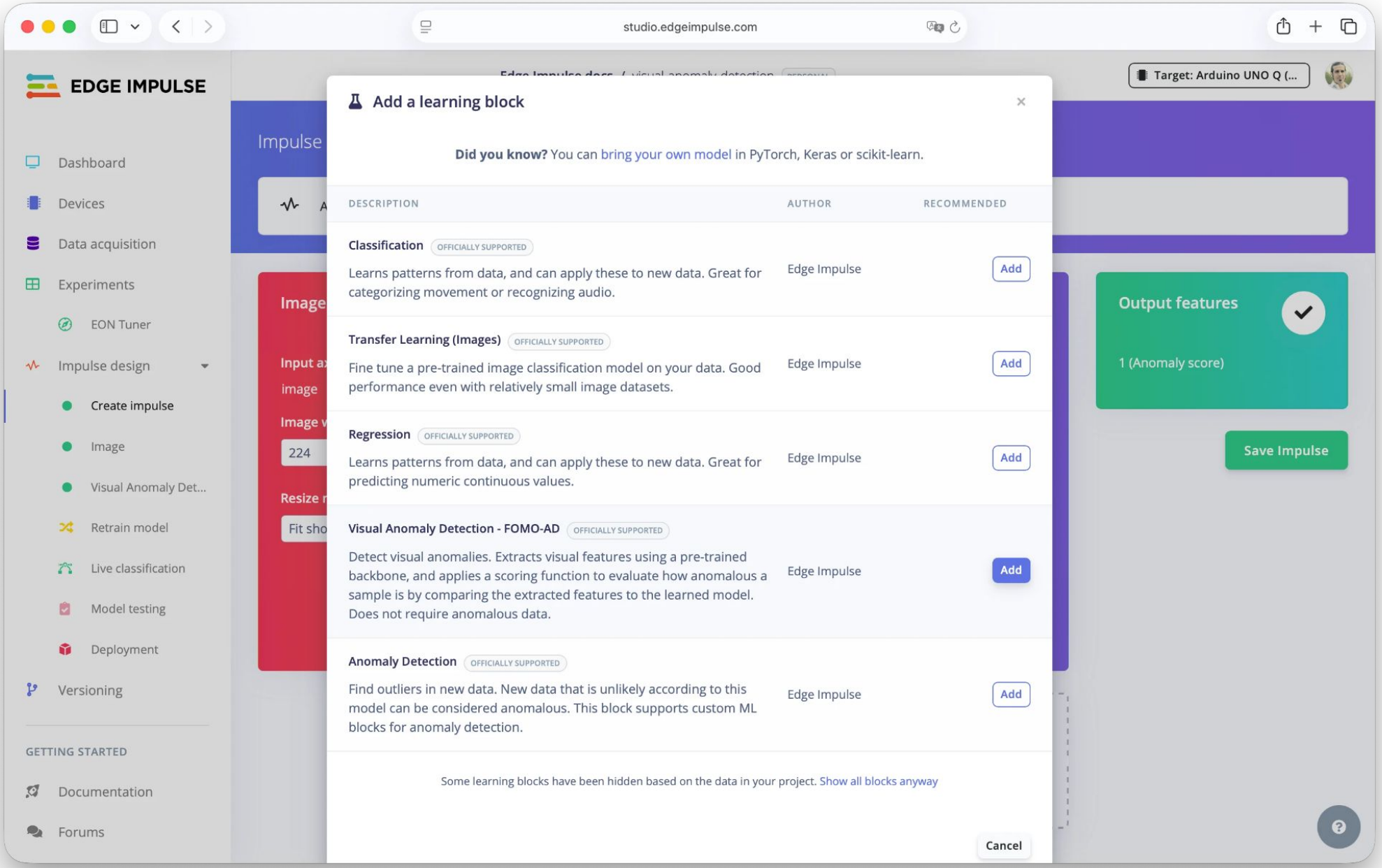

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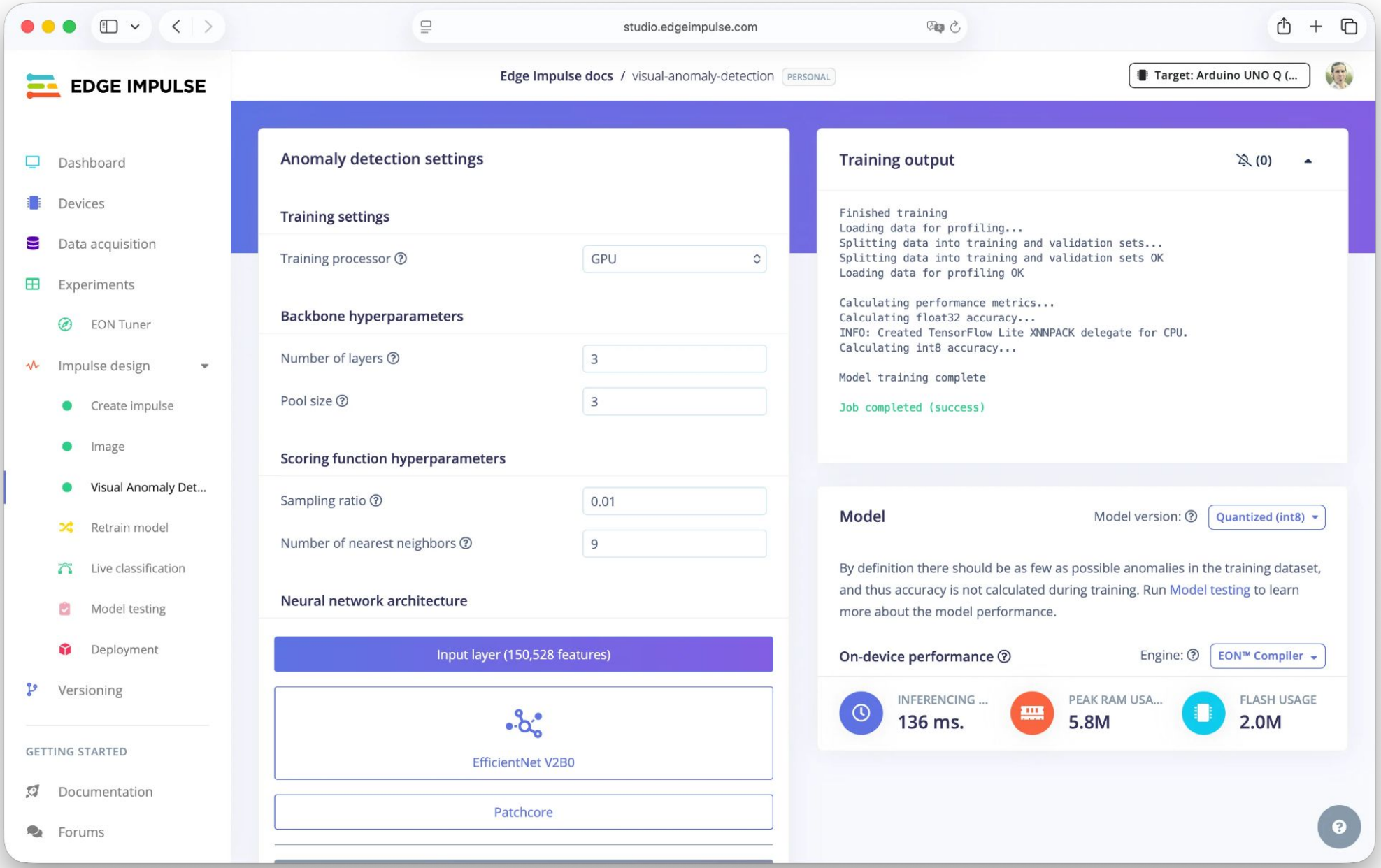

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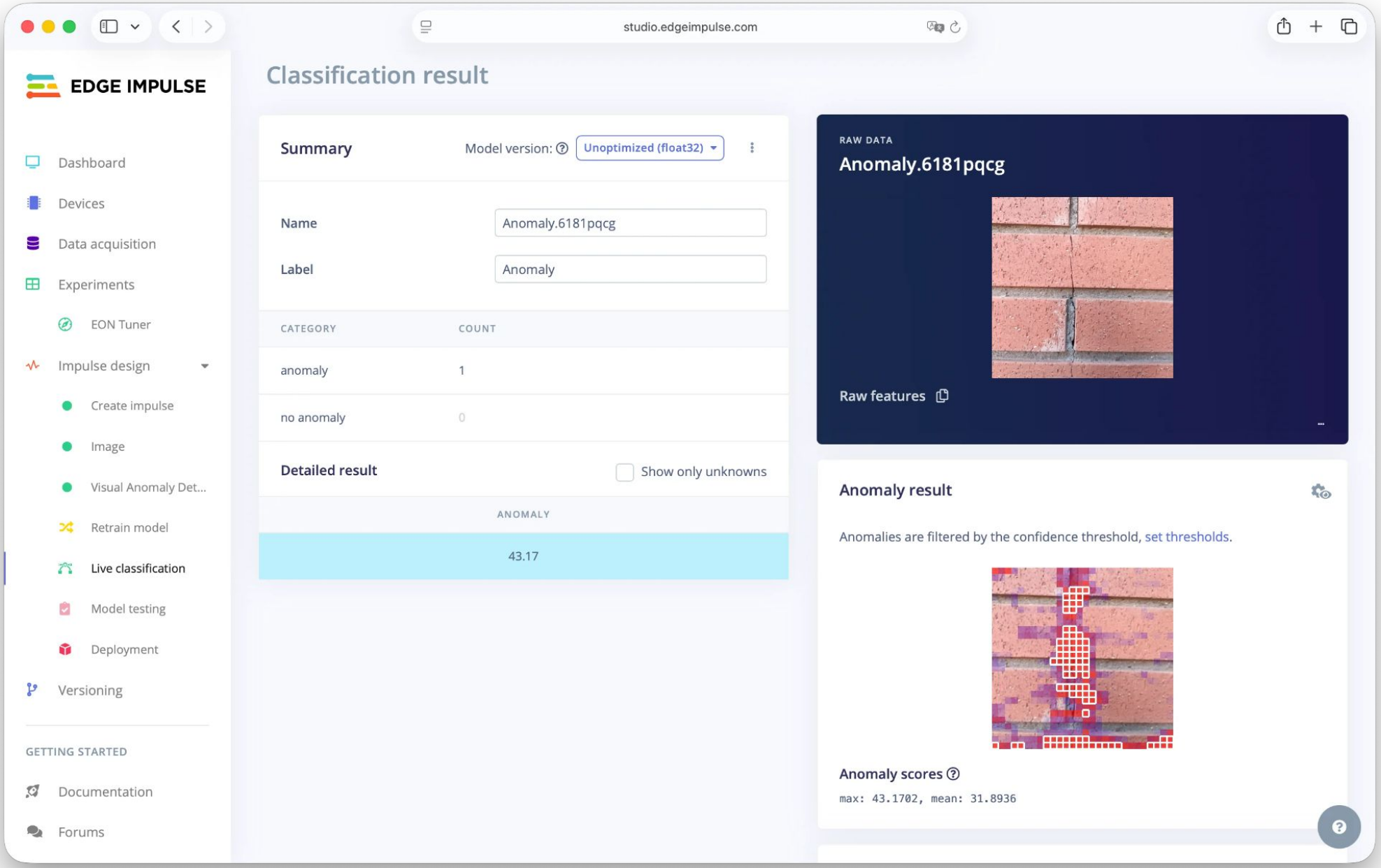

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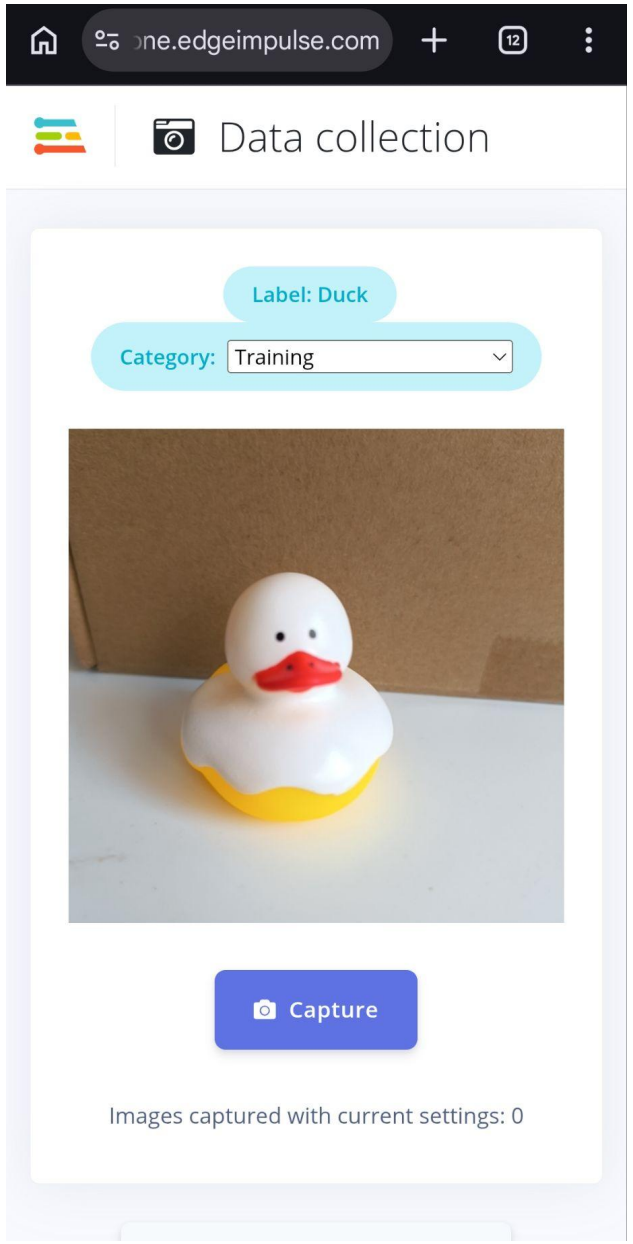

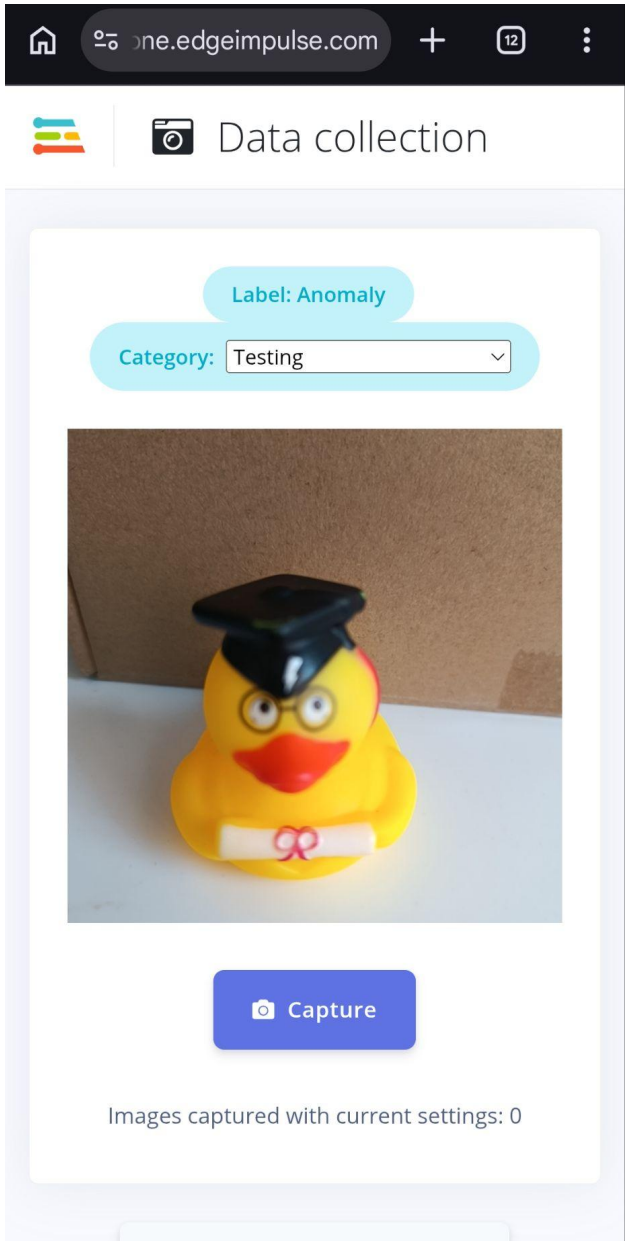

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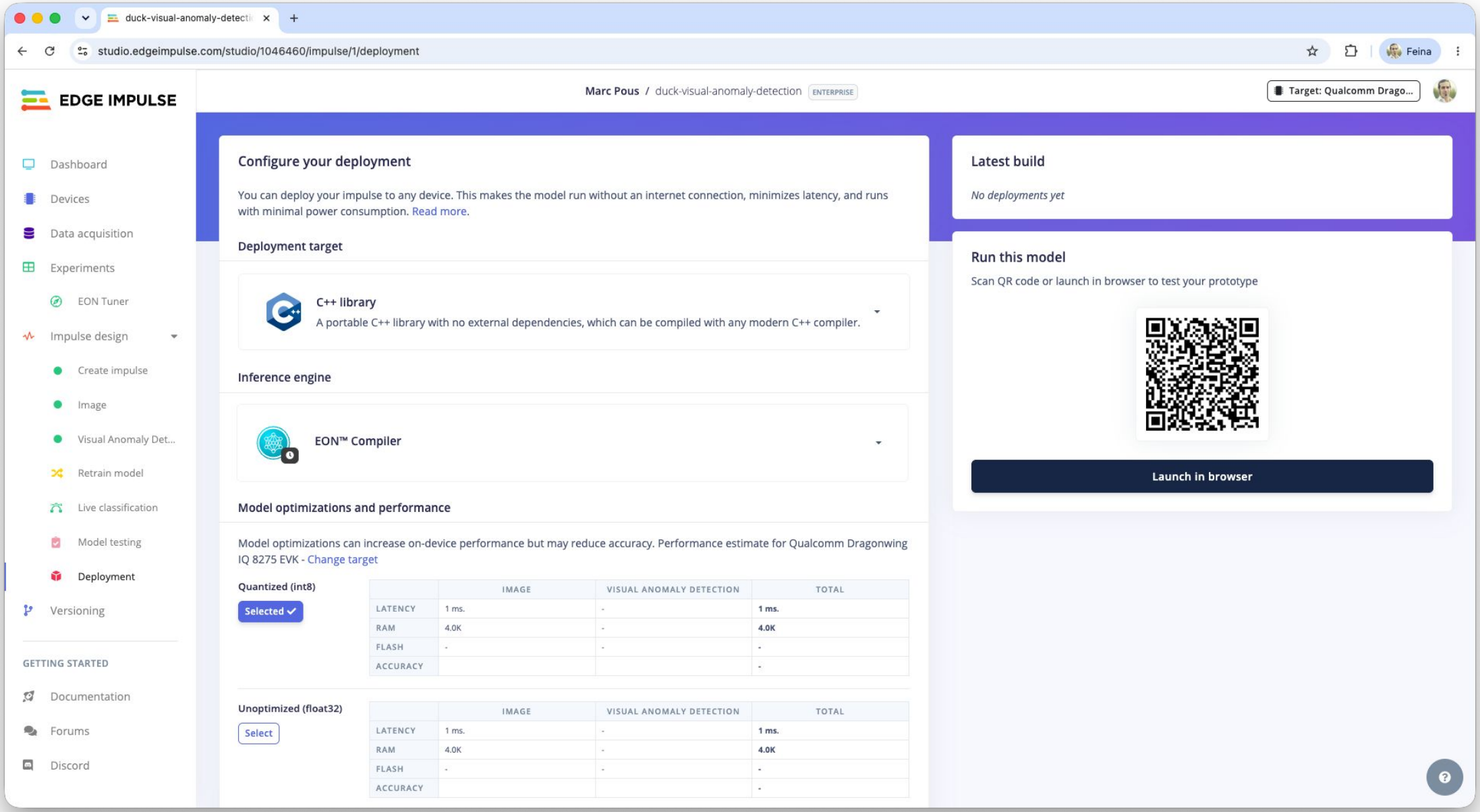

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Marc Pous / duck-...

Inferencing...

uncertain

Time per inference: 19 ms.

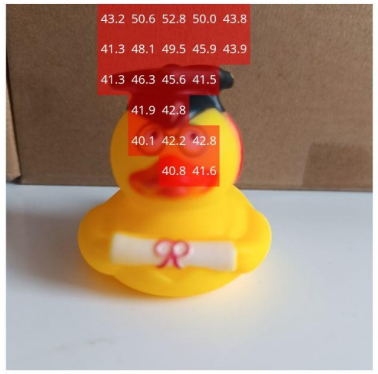
Marc Pous / duck-...

Inferencing...

anomaly

Time per inference: 21 ms.

Marc Pous / duck-...

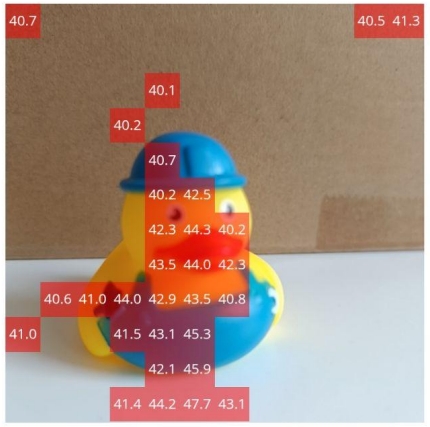


Inferencing...

anomaly

Time per inference: 20 ms.

Marc Pous / duck-...



Inferencing...

anomaly

Time per inference: 19 ms.

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Model Cascading

Object Detection → VLM

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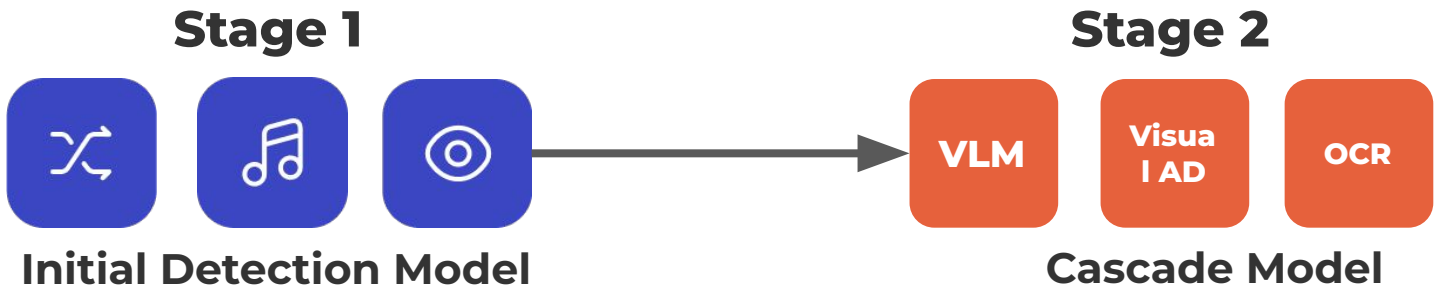
How does this work?



Any Sensor

Runs Continuously

Lower Power

Can narrow context for secondary models

Usually a specialised Edge Algorithm

Provides Deeper Context

Run when required

Runs on same device, edge server or cloud

Can be generalised (VLM/LLM) or a specialised second stage model (OCR, Visual AD)

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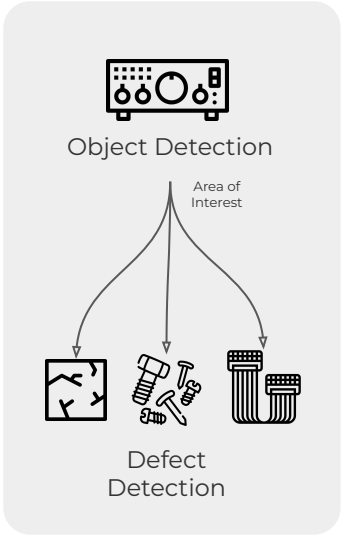
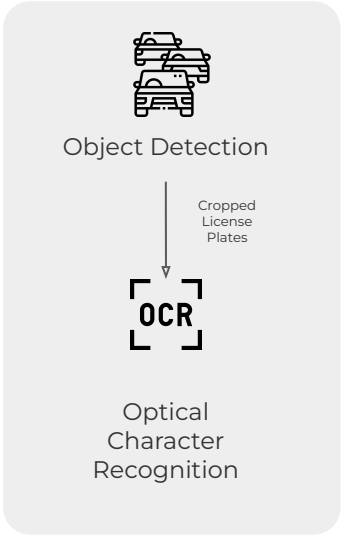
Examples of Model Cascading



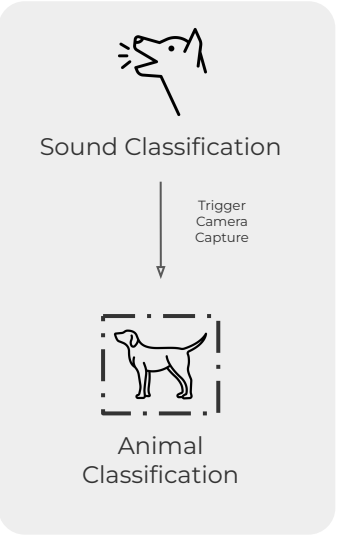
Production Line Monitoring

Run different defect detection models depending on the product



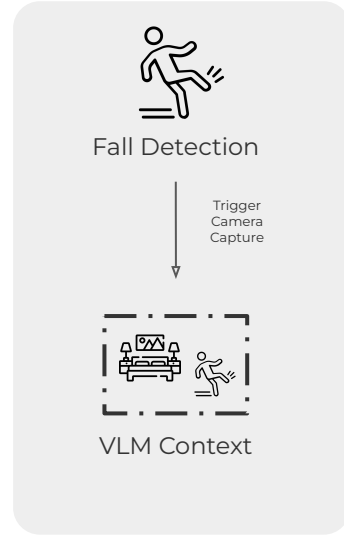
Vehicle Detection

Run different defect detection models depending on the product



Wildlife Monitoring

Use low-power sensors (microphone) to detect and wake, then use camera for context



Human Safety

Detect an event, then send to a generative model to provide context for first responders

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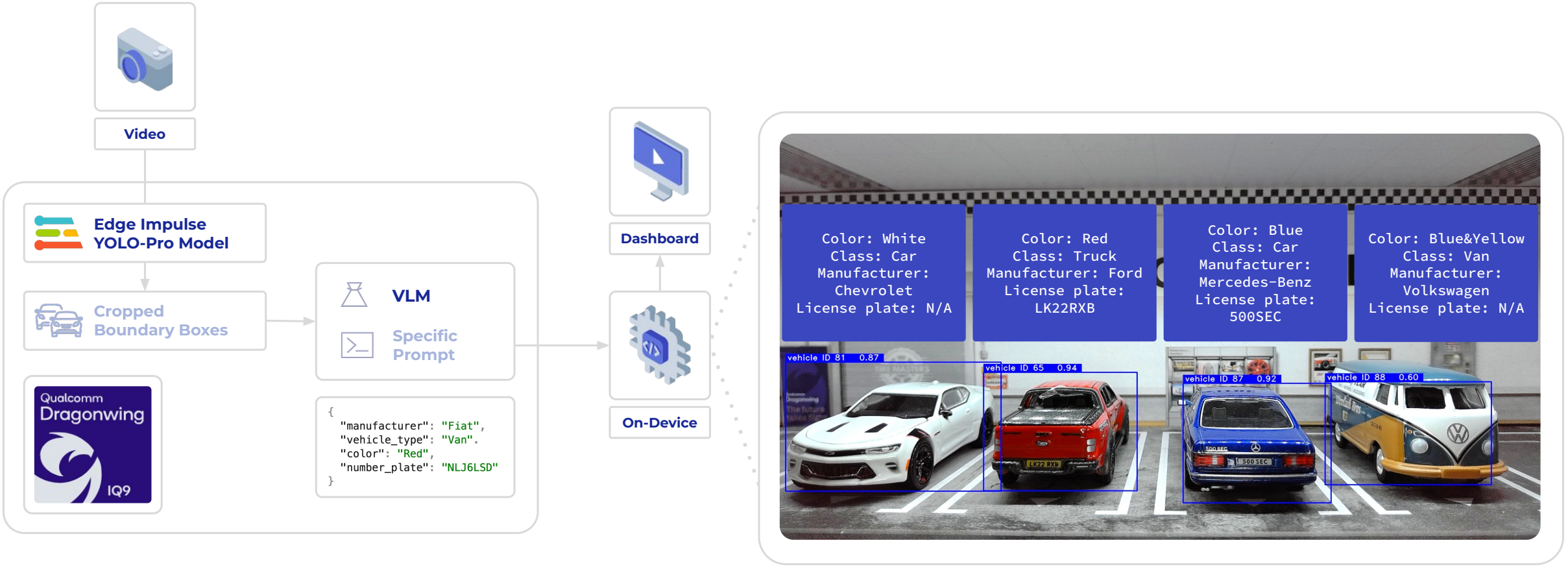
GenAI on the Edge: VLM Cascading



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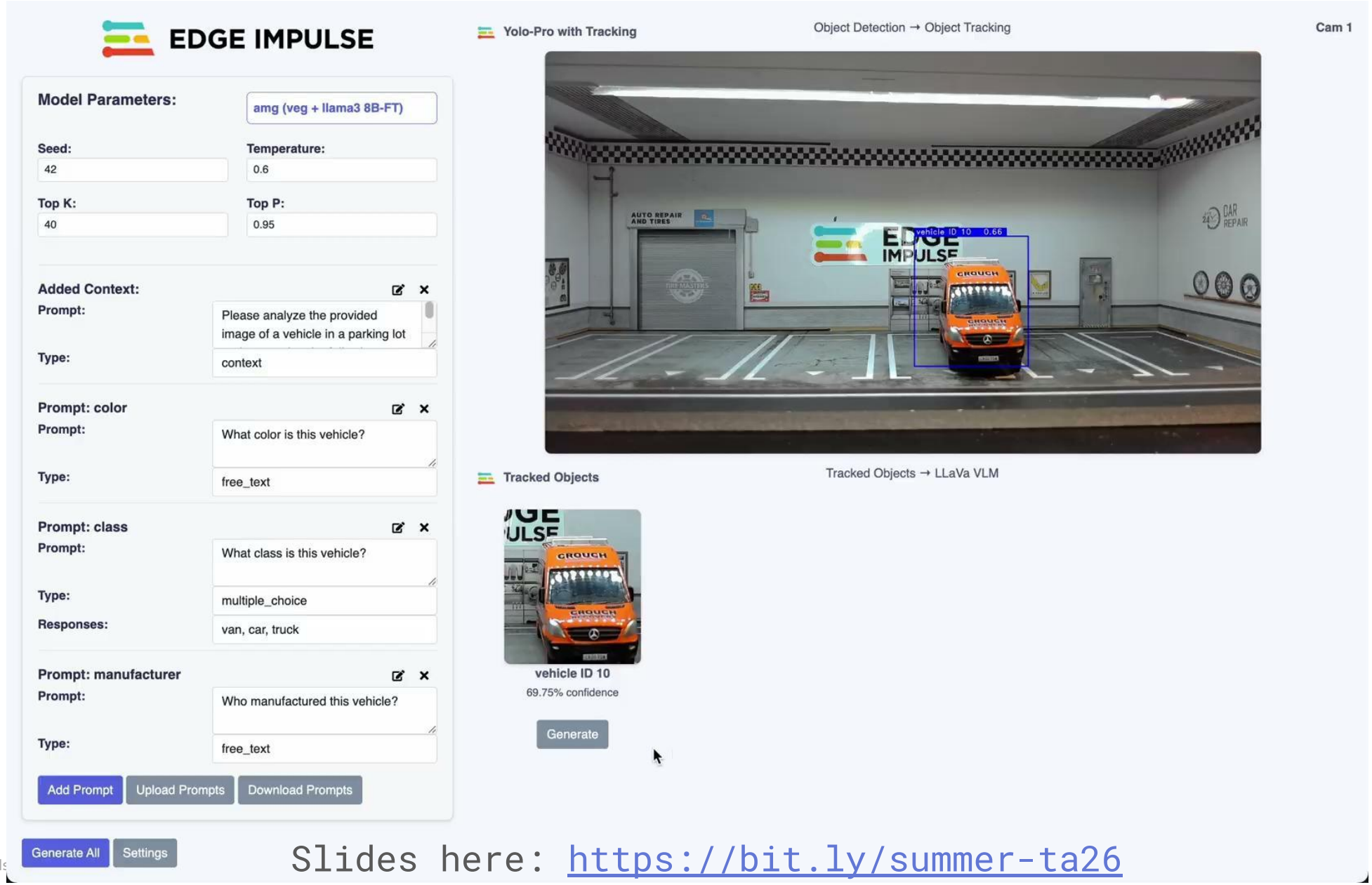

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Lessons Learnt!

1. **What is the AI, ML and Edge AI.**
2. **Train an Image Classification ML model.**
3. **Re-train an Object Detection ML model.**
4. **Test with the Arduino UNO Q.**
5. **Train a Visual Anomaly Detection ML model.**
6. **Learn about Model Cascading from Object Detection to VLM**

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One more thing

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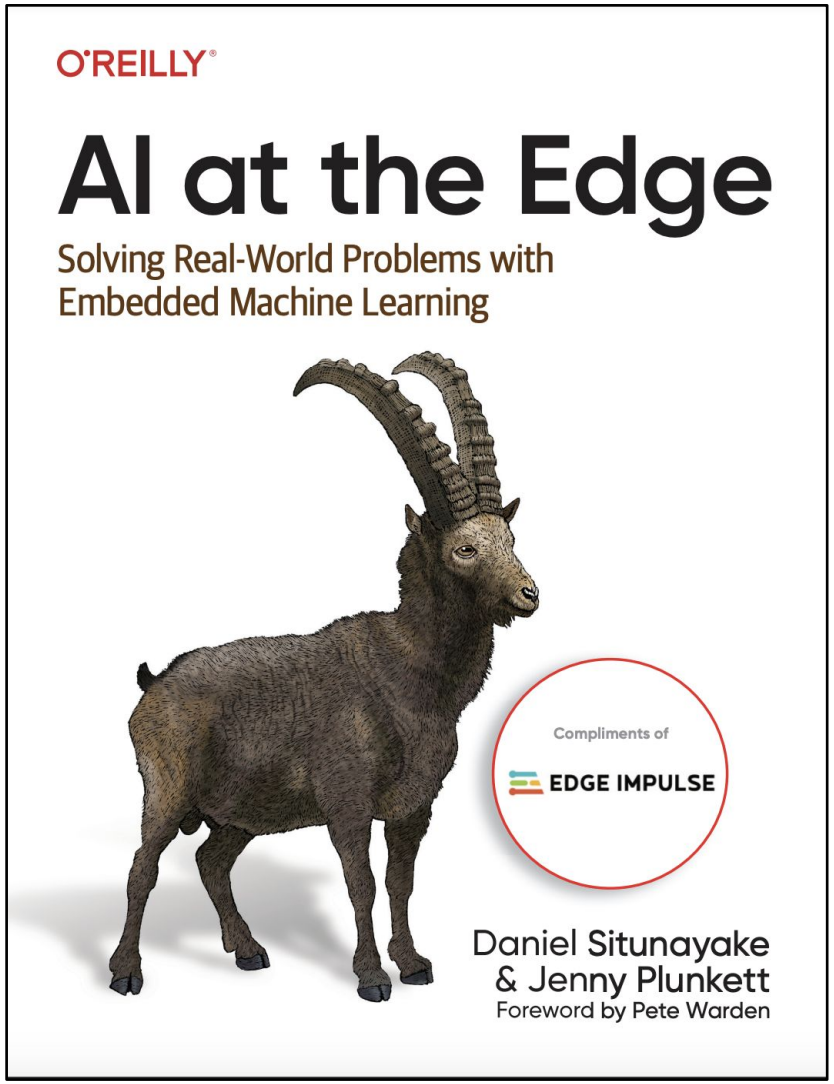

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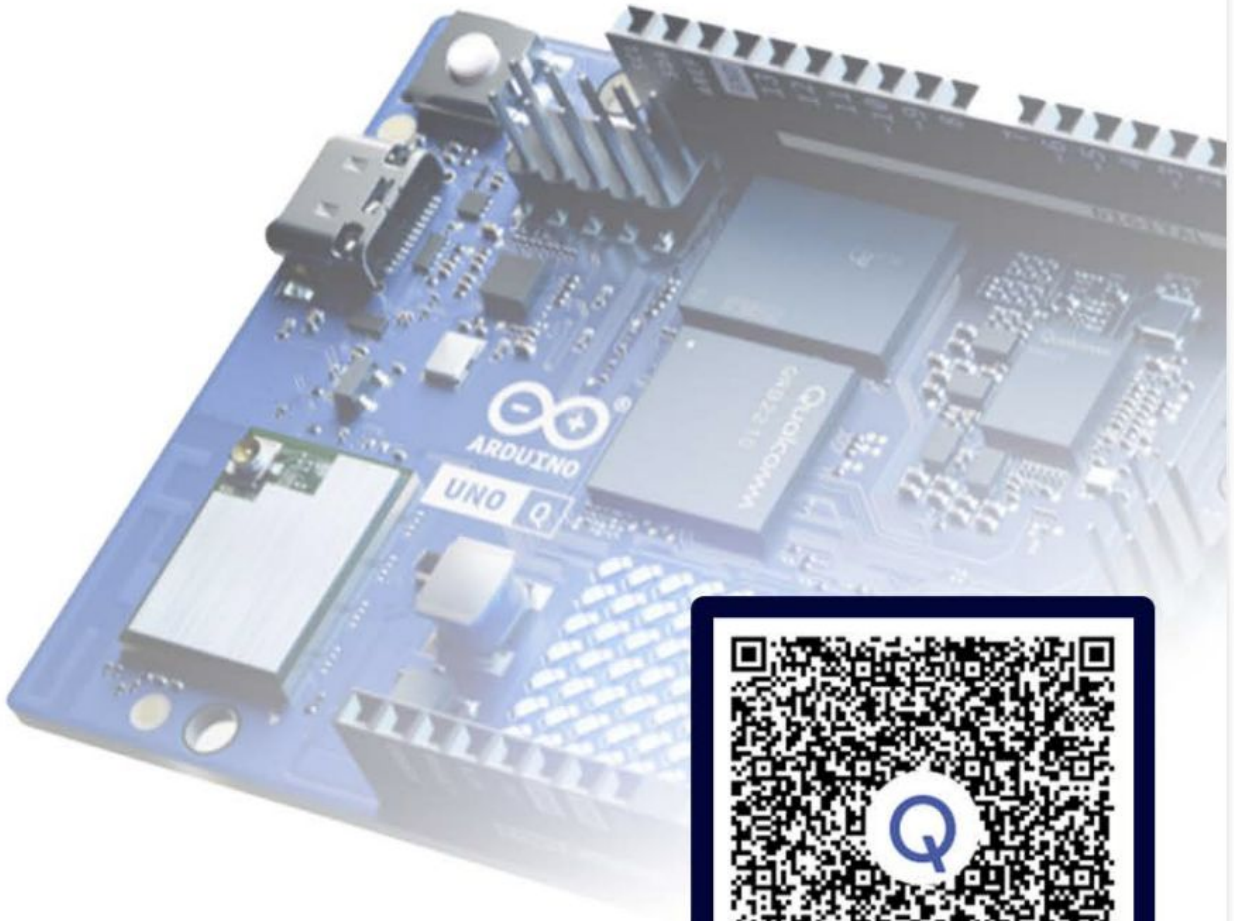

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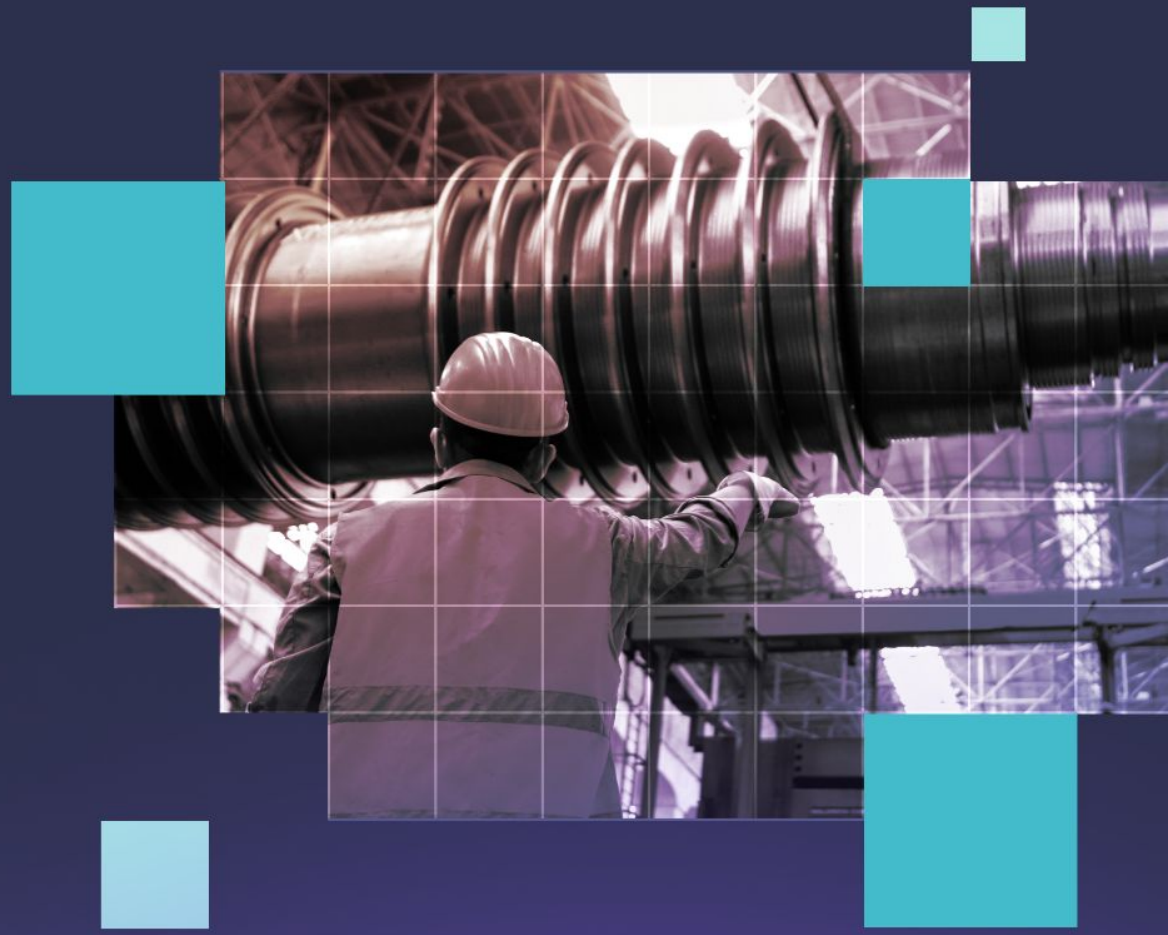

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